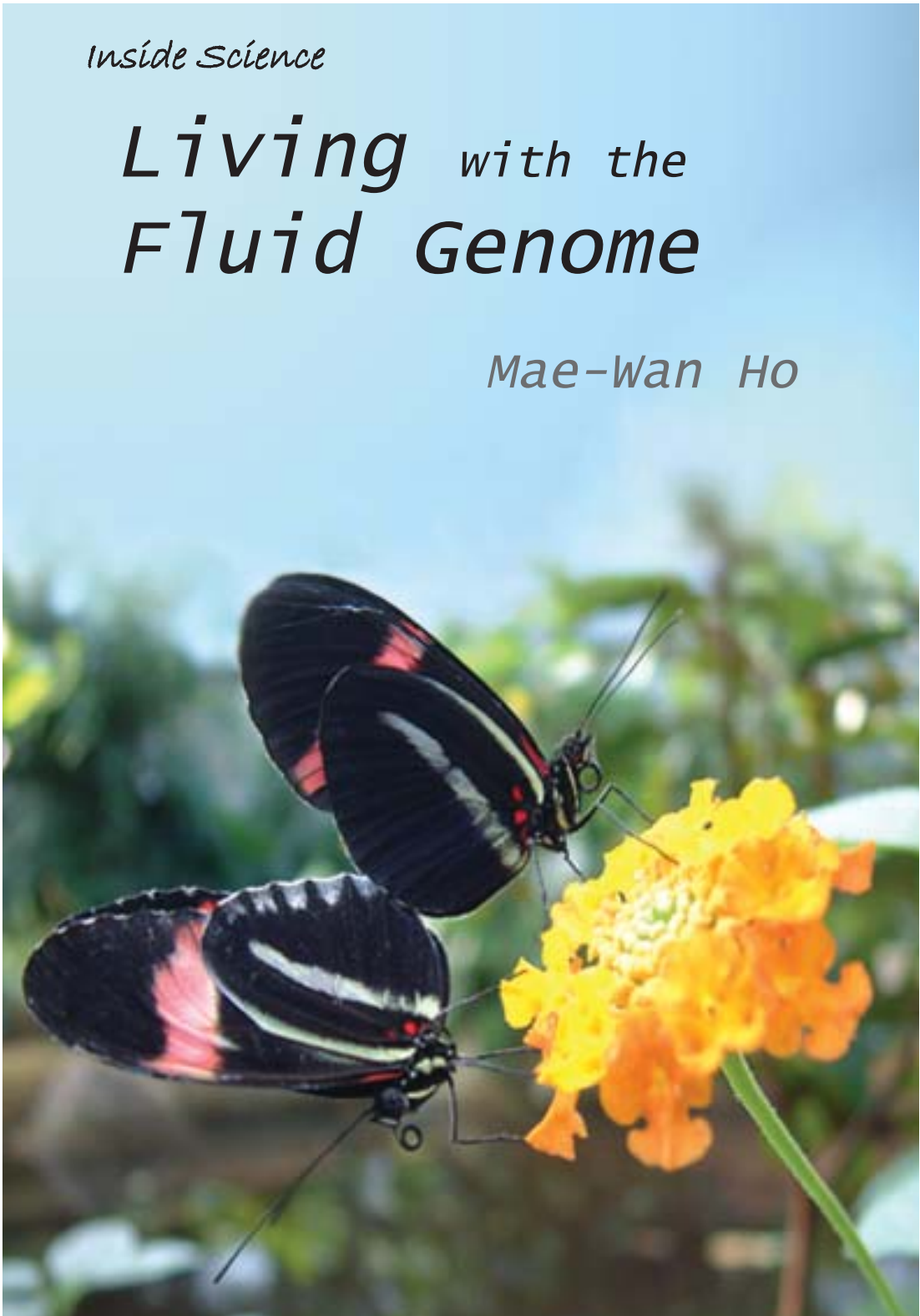


Inside Science

Living with the *Fluid Genome*

Mae-Wan Ho



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Living
with the
Fluid Genome

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Institute of Science in Society

TWN

Third World Network

Living with the Fluid Genome

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Preface

One of the most persistent dogmas in western science is genetic determinism, the belief that our genetic make-up, or our birth, ultimately determines who and what we are.

The modern science of genetics began with the twentieth century, though it is often said that genetics goes back to selective breeding, which probably arose with the first domestications of wild animals and plants by human beings some 10 000 years ago.

But the first agriculturalists, not unlike the indigenous farmers of today, would have used numerous varieties, each adapted to a different environment, whose aesthetic and cultural significance goes far beyond the one-dimensional measure of 'yield'. They would have discovered that one variety was never 'superior' to the others for all purposes or in all environments. In all likelihood, the 'genetic knowledge' that the first farmers acquired would have been much more multidimensional and ecological compared to genetic determinism.

The basic, unspoken assumption of genetic determinism is quite different. It is that all varieties or individuals with different genetic make-ups can be placed along a single line in order of superiority, ending with one that's *the* best. This same assumption is implicit in the idea of a supreme ruler mandated from heaven, the king, or god-king.

Throughout the ages, notions of 'bloodline' and 'noble birth' must have helped the ruling families retain political power within kin groups. The proverbial prince who, though forced to lead the life of a pauper as the result of political intrigue, nevertheless retained his princely powers, which enabled him to overcome all hardship to regain his kingdom as the rightful heir to his father, the deceased king.

It is not an accident that genetics is synonymous with the 'science of heredity' or the 'science of inheritance'.

Genetic determinism had also made it easy to condone, if not outright justify, slavery and racism. For, just as certain bloodlines are deemed inferior to others, so some 'races' are more 'primitive' or at a 'lower' stage of evolution than others. The 'master races', therefore, are meant to subjugate, conquer and even eliminate the rest, which include all who have different faces, speak with different tongues and live different lives.

This idea was first translated into science by Charles Darwin, the full title of whose famous book is, *The Origin of Species by Means of Natural Selection or the Preservation of Favoured Races in the Struggle for Life*.

It should come as no surprise that this same genetic determinist folklore was the guiding principle in the development of the modern science of genetics, which in turn reinforced the ideology, giving rise to the eugenics movement that lasted at least until the mid-1970s in the United States and Europe. Not only were 'inferior races' persecuted, 'inferior' and 'disabled' individuals were also considered 'unfit' and targeted for elimination.

Eugenics is surfacing again as human 'genomic' science, spawned by the sequencing of the human and other genomes, and is now promising to identify all the 'bad' genes that cause diseases and disabilities, so that they can be eliminated at conception or before birth, while the 'good' genes can be used for the 'genetic enhancement' of anyone who can pay for the privilege. This time round, eugenics will not be sanctioned by the state. It will be up to the 'global market' to decide. The poor will become a genetic 'underclass'. Social inequality will be redefined as, and transformed into, genetic inequality.

Fortunately, science as knowledge of nature is never just subject to our arbitrary whims and prejudices. We can delude ourselves, but only for so long. Nature has a way of fighting back, of puncturing our illusions.

One of the most important discoveries of modern genetics is the huge genetic diversity that exists in natural populations. Modern genetics has amply deconstructed the notion of race as a genetic entity, for the range of genetic variation *within* a population is much larger than the average difference between populations. Each individual is genetically unique, except for identical twins at the beginning of development. That is not all.

The story of the 'fluid genome', the main subject of this book, tells how geneticists came face to face with scientific findings that completely undercut the genetic determinist paradigm. It makes nonsense of all the eugenicist claims and promises, and exposes the futility as well as the hazards of genetic engineering. This important story is still largely untold, and is actively suppressed and misrepresented by the scientific establishment with its vested commercial and non-commercial interests in gene biotechnology.

As this book goes to press, the biotech corporate empire shows all the signs of crumbling, proving itself morally, financially *and* scientifically bankrupt. Yet our governments are appearing to be redoubling their efforts to prop up the sinking enterprise with huge public subsidies, and worse, attempting to force genetically engineered crops on their own citizens who overwhelmingly reject them, with the help of the scientific establishment.

Read on to find out why genetic engineering is hazardous, and why the whole biotech enterprise, from GM crops and gene drugs to health genomics, is a phenomenal waste of public finance and scientific imagination.

Most importantly, find out what it means to be liberated from the genetic determinist myth, and to be living with the fluid genome.

Mae-Wan Ho
February 2003
London

Acknowledgments

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Jonathan Matthews co-authored a paper that's the substance of Chapter 4. He has provided valuable insights into the political undercurrents of the genetic engineering debate.

Terje Traavik, with whom I have shared many 'road-shows', has greatly influenced my thoughts on the risks of horizontal gene transfer and 'naked' DNA.

Joe Cummins, my fellow traveller and collaborator since 1999 on genetics and biosafety, is tireless in reviewing and up-dating the scientific literature, and has taught me a great deal to make up for the deficits in my knowledge.

Teddy Goldsmith first introduced me to real ecology and has remained a constant source of inspiration in lively discussions over dinner at his home with him and his wife, Kathy.

Martin Khor, Director of the Third World Network (TWN), provided an unique insight into the shape of world politics that has informed my science ever since.

This book owes a lot to the enthusiasm and support of other colleagues in the Institute of Science in Society (ISIS) and TWN, past and present: Gurdial Nijar, Chee Yoke Ling, Lim Li Lin, Lim Li Ching, Chee Yoke Heong, Sam Burcher, Julian Haffeggee, Andy Watton, Angela Ryan and Nick Papadimitriou.

Lim Li Ching read and commented on several successive drafts of this manuscript, and Joe Cummins, Teddy Goldsmith, Martin Khor, Michel Pimbert Claire Robinson, Richard Strohman and Terje Traavik read and commented on the penultimate draft. The deficiencies that remain are entirely my own.

Finally, this book and much else would not have happened but for Peter Saunders, without whom nothing makes sense.

Dedication

To people everywhere who care

One

Why Science Matters

“Redefining the life sciences”

In July 1994, I was invited to a conference in Penang, Malaysia, on “Redefining the Life Sciences”, not in the least prepared for how radically it would change my life. The irony is that I had already 'redefined' a life science for myself some two years before, but had no idea I would be required to live up to it ever since.

Some 50 participants, consisting of scientists, social scientists, policy-makers and political activists, came from 15 countries; all at the behest of one or the other of the organizers: Martin Khor, Director of the Third World Network (TWN), Vandana Shiva from the Research Foundation for Science, Technology and Ecology in India, Tewolde Berhan Gebre Egziabher of the Institute for Sustainable Development in Ethiopia and Brian Goodwin, my then colleague at the Open University.

I learned how industrial agriculture was destroying the land and creating poverty on large scales. And just as people were recovering indigenous knowledge and plant varieties to regenerate the land and their livelihood, they faced the new threat of genetically engineered crops that would lock them back into the devastating agriculture of the green revolution, and worse.

Within a day, we were like friends who had lost touch for a hundred years. We drafted a *Scientists' Statement on the Need for Greater Regulation and Control of Genetic Engineering*, which was published by the Third World Network, and distributed widely.

That marked the beginning of my involvement in the great 'genetic engineering debate', which took me to numerous international and national forums and to local communities, where I learned, to my consternation and dismay, even more of what has been left out of the science curriculum in academia.

I soon found it necessary to widen the debate on genetic engineering to include science and science policy; and in 1999, co-founded the Institute of Science in Society (ISIS) with my husband Peter Saunders, to give more prominence to those issues.

It had quickly dawned on me how much science matters in the affairs of the real world, not just through practical applications like genetic engineering, but especially through a mechanistic world-view that makes people act unthinkingly, to shape the world to the detriment of human beings and all other inhabitants. I began to see how science can be used, often without conscious intent, to intimidate and control, to obfuscate, to exploit and oppress. I began to see how the dominant paradigm can generate a selective blindness in scientists, making them ignore and deny scientific evidence, or fail to interpret them correctly.

Gene biotechnology has all the hallmarks of bad science working together with big business against the public good, against public will and aspiration, and against the moral values of society.

What brought this bad science and big business together in the first place was the overriding belief that competition and the survival of the fittest result in progress and wealth creation. The 'fittest', in biological terms, are those supposedly endowed with superior genetic make-up. And as competition works in nature to improve the genetic make-up of species, it also seems right that by changing genes through genetic engineering, any desired organism could be created to serve our every need.

No one was prepared for the explosion of scientific findings that sent seismic tremors through the foundations of modern genetics just when genetic engineering and gene biotechnology got underway in the mid-1970s.

By the early 1990s, there was certainly sufficient evidence to suggest that the technology would not work, and is inherently hazardous. I said as much in a short paper written at the Penang conference, later published in *Third World Resurgence* with the provocative title, "Genetic engineering hope or hoax". Few people took that paper seriously, apart from my friends in TWN, who immediately saw the point, and have never wavered in their conviction since.

Genetic engineering of plants and animals - or as its proponents insist on saying, 'genetic modification', so as not to frighten the public - began more than 25 years ago. But neither the predicted financial boom nor the promised benefit to agriculture or health has materialised.

The myth of genetic determinism was well and truly exploded as the human genome map was announced in February 2001.

Craig Venter, breakaway 'shot-gun' (referring to a method of sequencing DNA) gene-sequencer, set up a private company Celera and threatened to overtake the international public consortium sequencing the human genome, the Human Genome Project. Eventually, a truce was called, and a ceremonious 'joint finish' declared by the Clinton administration in the White House in June 2000.

But Craig Venter's words eight months later sent the genomics stock market on a downward spiral: "We simply do not have enough genes for this idea of biological determinism to be right," he said, "The wonderful diversity of the human species is not hard-wired in our genetic code. Our environments are critical." Not surprisingly, Venter left, or was told to leave the company at the beginning of 2002, to set up his own non-profit organisation.

"I made a million dollars the hard way", Craig Venter was reported to have said, "I started with a billion dollars and worked my way down."

The GM enterprise is showing all the signs of sinking (see Box 1.1).

Genetically modified (GM) crops - crops with foreign genetic material inserted into them - first came on to the market in 1994 in the United States, in the form of the 'Flavr Savr' tomato, which had a gene to delay ripening. It was an agronomic disaster, and was quietly dropped several years later. There has been a string of other failures in GM crops that had been commercialised, and behind that, at least ten times the number that never made it to the market at all.

Comprehensive studies on a range of other GM crops carried out in US universities in the intervening years have shown that many yield less, perform poorly or inconsistently in the field, fail to reduce the use of agrochemicals, and result in drastically reduced overall income for farmers. This picture was borne out by a report from the US Department of Agriculture released in May 2002, which concluded, "GE [genetically engineered] crops do not increase yield potential and may reduce yields". And, "Perhaps the biggest issue raised by these results is how to explain the rapid adoption of GE crops when farm financial impacts appear to be mixed or even negative".

As if that's not enough, the GM varieties are genetically unstable, a major factor in the failures of GM crops (see Chapter 11); the new genes introduced into GM crops can make toxic and allergenic prod-

ucts; there's extensive GM contamination of conventional and organic crops; fields are plagued by herbicide-tolerant GM volunteers and superweeds; and antibiotic resistance marker genes and other bits of genetically modified DNA can spread to infectious bacteria and jump into the genome of cells including human cells.

There has been no success with GM animals either. Like GM plants, they do not breed true, and often suffer from gross abnormalities. That's why genetic engineers have resorted to cloning, i.e., creating genetically identical copies of an adult animal with proven qualities. But the clones turn out to be "a gallery of horrors". Cloned animals suffer from both gross and subtle abnormalities, and even the surrogate mothers carrying them can be badly affected.

Dolly, the celebrated cloned sheep born to the Roslin Institute in Scotland in 1996, was diagnosed with arthritis in 2002 and also developed lung disease. It had to be put down finally in February 2003 by its creator, Ian Wilmut, who had openly admitted that all cloned animals could be abnormal. We are now threatened with the commercial approval of these monstrosities as 'cloned meat' for our dinner table.

The United States Food and Drug Administration recently reported that the University of Illinois had marketed 386 pigs that were the progeny of transgenic animals. The recent incident was not an isolated one. In 2001, carcasses from transgenic pigs created at the University of Florida were sold for human consumption. The transgenic pigs were, in addition, treated with barbiturate drugs and chloroform. What else has been happening behind the scenes?

The same failures are documented in 'gene therapy', which is nothing but the genetic modification of human beings. The death of a healthy teenager in a clinical trial in 1999 forced a public enquiry, uncovering more than 650 'adverse events' and seven other deaths. This alerted the public to the hazards inherent to the technology, but was not enough to deter the gene-therapists. So disaster struck again. Cancer - a widely predicted risk - claimed its first gene-therapy victims between 2002 and 2003 (see Chapter 8).

The much-touted miracle gene-drugs are an ever-receding potential, as generic faults are being discovered in the handful of 'successes' (see Chapter 11).

Instead of cutting the losses and limiting further damages, the scientific establishment refuses to change direction, and our governments are organising massive bailouts of the biotech industry with public

funds at a time when our agriculture and national health are lurching from crisis to crisis under the failures of the old reductionist model.

Box 1.1

The GM debacle

A study carried out by the UK Soil Association and released in September 2002 concluded that GM crops were an economic disaster. They reportedly cost the United States an estimated US\$12 billion in farm subsidies, lost sales and product recalls due to transgenic contamination.

Massive failures of GM cotton, up to 100%, were reported in several Indian States, including failure to germinate, root-rot and attacks by the American bollworm, for which the 'Bt-cotton' were supposed to be resistant. A university-based study in India confirmed that the Bt-cotton was heavily infested (up to 80%) with the bollworm.

Biotech market shares peaked in autumn 2000, but have been falling sharply since, and performing well below the industrial average on both sides of the Atlantic. Tens of thousands have lost their jobs in mass layoffs from both the agricultural and genomics sectors. Most companies have suffered double-digit losses, and venture capital has dried up.

The cutbacks and spin-outs of agricultural biotechnology have been spectacular, and major retreats from funding academic research were staged in 2002.

Aventis was taken over by Bayer, which spun it out to Bayer CropScience. But Bayer has announced repeated job cuts in crop science amounting to a total of 15 000 over the next four years.

Monsanto suffered a series of setbacks in write-offs and lost profits after being taken over and spun out by Pharmacia. Pharmacia was later taken over by Pfizer on condition that it shed Monsanto. Just before Christmas, Monsanto's chief executive and president resigned, citing poor financial performance.

In September, Syngenta ended a three-year collaboration with UK's John Innes Centre that has hardly begun, abandoning the Syngenta Laboratory, part of a new Genome Centre complex completed eight months previously at the tax-payer's expense. Then, less than three months later, the corporation stunned the academic world by announcing the closure of its Torrey Mesa Research Institute

(TMRI) at the end of January 2003. TMRI, barely four years old, had led Syngenta's efforts to sequence the rice genome. And word is out that the controversial partnership between Syngenta and the University of California, Berkeley begun in 1998 will probably not be renewed when the \$25 million deal expires this year.

Meanwhile, evidence of the inherent hazards of GM has accumulated.

- Gene therapy claimed its first cancer victim last year, from among the only 9 cases of unequivocal successes in the world. The regulators pronounced the mishap 'rare' and regrettable, and gave the green light for similar trials to resume after several emergency meetings. The second case was discovered months later (see Chapter 8).
- GM soya-flour eaten in a milk shake and hamburger meal was found to transfer GM DNA to gut bacteria of human subjects, confirming fears that antibiotic resistance genes may spread to pathogens and new viruses and bacteria that cause diseases may be created (see Chapter 10).
- Multi-herbicide tolerant GM canola - that combined the single-herbicide tolerant traits created by different companies - has evolved rapidly in Canada and the United States, becoming serious weeds.
- Super-weeds tolerant to the herbicide Roundup are plaguing GM soya and cotton fields in the US.
- Both established seed stocks and indigenous landraces have been contaminated by GM varieties, posing serious threats to agricultural and natural biodiversity.
- Hazards associated with gene drugs may be generic. The body often produces antibodies against them, compromising their effectiveness, and in a minority of cases, causing serious illnesses and death (see Chapter 11).

Not surprisingly, there is worldwide rejection of GM crops.

Zambia made headlines in rejecting GM maize from the United States to alleviate the threat of famine in the country, on grounds that the safety of GM food is unproven at best and the threat of transgenic

contamination is all too real. Zambia received widespread support, and has galvanised African countries into uniting towards self-sufficiency and sustainability.

Media reports, such as the following, indicate the breadth and depth of opposition to GM technology:

All wheat buyers in China, Korea, and Japan have announced they will not buy the GM wheat that Monsanto is field-testing and planning to release commercially.

Italy's largest miller, Grandi Molini Italiani, has declared it will not import any GM wheat from the US, and may stop importing from other countries growing GM wheat.

Farmers and retailers in Switzerland have agreed *never* to produce or sell GM food.

Brazil's newly elected president intends to keep Brazil GM-free.

The elite French three-star chefs have launched a 'crusade' for a Europe-wide ban on GM crops and livestock.

China has cooled to GM crops after initial enthusiasm. Its Agriculture Ministry has just announced that China's northeast region of 127 counties will be developed into the world's largest producer of non-GM soya over the next five years, in a move to compete with foreign beans.

The two-way connection between science and society

Modern western science is distinctly mechanistic, tending to see nature in terms of separate parts that have no inherent connections with one another.

The machine metaphor in biology dates back at least to 17th century French philosopher René Descartes' concept of the body as a machine, separate from 'mind'. But it was Victorian England biologist Charles Darwin and his followers who completed the translation of the dominant social reality into scientific theory.

According to Darwin's theory of evolution by natural selection, species evolve and improve over time, thanks to the all-pervasive competition of one against all and all against nature.

Darwin was under the influence of the *laissez-faire* economic theory of Adam Smith, in which competition and the 'free' market - expanding behind the military might of the British Empire - were seen as the

key to economic success. Darwinism, and neo-Darwinism, its modern counterpart, in turn inspired the present neo-liberal economic theory, a more extreme version of Adam Smith's, as it involves unfettered competition that can only end in corporate capitalism, unmitigated by moral restraint.

The system of economic regulation and agreements set up after the Second World War has allowed transnational corporations to operate without responsibility or accountability. This same ideology is now driving economic globalisation that will destroy the livelihood of the poorest and threaten the survival of the global ecosystem. Many of the poorest countries already feel that they are targeted for elimination because they are surplus to requirement or unfit to survive.

This two-way connection between science and society is the clearest demonstration that science is not neutral or value free as is still widely assumed.

On a positive note however, this same connection between science and society opens the way to changing society through another kind of science, and that's what I hope we can do.

So let's begin again, with a brief account of the more authentic story of the gene that should have been told in all the text-books at least 10 if not 20 years ago, to see what morals can be drawn from the story.

Two

Reducing Life to Things

The modern science of heredity

The modern science of heredity began with Gregor Mendel, who experimented with garden peas in a monastery in Brno, Moravia, and published his theory that laid the foundations of modern genetics in 1859.

In the same year, Charles Darwin's *Origin of Species* took Victorian England by storm. He said that species were not created fully formed by the Christian God, but were evolved gradually through natural causes. According to Darwin's followers, Mendel's genetics would have completed Darwin's theory, which lacked a mechanism for heredity. But Mendel languished in obscurity; the book he sent to Darwin remained uncut and unread. It took more than 40 years to the beginning of the twentieth century before Mendel was re-discovered to inaugurate modern genetics.

The early history of modern genetics was dominated by the search for the *material* basis of heredity, *what* it was that makes offspring resemble parents, and different from other individuals. That was the first act of reduction and reification: it collapsed a whole way of life, an entire process of reproduction and development to some material entity that must be invested with explanatory powers that rival miracles.

The search for the material basis of heredity culminated in *deoxyribonucleic acid* or DNA. The high point came when James Watson and Francis Crick in Cambridge University worked out the double-helix structure of DNA, earning them a Nobel Prize that was shared with Maurice Wilkins at King's College, London, who showed Watson and Crick clinching X-ray diffraction data on DNA that belonged to Rosalind Franklin. Franklin died of cancer contracted from the X-rays of the diffraction chamber, a successful scientist at the age of 37, totally unaware that her data had been stolen by colleagues who would go on to win the Nobel Prize that should rightly have gone to her.

Starting with the basics

The genetic material, DNA is usually found in every cell. A *cell* is a unit of the living organism. Micro-organisms such as bacteria and protozoa have only one cell, whereas plants and 'higher' animals have many cells. The cells make up tissues and organs. The cell and its contents can be seen with the help of increasingly powerful microscopes (Fig. 2.1).

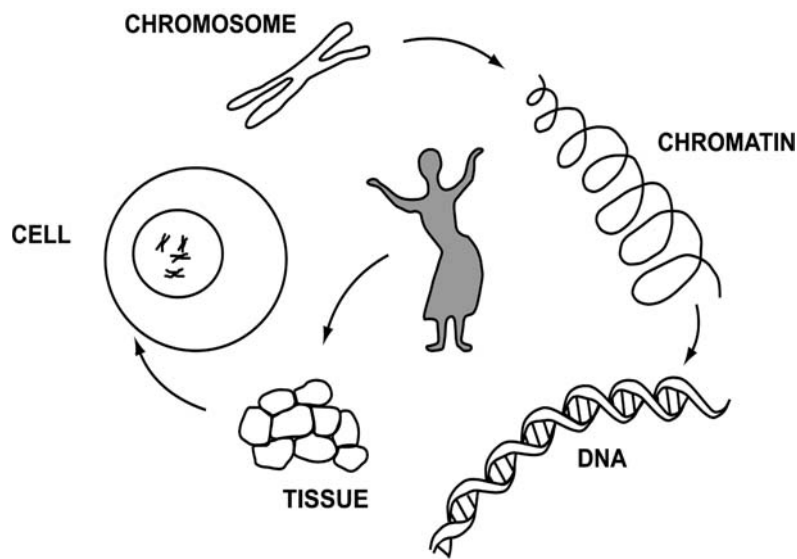


Figure 2.1. From organism to DNA.

Inside the cell of organisms such as the maize plant or the human being, the genetic material is enclosed in a spherical compartment, the *nucleus*, and packaged into long compact structures called *chromosomes*. The totality of all the genetic material packaged into chromosomes is the *genome*. Each species has a different genome. For example, there are 23 pairs of chromosomes in the human genome, one of a pair from each parent.

Bacteria have chromosomes that are *not* enclosed in a nucleus. They are referred to as *prokaryotes* to distinguish them from the *eukaryotes*, higher organisms whose genomes are enclosed in a nucle-

us. The *E. coli* bacterium, which lives in the gut of mammals and human beings, has only one chromosome in its genome.

Each chromosome is a very long molecule of DNA wound up and coiled around special proteins to form *chromatin*. (In animals and plants, each chromosome is duplicated but remains joined up at one point.) The DNA molecule itself, when stripped of all the bound proteins, consists of two strands wound around each other in a double helix. Each strand is made up of units joined end to end. There are four different units in DNA, labelled with the letters A, T, G, and C, which stand for the identifying *base* for each unit, adenine, thymine, guanine and cytosine, linked to a sugar (deoxyribose) that's linked in turn to a phosphate group (Pi). This unit is called a *nucleotide* (Fig. 2.2).

Pi - sugar - base

Figure 2.2. Structure of a nucleotide.

The bases of the two DNA strands pair up, with the bases sticking out flat at right angles from the backbone of each strand, which consists of alternating phosphate and sugar (Fig. 2.3).

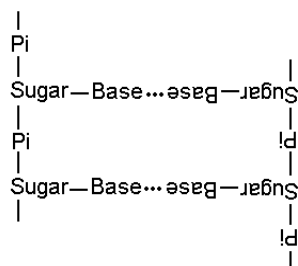


Figure 2.3. Structure of DNA showing anti-parallel strands.

Notice that the two strands of the DNA are *anti-parallel*, each going in opposite directions. To indicate this anti-parallel relationship, the direction of each strand is labelled as the 5' and 3' end respectively (the numbers refer to the bonding structure between the sugar and the phosphate in the backbone).

Significantly, the base-pairing between the strands is precise:

A pairs with T, and G with C. The double helix looks like a spiral ladder, with the paired bases forming the rungs of the ladder (Fig. 2.4).

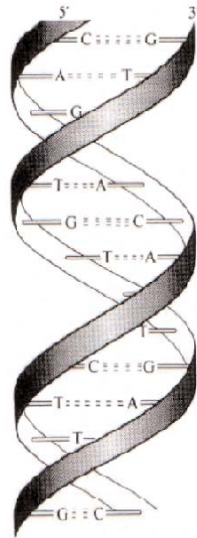


Figure 2.4. The double helix structure of DNA with complementary base-pairing.

On account of the specific base-pairing, the sequence of the bases on one strand is *complementary* to that on the other. So each strand is a template for making the other strand, and this provides the mechanism for exact replication, which is one of the functions of the genetic material, to account for constancy or sameness. Figure 2.5 depicts the 'semi-conservative' replication of DNA: the two strands of a DNA molecule unwind and split apart, each strand is a template for making a new complementary strand, and as a result, each of the two new DNA molecules consists of one old strand and one new strand.

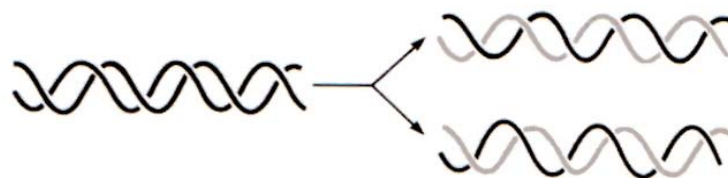


Figure 2.5. How DNA can replicate exactly.

DNA is the genetic material in all organisms. Many *viruses* - genetic parasites that depend on the cell to multiply and hence are not considered real organisms - make use of ribonucleic acid or RNA as genetic material. RNA is similar to DNA except that the sugar is *ribose* instead of deoxyribose, and in place of the base thymine (T), it has uracil (U). Furthermore, RNA does not usually exist in double-stranded forms. RNA is involved in *transcribing* (copying) the base sequence of DNA in the first step of protein synthesis (see later).

The genetic material is replicated and passed on from one generation to the next in reproduction, and is indeed a strong candidate for explaining why offspring should resemble their parents.

Besides providing for its own replication, certain stretches of DNA, called *genes*, specify the structure of proteins through a *genetic code* that translates from the base sequence of the DNA to the structure of a particular protein.

Proteins consist of *polypeptides*, which, like DNA, are made up of units joined end to end. Instead of nucleotides, however, the units in proteins are *amino acids* linked together through peptide bonds to give a polypeptide chain. Peptide chains fold up into a three-dimensional structure after they are synthesized, and two or more polypeptide sub-units generally make up a protein.

There are twenty different amino acids, and each polypeptide has its own sequence of amino acids that's suited for a given function. Proteins are responsible for practically all the activities that keep organisms alive. Most proteins are *enzymes* that catalyze chemical reactions so that energy can be extracted from foodstuffs and used to support the organism's activities or to build up its cells and tissues. Each enzyme acts on a specific chemical substance, its *substrate*, which it somehow recognizes from among the thousands of other chemical species, or

metabolites that are being broken down, built up or transformed within the body.

So the genetic code must translate a sequence of bases in the DNA into a sequence of amino acids in the polypeptide.

Working out the genetic code was another scientific *tour de force*. It turned out that three successive bases on one of the strands form a 'triplet', each triplet coding for one of 20 different amino acids that are strung together to make a polypeptide. There are 4^3 ($4 \times 4 \times 4$) or 64 possible triplets from 4 bases, so more than one triplet often code for the same amino acid, and there are triplets for 'start' and 'stop'.

To make a polypeptide encoded by the gene, the gene is first copied, or transcribed from DNA to RNA, through the same template mechanism, so that the base sequence in the resultant RNA is complementary to that in the gene. Only one of the two strands of the DNA double helix is the coding sequence. The RNA transcript is then processed - trimmed and modified - into *messenger RNA*, before its base sequence is *translated*, three at a time into the amino acid sequence of the protein (Fig. 2.6).

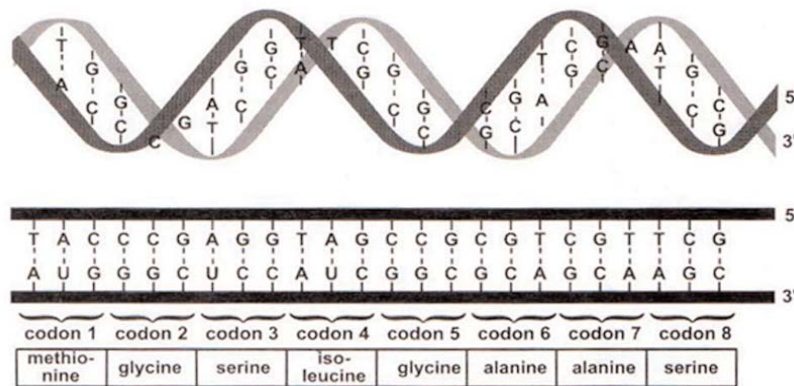


Figure 2.6. Genetic information flows from DNA to RNA to protein.

As said earlier, proteins perform all the vital functions in the organism; and it is the amino acid sequence of each protein (and its folded three-dimensional structure) that makes it especially suited for carrying out its specific task. So when the DNA base sequence is altered, it

could change the amino acid sequence of the protein encoded, which could in turn affect its function. These alterations are referred to as *mutations*, and neatly account for how differences and novelties may be generated.

Certainly, we know that mutations in single genes can have devastating effects. For example, sickle cell anaemia, common among Africans, and Tay-Sachs disease, common among Ashkanazi Jews, are both traced to mutations that change a single base in the gene, which leads to an amino acid substitution in the polypeptide encoded. There are many other changes in the base sequence of genes that do not result in significant differences to the function of the polypeptide encoded, however. That's because the physics and chemistry of the protein has not been sufficiently affected.

The picture built up from those early molecular investigations was that of a mechanical process: DNA base sequence is transcribed to a complementary sequence in RNA, which is then translated into a sequence of amino acids that make up a particular polypeptide chain. Francis Crick introduced the now infamous 'Central Dogma' of molecular biology: genetic information flows in one direction only, from DNA to RNA to protein, and never in reverse. It encapsulated the basic assumptions of genetic determinism in their modern form (see Box 2.1), reinforcing the general folklore that our genetic inheritance ultimately determines who and what we are.

Box 2.1

Basic assumptions of genetic determinism

- Genes determine characters in a straightforward, additive way: one gene-one protein, and by implication, one character. Environmental influence, if any, can be neatly sorted from the genetic.
- Genes and genomes are stable, and except for rare, random mutations, are passed on unchanged to the next generation.
- Genes and genomes cannot be changed *directly* in response to the environment.
- Characteristic acquired in life are not inherited.

Epigenesis and preformation

For centuries, *the* major debate in biology was whether the organism was already *preformed* in the 'germ', with development a mere unfolding of all the predetermined potentials, or whether the organism was built up *epigenetically* in a process of development that was subject to influences from the environment. One extreme doctrine of preformation maintained that the egg or sperm contained a minute person - a *homunculus* - that simply increased in size during development. This debate in biology is reminiscent of the theological debate between those who believed in *predestination*, the idea that one's destiny is ordained by God, and others who insisted, instead, that we are endowed with free will, so our destiny is in our own hands. Western science and Christianity have much in common.

The biological debate was resolved in favour of epigenesis when microscopes became powerful enough to show that organisms develop from a fertilised egg that's almost completely devoid of structures. Practically all the myriad structures and morphology of the organism are *generated* in the course of development.

Preformation and genetic determinism

But the idea of preformation still lingers on in genetic determinism: the claim that the process of development itself is specified in detail in the genes, in the so-called 'genetic blueprint'. Genetic determinism is a foundation concept of neo-Darwinism.

Neo-Darwinism says that organisms evolve by the natural selection of *random* genetic mutations. 'Random', in the sense that the gene mutations are not related to or affected by the 'selective force', or 'selective pressure' in any way. (Note the explicitly mechanistic language.)

Or as Harvard biologist, Ernst Mayr, the doyen and grand old man of neo-Darwinism said, evolution is a two-stage process: first the random variation, then the natural selection.

Another key assumption in neo-Darwinism is German biologist August Weismann's dictum that the 'germ plasm' is immortal and inviolable, rather like the eternal soul of Christianity that it has displaced. In modern language, the cells of the body - *somatic* cells - can mutate and change, but the genetic material in the *germ* cells (sperm and egg cells that give rise to the next generation) are somehow immune from all

environmental exigencies. There is said to be a 'Weismann's barrier' forbidding environmental influences from changing the genes directly, especially in the germ cells that give rise to the next generation.

Thus, so long as genes determine characteristics in a rather straightforward way, and the genes cannot be changed directly by the environment, then the only 'force' that shapes one's destiny as well as the whole of evolution is natural selection, the survival of the fittest.

This theory is very comforting for those who derive the most benefit from the *status quo*, and they are responsible for a huge 'Darwin industry' dedicated to 'explaining' why this is the best of all possible worlds, which is masquerading as mainstream science.

Three

Genetic Determinism & Corporate Capitalism

Genes for everything

From the late 1970s and well into the 1990s, as commercial genetic engineering expanded, there was a simultaneous proliferation of hypothetical genes for every imaginable human 'trait' that was to be 'explained' in neo-Darwinian terms, i.e., how it could have been 'selected for' by natural selection. Genes were in the air, especially as the Human Genome Project to obtain the genetic 'map' and the base sequence of the entire human genome was announced. Genes were as good for selling newspapers as for getting grants to do research, however banal.

There were genes for men philandering, for women preferring rich and powerful mates, for intelligence, alcoholism, criminality and even homelessness, reified attributes that overwhelmingly arise in social situations (Fig. 3.1).



Figure 3.1. Genes for everything.

One must bear in mind that all molecular biology could tell us about genes is that they code for proteins. How proteins contribute to the morphology and structure of organisms, let alone behaviour, requires other kinds of explanations in terms of physics, chemistry, mathematics, cell biology, physiology and ecology, for a start, that have simply been ignored or left out, or else reinterpreted to fit in with neo-Darwinian theory.

'Evolutionary' studies became a vast junkyard of just-so stories of 'selective advantage' attached to this or that bit of anatomy or behaviour, i.e., how it could increase the competitive, and therefore reproductive advantage of the individual possessing that gene that codes for that bit of anatomy or behaviour.

Many biologists, along with the general public, came to see the living world purely in terms of genes and DNA. There are no ecosystems, societies or communities, only selfish individuals competing against one another. There are no organisms either, only collections of genes all competing to replicate.

Even now, broadcasters, science journalists and sociologists are still asking for 'explanations' for art, language, consciousness, thought, whatsoever, solely in terms of its 'selective advantage' or 'what function it serves' to give the individual possessing that 'trait' the competitive edge in siring offspring. (And of course, men are much better than women at doing that, which explains why they are promiscuous, so we are told.)

It would be a mistake to ignore those stories, however. Not only do they constitute bad science in being devoid of intellectual content, thereby causing whole generations of biology students to fail to think deeply or logically, they serve to reinforce some of the most pernicious prejudices about human nature. And worse, they lead people to behave badly through the self-fulfilling prophecy.

“Selfish genes” rule

Oxford biologist Richard Dawkins has been the most influential defender and popularizer of neo-Darwinism and genetic determinism in Britain.

In his book, *The Selfish Gene*, he wrote,

“If you look at the way natural selection works, it seems to follow that anything that has evolved by natural selection should be selfish. Therefore we must expect that when we go and look at the behaviour

of baboons, humans and all other living creatures, we will find it to be selfish. If we find that our expectation is wrong, if we observe that human behaviour is truly altruistic, then we will be faced with something puzzling, something that needs explaining.”

He justifies that some pages later,

“We are all survival machines for the same kind of replicator - molecules called DNA - but there are many different ways of making a living in the world, and the replicators have built a vast range of machines to exploit them...”

So the vast complexity of the living world is nothing more than different ways for DNA to become replicated, hence the title of his book. All else are secondary details that need not concern us,

“...Exactly how [genes specifying proteins] leads to the development of a baby is a story which it will take decades, perhaps centuries, for embryologist to work out. But it is a fact that it does. Genes do indirectly control the manufacture of bodies and the influence is strictly one way: acquired characteristics are not inherited... Each new generation starts from scratch. A body is the genes' way of preserving the genes unaltered.”

E.O. Wilson, Harvard biologist whose specialty is ants, founded the discipline of Sociobiology in the early 1970s. Sociobiology - the modern counterpart of the Social Darwinism of the 19th century, which formed the ideological backdrop to the eugenics movement that led eventually to the persecution of indigenous peoples, the Jews and anyone else considered 'unfit' - attempted to explain all social behaviour, from ants to human beings, in terms of natural selection. Like Richard Dawkins, E.O. Wilson emphasized selfishness and the primacy of replicating and preserving the genetic material,

“True selfishness... is the key to a more nearly perfect social contract.... Human behaviour... is the circuitous technique by which human genetic material has been and will be kept intact. Morality has no other demonstrable ultimate function.”

Dawkins has written several more books on the same theme, ultimately reducing and reifying all of life and life's wonders to the replication of "selfish" genes. To this day, Dawkins has resolutely refused to learn any new genetics that contradicts his neo-Darwinian dogma.

Dawkin's faith in natural selection is nothing short of the theistic fundamentalism that he is so ferociously attacking. In a recent BBC radio programme, he accused the creationists (who oppose evolution,

and believe that God created every single species on earth) of preventing biology students from learning about natural selection, “the fundamental theorem of their own subject”.

The greatest danger of such DNA-centred biology is that people end up denying and explaining away all that's good in organisms and human beings. Altruism, love and compassion are seen as so many different forms of disguised selfishness. It serves to reinforce the dominant social reality of corporate capitalism that sanctions exploitation of the weak, the poor and politically dispossessed by the strong, the rich and the powerful. It is the case of the self-fulfilling prophecy.

Richard Dawkins has found it necessary to refute this charge by insisting that the human brain is so evolved that it can transcend our Darwinian origins. “We have the power to defy the selfish genes of our birth,” he declared. But to this day, he cannot tell us where this miraculous power comes from that enables us to “rebel against the tyranny of the selfish replicators” or against the innate competitiveness and aggressiveness of human nature.

Richard Dawkins and E.O. Wilson are not unusual among evolutionists. The cult of competition is endemic to evolutionary biology starting from Darwin himself.

Darwin owed a big intellectual debt to his contemporary, Victorian economist Thomas Malthus. Malthus' theory of population stated that human populations tend to increase exponentially, outstripping the means of subsistence. Hence, populations are inevitably kept down by wars, famines and diseases, and the influences of 'misery and vice'. From this grim theory, Malthus concluded that the poor laws of Victorian England, which conferred doles and bounties upon large families, were to be condemned, as they would exacerbate the very evils they were supposed to combat.

Malthus' theory has been severely criticised through the ages. Natural populations are actually held in check predominantly by social and cultural factors, and cannot be simplistically represented in the way that Malthus had done. Nevertheless, Malthus' vision remains a lasting influence, and there are echoes of it in the erroneous association, so frequently made, between famine and population explosion in the popular and not so popular media.

So-called 'population explosions' are often fictional. Famines have more to do with localised climatic fluctuations within a country that can be alleviated, but for the lack of public finance and infrastructure to pur-

chase food from regions in surplus to distribute to regions in need. This is so in the latest famines to hit southern Africa and Ethiopia. Many, if not most of the countries have surplus food in regions not hit by the drought. Their problems are currently being addressed by targeting aid in more effective ways.

The Zambian government, after rejecting GM food aid, has promptly released money to purchase and distribute food within the country. Europe and member states have responded generously, providing more than 382 million euros not tied to the purchase of GM food, or indeed other heavily subsidised food shipped from the North.

Forcing starving countries to accept GM food aid has been widely condemned across the world. UK's environment minister, Michael Meacher, said it was "wicked", particularly when there is an excess of non-GM food available.

As far as Darwin was concerned, Malthus' theory provided just the conditions for a relentless competition amid scarcity of resources, which would result in 'the survival of the fittest', the famous phrase coined by Social Darwinist Herbert Spencer, who had no qualms in re-importing Darwinian natural selection back to human society.

Many otherwise intelligent sociologists and self-professed Marxists today have become apologists, if not outright supporters of corporate capitalism and the particular model of 'globalisation' that the corporate powers are forcing on the world through international economic institutions such as the World Trade Organisation (WTO). The WTO is intent on removing all barriers to trade, finance and investment, so poor Third World countries can better be exploited. At the same time, unequal terms of trade, corporate and farm subsidies and 'intellectual property rights' are awarded to protect the profits of the rich and powerful countries, chief among which are the United States and Britain. Herbert Spencer and other Social Darwinists of the 19th century would have been very comfortable with this regime that promotes 'market competitiveness' above all, by means fair or foul.

I am reliably informed that Dawkins' fans include both Robert Shapiro, ex-, ex-, ex-Chief Executive Officer (CEO) of Monsanto, and Bill Gates, ex-CEO of Microsoft, well known for his aggressive approach to marketing, and to wiping out competitors. Bill Gates was reputed to have run his company along Darwinian lines, with different groups and individuals competing against one another to produce the best in the shortest time.

Bill Gates endowed a Chair for the Public Understanding of Science in Oxford University for Richard Dawkins. Dawkins has spared no effort since in propagating his neo-Darwinian views, and became Fellow of the Royal Society several years ago, an honour reserved for the best scientists.

The cosy relationship between the scientific establishment and corporate business runs deeper than ideological kinship. An 'academic-industrial-military complex' has matured with the rise of gene biotechnology that's increasingly active in suppressing scientific dissent in the genetic engineering debate, threatening the survival of science and endangering lives, as you shall see in the next chapter.

Four

Engineering Life and Mind

Science to order

During most of the period of the Cold War, nuclear (atomic) physics was the big science that got the lion's share of public funding. The atom bomb is the ultimate deterrent to war, we are told, as it is so utterly destructive.

Within the past two decades, giant biotech corporations have been increasingly dictating the kind of science and scientific research that gets done, which ensures their survival and prosperity, so they can the better exploit the masses for further gain.

Under the banner of the 'free market' and 'free choice', we are losing our right to self-determination and self-sufficiency in every aspect of our daily lives: our food, health, social mores, the way we choose to live and most seriously of all, our right to think differently from the corporate establishment.

The suppression of scientific dissent is one of the most serious and visible signs of the 'academic-industrial-military complex' at work. It goes against the very grain and essence of what science ought to be: the open, disinterested enquiry into the causes of natural processes. Of course, the scientific establishment has never lived up to this ideal, and it now has big industry and big government to bring out its worst

In the summer of 1998, senior scientist Arpad Pusztai from the Rowett Institute in Aberdeen, UK, told the public in a TV interview that feeding young rats GM potatoes appeared to harm them. As a result, he lost his job, his research group was disbanded, and a gagging order placed on him, on pains that his colleagues, including his wife, biochemist Susan Bardocz, would also be sacked. Bardocz, and Pusztai's collaborator, Stanley Ewen, were both retired early subsequently.

Pro-biotech scientists and Fellows of the Royal Society - *the* top, and hitherto most respected, society of scientists in the UK - were unanimous in their condemnation. Sir Robert May, the then Chief

Scientific Officer, said Pusztai had violated every cannon of scientific rectitude. He had 'spilled the beans' before the scientific findings went through the peer-review process for publication, causing undue public alarm and damaging the biotech industry. His integrity as a scientist was called into question. The Royal Society issued a hasty official report discrediting Pusztai's findings.

Suppression is particularly fierce when it comes to getting adverse results published, as Pusztai and Ewen discovered when they submitted a paper reporting part of their results to *The Lancet*. The anonymous review process took a year. On the eve of the paper's eventual publication, another prominent Fellow of the Royal Society, Peter Lachmann, allegedly threatened *The Lancet's* editor in an attempt to block it.



Figure 4.1. The Pusztai affair.

A campaign to discredit Pusztai has continued to this day. And he is no longer alone. Other scientists who have reported adverse findings on GM are either not heard of again, or subjected to the same

concerted attack by the pro-GM establishment (see below).

For every Pusztai, there probably are at least ten other casualties that did not get public airing. Add to that numerous cases of censorship from employers and even self-censorship, scientists keeping quiet on damning findings or arguments for the sake of keeping one's job, or simply not wanting to 'get involved' and complicating one's life.

In September 2000, trade union leaders warned that the integrity of British science was being threatened by "a dash for commercial cash". The Institute of Professionals Managers and Specialists carried out a survey of scientists working in government or in recently privatised laboratories. One-third of the respondents had been asked to change their research findings to suit the customer's preferred outcome, while 10% were pressured to bend their results to help secure contracts. Britain's handful of top research universities are dependent on private funding, often amounting to 80-90% of the total research budget.

The four unions representing scientists and technical staff in Britain launched a charter, saying that research must be guaranteed "by peer review, open publication and by autonomy over a significant proportion of its resources."

But the Royal Society, the UK Parliament and the House of Lords merely got together with a transmogrified PR company, Social Issues Research Centre (SIRC), funded by the food industry, and the Royal Institution (with a long and distinguished history of presenting science to the public beginning with pioneering physicist Michael Faraday), to issue a set of 'guidelines' for journalists and scientists clearly aimed at preventing the public report of 'adverse' findings.

In January 2001, a new Science Media Centre was announced, supported by Science Minister Lord Sainsbury, who has strong commercial interests in GM technology, to be housed in the Royal Institution headed by Baroness Susan Greenfield, Oxford neurobiologist well known for popularising brain science and also an advisor to SIRC. Its aim is to help "sceptical and impatient journalists" get their stories right on controversial issues such as "animal research, cloning and genetically modified food".

In June 2002, this Science Media Centre was caught trying to suppress a BBC drama, "Fields of Gold" that attempted to draw attention to horizontal gene transfer, the most insidious hazard of genetic engineering (see Chapters 8-10).

It is notable that in the entire Pusztai episode, corporate business has been quite invisible (though there had been unconfirmed rumours of phone calls from Monsanto to the Rowett Institute); academic institutions seem all too willing and ready to punish offending academics themselves, and not ashamed to be seen doing so.

In my own case, it was indeed a phone call from Brian Cass, chief of Huntingdon's Life Sciences whose clients included Novartis (now Syngenta) that resulted in my being hounded out of the university I had worked in for nearly 25 years.

Environmental journalist George Monbiot gave more examples in his book, *Captive State, The Corporate Takeover of Britain* of similar treatments universities mete out to academics daring to dissent from the corporate agenda or to criticise it.

But the phenomenon is far from new. I count myself among those at the receiving end of 'intellectual suppression'. Brian Martin, theoretical physicist turned social scientist in the University of Wollongong, Australia, has written volumes on the subject of how the academic establishment punishes individuals who challenge conventional wisdom or break with the orthodox opinion in any way. They are subjected to many kinds of abuse, cease to get grant support for their research, fail to get tenure or are passed over in promotions.

Corporate takeover of science brokered by governments

Governments have sold science and scientists to the corporations. The sell-out began under the Conservative government in Britain, and with science research funding which effectively controls what kinds of science could be done. The 1993 science white paper, *Realizing our Potential*, intended to "produce a better match between publicly funded strategic research and the needs of industry". The research councils, which distribute most of the public money for science, would be obliged to develop "more extensive and deeper links" with industry. They would be required "to recruit more of their senior staff from industry".

The Labour government extended those reforms enthusiastically. Its 1998 white paper on "competitiveness" launched a 'reach-out' fund to encourage universities to "work more effectively with business". The role of the higher education funding councils, which provide the core money for universities, was redefined "to ensure that higher education is responsive to the needs of business and industry".

Lord Sainsbury pointed out last year that British universities spun

off 199 companies in 2000, up from an annual average of 67 in the previous five years and a mere "handful" before that. The UK ratio of companies to research spending was more than six times higher than in the US. "It's a dazzling record," he said, but lamented UK's failure to celebrate such a "stunning change in the entrepreneurial attitudes of our universities".

The key to delivering Lord Sainsbury's "good science", redefined in terms of commercial productivity, is the higher education funding councils, which provide the core money for universities.

Thus, it comes as no surprise that the Biotechnology and Biological Sciences Research council (BBSRC), the main funding body for Britain's academic biology with an annual budget of £190 million (in 2000), is chaired by Peter Doyle, the former executive director of the biotech corporation, Zeneca; nor that its main committees are stuffed with corporate representatives: Syngenta sits on 3, AstraZeneca on 2, GlaxoSmithKline on 3, Pfizer on 4, Unilever on 2, not to mention Genetix plc, Lilly and Merck Sharp & Dohme.

The BBSRC was established in 1994 to replace the biological program previously run by the Science and Engineering Research Council (SERC). Whereas SERC's mandate was to advance science of all kinds, the BBSRC's purpose is "to sustain a broad base of interdisciplinary research and training to help industry, commerce and Government create wealth".

Gene biotechnology research has replaced nuclear physics as the big science. In January 1999, the BBSRC set aside £15 million for "a new initiative to help British researchers win the race to identify the function of key genes". In July the same year, £19 million was to be spent on new research facilities to "underpin the economic and environmental sustainability of agriculture in the UK" through "work on genetically modified crops". In October, £11 million was allocated to projects that would enable the UK "to remain internationally competitive in the development of gene-based technologies". Every year, the Council gives more than £10 million in grants to the John Innes Centre in Norwich, the genetic engineering institute which houses the Sainsbury Laboratory and had a research alliance with Zeneca and Dupont, and lately, Syngenta, although Syngenta had terminated this partnership abruptly in 2002 (see Chapter 1).

In 2000, an extra £1.9 billion was committed to 'health genomics research' over the following five years. That was in addition to the

Government's projected spending of £675 million on university infrastructure through the Science Research Investment Fund, which included high tech facilities for studying genes and proteins. All that was supposed to help industry discover new lucrative drugs. However, industry has already warned that drug discovery through genomics was going to bankrupt pharmaceutical companies (see Chapter 12), and it is likely to prove right.

But as the dream of miracle drugs is receding and the biotech corporate empire showing all the signs of collapse, the Blair government propped it up with even more subsidies in 2002. It introduced further new tax credits for research and development: "a £400 million boost to innovation, affecting £11 billion of expenditure by 1 500 large companies in the UK". Even that has failed to lift the gloomy prospects.

In the United States, successive governments have brokered the corporate take-over of science in a similar way, going back to the 1862 federal legislation that created the land-grant universities. But this accelerated with the passage of the Bayh-Dole Act in 1980, which for the first time, allowed universities to patent the results of federally funded research. The Business-Higher Education Forum, a coalition of corporate and academic leaders, and similar groups lobbied to get universities joined up with the marketplace.

Since then, the US Congress has passed numerous other laws to cement university-industry ties, including generous tax breaks for corporations investing in academic research. From 1980 to 1998, industry funding for academic research expanded at an annual rate of 8.1%, reaching \$1.9 billion in 1997, nearly 8 times the level 20 years ago. Before Bayh-Dole, universities produced roughly 250 patents a year, many were never commercialised. But in 1999, more than 120 US research universities filed a total of 7 612 patent applications. Licenses to industry generated \$641 million for the universities, about \$40 billion in economic activity overall in the heyday of the biotech bubble.

Another factor driving academic institutions into the maws of big business is the cut in public research budgets, which enables corporations to buy up departments and whole institutions for new ideas and prestigious labour at bargain basement prices.

Postdoctoral fellows earn as little as \$15 000 a year working, at times, round the clock. (I know, having worked in and visited several academic institutions in the United States.) Projects in a university lab typically cost about half what it would at a drug company. And the pub-

lic has greater faith in research at universities than in private labs.

Swiss drug giant Novartis is paying \$24 million over six years to the University of Maryland's Psychiatric Research Center for access to its brain tissue bank and one of its labs.

The University of California at Berkeley, similarly, had suffered decades of financial cutbacks when Novartis agreed in 1998 to pay \$25 million to be allowed to sift through the research of the department of plant and microbial biology, and license up to about one-third of the research results. Along with that, the company gained the right to sit on a committee deciding on the research of the department.

When Berkeley plant geneticists Ignacio Chapela and David Quist uncovered the transgenic contamination of maize landraces growing in remote regions of Mexico and reported it in *Nature* in November 2001, they were subjected to vicious attacks and intimidation rivalling the Pusztai episode in Britain, and orchestrated from within their own Department, reportedly aided and abetted by Monsanto.

Novartis also pays \$20 million a year for some research at the Scripps Research Institute in California and over the past decade has paid up to \$100 million for research at the Harvard-affiliated Dana Farber Cancer Institute.

But big-time spending by biotech giants has come to an abrupt end (see Chapter 1).

Science behind closed doors

The academic-industrial complex is not open to public scrutiny.

At the 2000 World Economic Forum in Davos, *the* gathering of corporate bosses that run the world, Bruce Alberts, President of the US National Academy of Sciences (NAS), was reported to have got together behind the scenes with a group of a dozen other presidents of national science academies to create an International Academy Council (IAC) to provide "impartial scientific advice" to governments and international organizations on issues such as genetic engineering, threatened ecosystems and biodiversity.

Bruce Alberts also chairs The National Research Council (NRC), which has a full-time staff of 1000 and a \$ 200million annual budget. Through the NRC, the NAS conducts studies and prepares about 200 reports annually, largely under contract to federal agencies.

In flagrant violation of the rules of open government - the 1972

Federal Advisory Committee Act - which Alberts still vehemently opposes, NRC committees and panels meet secretly in closed sessions. They do not disclose their minutes or conflict of interest statements, and fail to require that their membership reflect balanced representation of divergent interests and viewpoints.

A blatant conflict of interest arose in the composition of the NRC biotechnology panel set up in March 1999, with disproportionate representation of experts directly linked to industry. The panel's executive director, Michael Phillips, was secretly negotiating for a senior position in the Biotechnology Industry Organization (BIO), and joined the industry some 3 months later.

The NAS report on GM crops was released in April 2000 amid fresh controversy. BIO - the industry's lobby - was delighted by the report, claiming in a press release that GM foods "are thoroughly tested and safe". Critics, however, rejected the report. US Senator Dennis Kucinich called for the study to be scrapped because the panel was "tainted by pervasive conflicts of interest".

The NAS is not the only scientific body found wanting in assessing the safety of GM foods. Less than a year later, the NAS, in turn, criticised the federal reviews of GM crops carried out by the US Department of Agriculture (USDA), branding it "superficial", for placing too much emphasis on protecting corporate trade secrets, and lacking in both transparency and rigour. It called on the USDA to seek evaluation by scientific experts outside the agency and to solicit greater input from the public.

Corporate science kills

Corporate control of biomedical research is particularly strong, as it involves huge profits. The pharmaceutical industry has been the most profitable industry by a wide margin. And the federally funded US National Institutes of Health may be the drug industry's biggest benefactor. This government agency alone spent more than \$23 billion on research in 2002.

Biotech drug giants like AstraZeneca, GlaxoSmithKline and Eli Lilly are reported to be funding annual meetings of professional groups such as the American Psychiatric Association, saturating the conference with corporate logos and free gifts and paying for sessions "to control which scientists and papers were presented and to help shape the presentations".

Meanwhile, academic institutions have been getting into an orgy of incestuous relationships with industry. Scientists find themselves testing drugs they have invented, sitting on committees approving the drugs and holding financial stakes in companies that stand to profit from them. Corporate science is endangering lives.

In 1999, teenager Jesse Gelsinger died in a gene therapy trial at the University of Pennsylvania. Government regulators cited the researchers for numerous safety violations. The scientist overseeing that trial and the medical centre had financial stakes in the therapy tested.

A litany of misconducts has come to light since, as reported in a series of articles in the *Baltimore Sun* and elsewhere in 2001.

Doctors at the Fred Hutchinson Cancer Research Centre allowed a cancer experiment to go on for years, even though patients were dying at a higher rate than with standard therapy. The Centre and some of its physicians had financial stakes in the treatment.

A University of Pittsburgh scientist funded by several drug companies was accused in a lawsuit of manipulating a study of children's ear infections and contributing to the dangerous overuse of antibiotics.

The FDA reprimanded a Tufts University researcher for improperly treating a cancer patient with a gene therapy that may have caused his tumour to double in size. Both the scientist and a Boston medical centre held large stakes in the company developing the treatment.

In July 2001, the US government ordered a suspension of all clinical trials in Johns Hopkins following the death of a previously healthy 24 year old volunteer in an asthma experiment.

Researchers at the University of Alabama at Birmingham carried out tests on cancer patients for years, with a drug they knew was useless. But they had financial stakes in a company making the drug, so they falsified the data to manipulate the market.

But the governments are far from impartial in imposing the necessary regulation. The United States Food and Drug Administration (FDA) currently stands accused of withholding tens of thousands of documents on hundreds of side effects and dozens of deaths linked to xeno-transplantation - transplanting animal organs and tissues to humans.

The FDA was in the news again in June 2002 when it re-approved the drug alosetron (Lotronex) belonging to GlaxoSmithKline. The drug, prescribed for 'irritable bowel syndrome' and related disorders was withdrawn on account of links to severe complications including deaths.

When the FDA committee met in April 2002, there were 113 reports of serious complications of constipation, 84 of ischaemic colitis, and six of small bowel ischaemia. This accounted for more than 100 cases of hospitalisations, 50 cases of surgery and at least 7 deaths; and as the FDA staff point out, only 1 to 10% of such events are reported.

A senior scientist on the FDA's advisory committee had decided to speak out, warning of more deaths if the drug is re-approved.

The drug giants funded FDA to the tune of \$162 million in 2002, almost half of the cost of reviewing drugs. In return FDA must meet tighter deadlines and improve "responsiveness to communication with industry sponsors during the early years of drug development", with disastrous consequences.

Thirteen dangerous prescription drugs have been withdrawn from the market in the last decade, but not before hundreds of patients died and thousands were injured. Yet no congressional committee has investigated why the FDA approved these dubious medicines or why they were not withdrawn right away.

In May 2002, Congress opted instead to renew the arrangement that's a major source of the problem.

Scientific journals closed to debate

Almost as a counsel of despair, the top biomedical journals announced in August 2001 that they would reserve the right not to publish drug company-sponsored research "unless the researchers involved are guaranteed scientific independence". A week later, Britain's top science journal, *Nature*, said it expects all its authors to declare "any competing financial interests".

But the journals have another important role, to encourage open scientific debate. Unfortunately, top journals such as *Science*, *Nature* and *New Scientist* are dependent on corporate sponsorship, and appear reluctant to give voice to scientists dissenting from the corporate view, while granting undue and apparently unlimited access to their pages to pro-biotech scientists and other supporters of the industry.

Nature, having published a paper from the two Berkeley scientists reporting on the transgenic contamination of Mexican maize landraces (see above), seemed to have succumbed to pressure from the pro-GM lobby. It 'retracted' the paper several months later, a move that was unprecedented for a paper that was neither wrong in its major findings, nor fraudulent. The scientists' findings have since been confirmed.

Nature has subsequently also rejected a paper from Mexican government scientists, which confirmed the Berkeley scientists' results.

These journals have turned down nearly all my attempts to get the hazards of horizontal gene transfer (Chapters 8 -10) debated within their pages, even when I was openly attacked there.

Science advisors betraying public trust

Britain's Food Standards Agency (FSA) has been given the responsibility of overseeing the safety of GM foods, after the public lost confidence in its predecessor, the Ministry of Agriculture, Fisheries and Food (MAFF). The FSA is chaired by Oxford zoologist, Sir John Krebs, a prominent sponsor of the SIRC 'guidelines' aimed at suppressing adverse reports on GM. Krebs himself has made no secret of his pro-GM, anti-organic stance.

There is still no open public debate on the full range of scientific evidence of actual and potential hazards of genetic engineering. The promised national debate in Britain has turned into a farce.

As its contribution, the government funded Natural Environment Research Council sponsored an online debate organised and hosted by *Spiked*, a website run by a group that is fanatically pro-GM. The views presented as part of the "debate" more than reflect that bias in a way that mirrors what the Government is doing on the national stage.

The Government's "public debate" is happening in parallel with a scientific review and an economic cost/benefit study.

The cost/benefit study, carried out by the Prime Minister's Strategy Unit, put out a "Scoping Note" for public consultation, which read more like an excuse to present a mass of speculative 'benefits' of GM crops.

The scientific review is conducted by just three scientists, the avowedly pro-GM Sir John Krebs of the Food Standards Agency, the Government's chief scientist David King, and the chief scientific adviser for the Department for Environment, Food and Rural Affairs (DEFRA). The 'independent' Science Review Panel consists of an overwhelming number of fervent GM supporters, including representatives from Monsanto and Syngenta. It has been accused of being rigged to avoid debating the scientific evidence, so as to keep up the pretence that anti-GM is anti-science.

And ahead of the great 'national GM debate', the government's science advisors have effectively given the green light to commercial growing of GM crops in a summary dismissal of mounting evidence that

GM crops are hazardous to health and the environment.

The Advisory Committee on Releases to the Environment (ACRE) and the Advisory Committee on Novel Foods and Processes (ACNFP) are still insisting there is "no evidence" GM crops pose greater risks than their non-GM counterparts in their response to a public hearing. The hearing, which began more than two years ago, was triggered by widespread opposition to the government's proposal to put the GM maize Chardon LL on the national seed list.

During the public hearing, over 15 box-files of written material were submitted and months of oral presentations heard, much of which consisting of scientific evidence of GM hazards as well as challenges to the poor, substandard and incomplete scientific research accepted by the committees for approving GM crops and products.

This latest reassurance from our government's science advisors was no news to anyone. No mainstream media bothered to report it. The public is right to suspect the worst as less than a month before, the British Medical Association (BMA), not known for adopting radical positions in scientific matters, repeated its call for a moratorium (first time in a report published in 1999) on GM crop trials, expressing concerns over the impact of GM foods on long-term health. The BMA has a membership of over 120 000 representing more than 80% of doctors in Britain.

Our government's science advisors have been caught playing fast and loose with scientific evidence, abusing public trust, and in blatant violation of both good science and the precautionary principle.

Scientific establishment peddling GM

Meanwhile, the Royal Society appears to be doing its best to help corporations present their case for GM. It issued a press release on a paper to be published in the Royal Society Proceedings, reporting research funded by Monsanto that's supposed to show how GM sugar beet is good for biodiversity - provided weeds are left to grow for a while before they are killed by herbicide - which, unfortunately and predictably, compromises yields for the farmer. So would farmers do that rather than spraying at or soon after planting?

The research is obviously contrived, and had been thoroughly deconstructed by the Soil Association when it was first touted several years ago. The Royal Society is unashamedly peddling bad science for Monsanto, after it had gone out of its way to destroy Pusztai and his

work that had satisfied all the usual standards of good science. Worse, a new smear campaign on Pusztai and other honest scientists may be in the offing.

The Royal Society has established a group to analyse the coverage of science in the media and public reactions, in an attempt to minimise "harm" from "inaccurate and misleading communication of the results of scientific research, particularly when they have implications for human health". Among the cases to be examined by the committee are research reported by Pusztai, and by Ignacio Chapela and David Quist on transgenic contamination of Mexican maize landraces. So we haven't heard the last.

As its contribution to the national GM debate, the Royal Society sponsored two public events. The first was a conference on the 'health benefits' of GM crops. These were Syngenta's 'golden rice' and edible vaccines in plants, now a major push in the 'next generation' GM crops.

The panel made little or no attempt to discuss the risks, so much so that a member of the audience commented that he had never been to a discussion meeting in which the entire panel as well as the chair were supporters of GM.

Golden rice, genetically engineered to make pro-vitamin A, is said to cure blindness due to vitamin A deficiency, widespread in the Third World. It was hyped in the journal *Science* a few years ago, and thoroughly discredited by critics who pointed out it would be much cheaper and healthier to get vitamin A from other sources such as green leafy vegetables. The amount of pro-vitamin A produced in golden rice is very small - Greenpeace calculated that one had to eat 9 kilograms of cooked rice a day to satisfy the daily requirements for vitamin A.

This renewed hype of golden rice was especially pathetic when scientists in the Philippines, using traditional breeding methods, have just developed a 'dream rice' containing vitamin A, zinc and more than twice the normal amount of iron. A feeding trial on 30 anaemic Filipino women showed that their health improved significantly within three months.

As for edible vaccines in plants, they made headlines in November 2002 when the US government ordered the biotech company ProdiGene to destroy 500 000 bushels of soya beans contaminated with GM maize that was engineered to produce a drug not approved for human consumption. A day later, it was revealed that ProdiGene did the same thing in September, when the US Department of Agriculture

ordered 155 acres of nearby maize to be incinerated for fear of contamination. The company was fined \$3 million, and the food industry, which had previously embraced GM food, is now demanding that there should be no food crops engineered to produce pharmaceuticals.

A big question mark remains over the safety of these edible vaccines. Could the genes of dangerous viruses and bacteria incorporated into the plants recombine with other viruses and bacteria to generate new, deadly pathogens? Such recombination has been demonstrated in many laboratory experiments, which we have reviewed in a scientific paper on the hazards of transgenic plants containing the CaMV 35S promoter (see Chapter 8).

The Royal Society's second event, a discussion meeting on GM crops, was not much better. The speakers selected were such that the discussion on hazards was restricted to those that could result from ordinary cross-pollination between GM crops and conventional crops. This left out whole areas of hazards unique to GM crops, such as the potential for toxins and allergens to be introduced via the foreign genes into our food crops; the instability of transgenic DNA and the potential for horizontal gene transfer and recombination (see Chapters 8-11).

Towards the end of the conference, two pro-GM scientists in the audience got up and one after another said that the tight regulation of GM was irrational given "the lack of evidence that GM crops pose any extra risks". One of the scientists was Phil Dale, who sits on ACNFP, and whom I had encountered many times before.

I felt obliged to speak up then. I pointed to critical scientific evidence and scientists that have been excluded from the conference, and ended by saying that the Royal Society has completely lost credibility and public trust.

Peter Lachmann, the same notorious Fellow of the Royal Society who allegedly tried to block Stanley Ewen and Arpad Pusztai from publishing in *The Lancet*, then got the floor. He stood up, looked around, and said that as Bob May, President of the Royal Society, who had chaired the previous session, was not around, it was up to him to defend the Royal Society; which he did by attacking me.

"Dr. Ho never did any work of her own on GM," he said. "All her papers refer to other people's work and are very controversial. And her other works are all also very controversial." The Royal Society was indeed doing its job properly, he insisted, presumably by excluding the likes of me. That brought instant applause from the biotech lobby and

the sycophants who had stuffed the hall.

The obvious riposte to Lachmann is that the pro-GM scientists would do well to read other people's work properly, and better yet, communicate the findings to the public without misrepresentation and obfuscation. Then people like me wouldn't have had to waste precious time that could have been devoted to infinitely more interesting and rewarding pursuits.

As for referring to my work as "very controversial", that exactly made the point for me. The orthodox establishment that Lachmann represents feel quite threatened by any new findings that upset or cast doubt on the dominant paradigm, and that certainly includes most of what this book contains. An image flashed through my mind just then. A young woman telling me a few years ago how, as a student in university, she and her friends read a book I had co-edited with Peter Saunders on evolution, *Beyond neo-Darwinism*, only "in bed and under the covers".

Before I could reply, a slight commotion spread from the back of the hall. Up sprang Bob May, to almost everyone's surprise. "I sat at the back on purpose," he said, "so I could follow what was going on without being noticed."

Instead of backing up Lachmann, he said, "The discussions should range widely, not only on science but on non-science issues." He was not applauded.

I left things at that. Lachmann had gone too far, even for Bob May. I was glad May heard us loud and clear.

At the end of the conference, when my colleagues from ISIS tried to distribute our magazine, *Science in Society* - which is about the only source of critical scientific information available to the public - they were told it was not allowed; this, from an institution that's freely lending its full trappings and gloss of respectability to corporate propaganda.

The Royal Society has been reported to the Charities Commission by a number of non-government organisations for abusing its charter. They accuse it of "attempting to rig" the GM Science Debate, and claim that "senior figures within the organization are undermining its former good name and reputation by using crude propaganda techniquesto attack scientists who have the temerity to question the safety of GM foods and crops."

Five

The House of Cards Came Tumbling Down

Turning genetics upside down

It is not just crude commercial incentives that have prompted the scientific establishment to engage in suppressing dissent and adverse scientific evidence. There are deeper reasons why many in the scientific establishment are defending the orthodox opinion.

One has to appreciate that the assumptions of genetic determinism, in their more neutral form, have been the bread and butter of mainstream biology for at least 100 years, rather the way that Newtonian mechanics had been the foundations of physics before quantum physics came along and threw the tidy Newtonian universe into marvellous disarray. Quantum physics has been around for more than a century and is now generally accepted. But its profound implications have yet to reach the consciousness of the scientific community, let alone the general public.

The fluid genome began with scientific findings dating from the mid to late 1970s. By then, techniques for manipulating and recombining DNA had developed sufficiently for commercial genetic engineering to begin. It is no accident that a mechanistic, market-obsessed culture should reify *processes to things*, for things, unlike processes, can readily be turned into commodities. So that's what DNA has become. In retrospect, the patenting of genes and organisms was a logical conclusion to the search for the *material* basis of heredity.

It so happens that the same techniques that make genetic engineering possible are also powerful research tools that enable geneticists to dissect the genome and monitor changes in the genetic material in ways that were not possible before. Within a decade, the whole of genetics would be turned upside down.

But the doctrine of genetic determinism continued its stranglehold on mainstream biology. A typical popular book on genetic engineering well into the 1980s, still perpetrated this discredited ideology,

“DNA transmits its message to the cell, forcing it to listen...the nucleus [containing the DNA] is the ultimate source of all decisions and changes of direction in a cell's existence.”

Did this writer have corporate bosses in mind when referring to DNA, with the rest of the cell constituting the work force?

Under the pretext of promoting public understanding of science, the food industry, supported by government scientists, publishes lavish pamphlets, promoting genetic engineering based on genetic determinism. We are told:

“Research scientists can now precisely identify the individual gene that governs a desired trait, extract it, copy it and insert the copy into another organism. That organism (and its offspring) will then have the desired trait.”

And the same comes from the major research council, BBSRC, that funds public research in the UK, but has been controlled by industry for years (see previous chapter):

“The key to these new biotechnologies is the ability to identify, isolate and manipulate the individual genes that govern specific characteristics or traits in plants, animals and micro-organisms. We can alter genes and so adjust the characteristics they code for, and we can move specific genes from one organism to another in a very precise manner. As a result, specific characteristics can be transferred from one individual to another with a level of control not imaginable a few decades ago.”

The crudest version of genetic determinism was perpetrated by James Watson of DNA double-helix fame, when promoting the Human Genome Project to sequence the entire human genome:

“We used to think that our fate was in the stars. Now we know, in large measure, our fate is in our genes.”

Now, read on to find out why all those statements are either outright falsehoods, or at best, half-truths.

Scientific findings that shook the world

The initial crack to the genetic determinist edifice appeared before genetic engineering research really got underway. Howard Temin and David Baltimore in the United States independently discovered *reverse transcriptase*, an enzyme that does the reverse of transcription - making a copy of complementary DNA from an RNA sequence.

Reverse transcriptase was initially found in retroviruses, such as the ones associated with AIDS and certain cancers, which have RNA as their genomes; and the complementary DNA of the viral genome, made by reverse transcriptase, is inserted into the genome of the cell before the virus can replicate. This was the first contradiction to Francis Crick's Central Dogma that genetic information flows in one direction only, from DNA to RNA to protein (Chapter 2).

Then came a torrent of astonishing revelations. *Overlapping genes* were the first to contradict the idea that one gene specifies one protein. Apparently, the base sequence in the DNA can be read in more than one 'reading frame' (shifted one or two bases from the usual) to give different triplets that code for entirely different proteins.

Then *interrupted genes* hit the headlines. These are genes in which the coding sequence is broken up into short stretches (*exons*), interrupted by long non-coding sequences (*introns*). The introns are spliced out from the RNA after transcription, and before translation (Fig. 5.1).

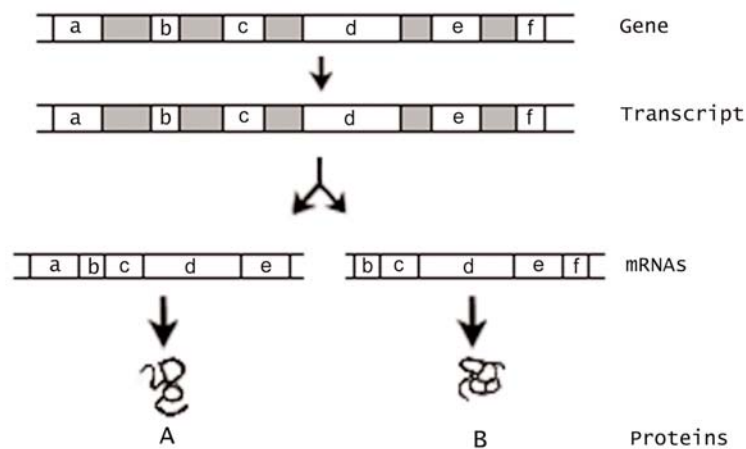


Figure 5.1. Alternative splicing of interrupted genes gives many proteins from one gene.

After that, surprises came thick and fast. The old static picture of genes and genomes determining the characteristics of the organisms

in linear causal chains has had to be consigned to the dustbin, once and for all.

Genes and genomes are nothing if not dynamic and flexible as they respond to constant feedback from the environment that changes both their function and structure.

Let's first look at the function of genes.

Genes that regulate the expression of other genes

The regulation of gene function has been an active research area at least since the late 1950s, when French geneticists Francois Jacob and Jacques Monod discovered the *lac operon*, a unit of gene regulation in the *E. coli* bacterium, which senses and responds to the presence of lactose (the sugar in milk) in its environment. Since then, many operons have been discovered in bacteria and other organisms, each of which senses and responds to a specific signal in the environment to ensure that appropriate genes are active when and where they are needed. Jacob and Monod also received a Nobel Prize for their work.

An operon consists of short stretches of DNA, *regulatory elements* linked to several genes that code for proteins. The *lac operon* has two kinds of regulatory elements (Fig. 5.2). The *promoter*, *p*, which anchors the enzyme, RNA polymerase, that transcribes or copies of the base sequence of DNA into a complementary sequence of RNA, and the *operator*, *o*, which blocks transcription when a regulatory protein, the repressor, binds on to it. The regulatory elements are located 'upstream' of genes that code for a number of enzymes with related functions. In the case of the *lac operon*, the enzymes, Z, Y and A are all involved in metabolising lactose.

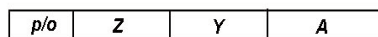


Figure 5.2. The *lac operon*.

The regulatory elements associated with the genes are very important. They enable the proteins to 'talk ' to one another, to coordinate their action, as well as respond to the environment; to decide when, where, by how much and for how long each gene is expressed, i.e., when the protein specified by the gene is made in each cell of the body.

Apart from promoters and operators, there are *enhancers* in higher organisms that act as regulatory elements. The enhancers in higher organisms could be quite some distance 'upstream' or 'downstream' of the genes they regulate, or they may be within the introns of the genes themselves.

Many regulatory elements are themselves strings of 'motifs' or short sequences that are specific binding-sites for different regulatory proteins. There are many regulatory proteins. Apart from repressors that block transcription, there are others, *transcription factors* and *activators* that are necessary for the transcription and hence the expression of the downstream gene.

The complexity of transcription is only recently being unravelled. For example, some thirty or more transcription factors are common to all cell types and are essential for the transcription of many, if not all, genes. More than 60 other factors are specific to different kinds of cells, or are made in response to special stimuli such as heat shock, specific metabolites, or hormones, i.e., they are themselves subject to environmental regulation.

Distinct combinations of the same sequence motifs make up the regulatory elements of different genes. For this reason alone, the functions of genes are mutually entangled in complex networks. It is almost impossible to change the function of one gene without affecting numerous other genes.

All possible mappings between genes and proteins

According to the Central Dogma, one gene codes for one protein. But it soon transpired that all possible specifications exist: one-to-one, one to many, many to one, and many to many. I have mentioned overlapping genes that give more than one protein from the same DNA sequence, and alternative splicing of interrupted genes that gives multiple proteins from the same RNA transcript. There are yet other ways in which the organism expands its repertoire of proteins from a much smaller number of genes.

The same gene can be read at different starting points to give different transcripts that result in distinct proteins.

Some genes are represented in *super-families* of hundreds or thousands of similar sequences, all encoding slightly different versions of the same proteins.

The extreme case of many genes making many proteins was first

discovered in the immunoglobulins, antibody proteins that protect the body from infections and other foreign substances. Multiple, slightly different versions of several genes, separated by large distances on the same chromosome, have to be joined together by rearranging the chromosome itself so as to form a continuous coding sequence in the genome of blood cells that synthesise the antibody proteins. This generates an initial diversity of millions of different antibodies that allows the organism to cope with a wide range of unexpected invaders. In addition, a complicated process of somatic *hyper*-mutation and recombination is involved in expanding the repertoire to make highly specific antibodies in the later stages of an immune response. More will be said on this fascinating system in the next chapter.

The gene transcripts are subject to extensive processing, including alternative splicing to produce different proteins from the same gene in different cells, and in the same cell at different times. One explanation for there being “too few genes” - 28 000 by the latest count (December 2002) - in the human genome, says Barry Commoner, a long time critic of genetic determinism now in his 80s, is that up to 59% of human genes are interrupted, and are now known to be alternatively spliced. Alternative splicing of a single gene in cells of the inner ear of chicks (and of humans) could give rise to 576 variant proteins. But that number is dwarfed compared to a gene in the fruit fly that could potentially generate up to 38 016 variants by alternative splicing.

Changing the message encoded in the gene

Most outrageously, the message encoded in the gene can be changed after it is transcribed. It is as though you have sent a letter or an e-mail to a friend inviting him to dinner, but someone in the post-office or someone hacking into your e-mail changes your message completely, so your friend thinks you are breaking off relationship with him.

It turns out that the gene transcript can become 'edited', by chemical modifications, or by addition of bases, to change the base sequence of the RNA transcript, so that it is translated into a protein entirely different from the one encoded. This process of 'RNA editing' was first discovered in the *Trypanosome*, a parasite that causes sleeping sickness, but it has since been found in mammals, amphibians, plants, protozoa and viruses. There is evidence that the editing is necessary for the organism to function properly. RNA editing depends on special enzymes called RNA *editases*, which exist in different forms

specific to different editing jobs. Moreover, rather like the process of transcription, many additional proteins are involved in forming a big complex of molecules that does the editing, an *editosome*, comparable to the *transcriptosome*.

Map that leads nowhere

The result of these complexities is that there are at least 100 times more proteins than genes. So many different mechanisms intervene between the gene sequence in the DNA and the precise protein made under different circumstances in individual cells that the more we know, the more bewildered we become. Knowing the genes and gene sequences does not offer much at all in the way of understanding how to get from genes to proteins, let alone organisms.

It must be emphasized that so far, only the DNA that codes for proteins in the human genome has been mapped, and the mapping is still incomplete. The coding sequence is less than 5% of the DNA in the human genome. More than 95% of the DNA is referred to as 'junk' DNA because no one knows what functions it serves, and it cannot be sequenced by current technology. The human genome is not special in that regard, most genomes, apart from those of bacteria, consist of large amounts of such 'junk' DNA containing mostly repeated sequences that do not appear to be genes, though they could be important regulatory signals; we simply don't yet know for sure (but see Chapter 11).

The situation is well described by science writer Tom Bethel. The genome map leads "nowhere", he said; so much for genetic determinism.

"Imagine that an intelligence service were to discover some unintelligible messages being sent by a spy. At first the intelligence agents naturally assume they are looking at a code. They assume the task of decoding will be straightforward. But on closer analysis it turns out that the message means one thing if the signal has been received and acted upon, another thing if it has been received and not acted upon, another thing if the receiving apparatus is not switched on, and so on. Rather than just a code the message is a bit like a set of rules for a rather complex interactive game. There are feedback loops, and circuits within circuits, and a lot of things happening inside the cell but outside the genome...."

The gene does not even determine the amino acid sequence of

the protein encoded. Instead, the precise protein made depends on influences from the context - the state of the cell and the organism as a whole - propagating back to affect transcription at all stages, including whether the gene gets to be transcribed, or even, becomes silenced. The causal loops from genes to proteins are circular, multidimensional and entangled. *There are no linear, one-way chains of command proceeding from the genes to proteins.*

One of the most significant discoveries within the past decades is about stem cells, which are found not only in the embryo, but also in every organ of the adult, including the brain. These cells retain the potential to multiply indefinitely, and differentiate into numerous cell types. Stem cells in the adult were long known to exist in the bone marrow, which normally replace all kinds of blood cells. But they have been found to differentiate into liver, brain, and muscle cells. Similarly, adult brain cells normally differentiate into all cell types in the brain, but when removed from their usual environment they can differentiate into muscle and liver cells. Stem cells are offering the possibility of cell and tissue replacement. They also give us a tantalising glimpse of how dramatically and crucially environmental cues determine and change the developmental 'fate' of cells. Each differentiated cell expresses a different repertoire of genes and proteins at different times according to feedback from their microenvironments, which include influences from other cells.

Six

The Fluid Adaptable Genome

There is no holding the genome still

Those who want to salvage the genetic determinist dogma, by appealing to a constant genome that nevertheless prevails amid the bewildering complexity of gene-protein relationships, are in for a shock. The structure of genes and genomes are both subject to small and large changes in the course of normal development, and in response to feedback from the environment. So much so that molecular geneticists had coined the descriptive term, 'the fluid genome', by the early 1980s.

As succinctly put by Gabriel Dover and Dick Flavell, two scientists who played key roles in defining the new genetics,

"The application of new molecular techniques reveals that, beneath the level of the chromosome, the genome is a continuously changing population of sequences. Mobility, amplification, deletion, inversion, exchange and conversion of sequences create this unexpected fluidity on both an evolutionary and developmental timescale."

In retrospect, the fluidity of the genome should occasion no surprise. Attempts to clone animals in recent years, by transplanting the nucleus of a cell from an adult into an egg that has had its nucleus removed, give a dramatic demonstration of how the genome can respond to the egg cell to become substantially 'reprogrammed', i.e., chemically modified and reorganised. The 'reprogrammed' nucleus is capable of supporting the development of the egg into an embryo. Unfortunately, the 'reprogramming' is far from perfect, resulting in overwhelming cloning failures and defects.

Nevertheless, numerous processes contribute to the fluidity of the genome, all of which appear to be regulated in much the same mysterious way that the flexibility of gene function is regulated. I have mentioned the chromosomal rearrangements that are necessary for generating a diversity of antibodies. These are a routine part of the surveillance function of the immune system. There are other mechanisms.

One reason that genomes are so big relative to the actual genes they contain may be because they are continuously turning over and changing as the result of many processes operating on developmental and evolutionary timescales that end up accumulating useless DNA as a kind of excess baggage that the genome is too busy to throw away. These processes move genes around, mutate, rearrange, recombine, replicate sequences, delete or insert sequences, and more remarkably, exchange and convert sequences. Sequences in the genome can be amplified or contracted thousands or hundreds of thousands of times as part of normal development, or as the result of environmental challenges. The genome can even undergo large reorganisations or rearrangements to form new chromosomes.

Genomes can duplicate wholesale. The biggest genomes in the world belong to single-celled ciliate protozoa, which tend to selectively amplify, delete sequences and drastically rearrange their genomes as part of normal development.

Richard Flavell envisaged cycles of 'DNA turnover' in the genomes of all species, involving all the above processes, which provide major sources of variation for evolving new species. This was considered quite heretical at the time, because it meant that constant, incessant variation was driving evolution rather than natural selection.

Let's look at some of the processes in more detail.

Jumping genes and viruses

Jumping genes are called *transposons* or mobile genetic elements. They are units of DNA that encode enzymes - *recombinases* and *integrases* - that can cut out the unit and reinsert it in a different location in the genome. In the process, some elements, *replicative transposons*, can also multiply copies of themselves and spread throughout the genome, or spread to other genomes (see below).

It was plant geneticist Barbara McClintock who discovered transposons more than 50 years ago; and for that, she was awarded a Nobel Prize belatedly in 1983. McClintock was studying a number of very unstable genes in maize strains that mutated spontaneously at high frequencies. This was due to transposons that jump in and out of the genes, disrupting their function.

One class of transposons, *retrotransposons*, depend on reverse transcription to move and duplicate themselves. These are very similar to retroviruses from which they may have been derived. Retroviruses

include the Human Immunodeficiency Virus (HIV) associated with AIDS, and other viruses associated with various cancers.

Many relict virus-like sequences that have lost their ability to move independently are present in all genomes including the human genome. About 50% of the human genome is now known to consist of virus-like sequences and transposons, many with reverse transcriptase. Relict sequences can be mobilised under certain circumstances by the recombinases and reverse transcriptases encoded in other elements. The 'crippled' vectors (gene carriers) of genetic engineering, made from parts of different viruses, transposons, and other genetic parasites, may be similarly mobilised with the help of the enzymes encoded in transposable elements, which is why these vectors are dangerous (see Chapter 8).

Transposons, relict viral sequences and other inserts into genes can disrupt gene function. In addition, transposons and other sequences, including viral vectors, carry strong promoters that can lead to over-expression of genes associated with cancer. The first cancer victims of gene therapy (see Chapter 1) came about because the retroviral vector integrated in the wrong place, causing a gene to over-express in a white blood cell, that in turn, caused that cell to proliferate out of control, resulting in leukaemia (more in Chapter 8).

Transposable elements are responsible for much of the fluidity of genomes. Transposition - movement of transposons - not only involves changes in the position of the transposable elements themselves, but in the case of replicative transposons, the process can spread copies of the transposons around the same genome or to other genomes by infection. One transposon, the *P* element, has spread to all *Drosophila* (fruitfly) species within a period of 50 years. *Drosophila* strains in the laboratory, which were established before the spread of *P* elements, were free from the element. A tiny mite, parasitic on many *Drosophila* species, is thought to be responsible for spreading the *P* element across species barriers. Transposable elements are responsible for many 'spontaneous' mutations in maize, in *Drosophila*, as mentioned above, and also in human beings. Transpositions lead to sequence duplications, deletions and chromosomal rearrangements. The genetic upheavals caused by transpositions can be quite considerable.

Transposable elements in the genome do not always move. In fact, even those elements that are capable of moving mostly stay put. Transposition is regulated, so transposons may change from active to

an inactive state and *vice versa* (I shall say more on the regulation of transposons in Chapter 11). Inactive transposons are correlated with an increased level of methylation, a chemical modification of the DNA. Methylation is strongly implicated in 'gene silencing' - the failure of gene expression - first discovered in the foreign genes (transgenes) introduced into genomes by genetic engineering.

Transposition is often triggered by environmental stress and also by hybridisation, i.e., crossing individuals from different species. Some geneticists see this to be a potentially 'adaptive' function; for genetic variants could be generated to enable the organism to overcome the environmental challenge, or the physiological effect of mixing two different species. But the overall result of stress-induced transposition is to increase the rates at which genes become inactivated as the transposons insert randomly into them. There is now also evidence that transposition can be triggered by the integration of foreign genes into the genome, as during the creation of transgenic organisms (see Chapter 11).

All that has implications for human health. Could it be that one of the effects of stress in human beings is to increase transpositions in cells and cause harmful mutations including cancer? Certainly, most 'spontaneous' mutations turn out to be caused by insertions of transposable elements into the gene.

Stress is also known to induce proviral sequences in the genome to develop into infectious viruses in both plant and animal cells.

Genetic concertinas

Sequences in the genome can expand and contract like a concertina as part of normal development. For example, there are genes that encode ribosomal RNAs. These are RNA molecules that make up *ribosomes*, the molecular complexes responsible for translating messenger RNA into proteins (see Chapter 2, Fig. 2.6). These ribosomal RNA genes undergo waves of expansion during the maturation of the egg cell, and again in the fertilised egg, of *Xenopus*, the African clawed 'toad' (which is really a frog). Eventually, more than 70 percent of the DNA in the egg nucleus ends up coding for ribosomal RNAs. Ribosomal RNA genes already exist in the genome in numerous copies as multigene families.

Gene amplifications and contractions also occur in many plants that switch from juvenile to adult phases, and these too, involve

repetitive sequences.

The potential for expansion and contraction of sequences is very widespread. Expansion of sequences happens readily in mammalian cells exposed to drugs or cell poisons, as during chemotherapy against cancer. One of the best-studied examples is methotrexate-resistance. Methotrexate inhibits the enzyme dihydrofolate reductase, which is necessary for making DNA. Cells treated with methotrexate cannot make DNA and are therefore unable to multiply.

However, in the course of drug treatment, the cells soon overcome the inhibition by one of three mechanisms: increasing activities of membrane proteins that pump the drug out of the cell, mutating the enzyme so it is no longer sensitive to inhibition of the drug, or overproducing the enzyme so as to overcome the inhibition of the drug. Overproduction of the enzyme is accomplished by expanding the section of the chromosome containing the gene for the enzyme up to tens or hundreds of thousands of times. The expanded or amplified region is about ten times longer than the gene itself, probably comprising introns of the gene and regulatory elements. Amplifications are accompanied by gross changes in the structure of the chromosome, and may involve chromosomal rearrangements, and even extra chromosomes appearing, which have a wide range of abnormal structures.

It is important to stress that the same genome changes are *repeatedly* generated by particular drugs in particular cell lines. They are physiological responses that can occur in every cell, and have nothing to do with selection of random mutations.

Similar genetic and genomic changes are inducible in insects exposed to insecticides and plants exposed to herbicides, which render the organisms resistant to the poisons. Again, these are non-random physiological responses common to most, if not all individuals in the populations.

Unfortunately, the researchers involved have invariably mistaken them for examples of selection, or natural selection; at least partly because to admit non-random variation is considered such a great heresy against neo-Darwinism that few have the courage to risk being ostracised by the establishment.

The extent to which genomes of both somatic and germ cells can change in response to the environment should make geneticists extremely cautious about 'cloning' from the cells of adult animals. The high failure rates of such somatic cell nuclear transplant is most likely

connected with the genetic changes that accumulate in the genomes of somatic cells.

Hyper-mutations

The same kinds of stimuli that cause gene amplification can also cause *hyper*-mutations. These are mutations that occur at very high frequencies, up to one in a thousand; and are a thousand to a million times higher than the 'random' mutations that geneticists had studied for decades before in laboratory cultures of bacteria.

Hyper-mutations are involved in blood cells that make antibodies in the later stages of the immune response, in which antibodies with high affinities for the foreign substances (antigens) are produced (see next chapter).

Hyper-mutations occur in response to stress or other environmental challenges and show selectivity for mutating certain genes that enable them to overcome the stress.

Changing genes to order

One interesting class of environmentally induced changes to genomes is associated with a spectrum of heritable modifications that appear in flax and other plants treated with various mixtures of fertilisers.

As is well known, plants do not have separate germ cells and somatic cells, for every somatic cell is capable of developing into a germ cell, so genetic modification of somatic cells in all plants will be inherited in the next generation. That is the basis of plant genetic engineering in plant cell culture, where the modified cells can readily be stimulated to grow into plantlets by adding plant hormones.

The phenomenon of heritable modifications after treatment with fertilisers, discovered in the 1950s, was not analysed using molecular techniques until some thirty years later, by Chris Cullis in the then John Innes Institute, a government-funded institute where researchers were free to pursue scientific interests not immediately tied to wealth creation. It has since become the John Innes Centre, privatised, but still publicly funded to a large extent, and the top UK research centre for, and promoter of, plant genetic engineering.

Chris found that a 1.5% solution of the fertiliser, ammonium sulphate added to the compost at the time of sowing gave a large type of plants, L, whereas a 1.5% solution of a triple-superphosphate with low

pH compost, gave a small type, S. The L and S 'genotrophs', as they were called, differed in many characteristics, from each other and from the 'parental' type from which they were derived, including size, hairiness, distinct types of enzymes expressed, as well as in genome size and in the copy number of repeated ribosomal RNA genes. The changes were stably inherited in subsequent generations in the absence of the inducing environment, under the usual conditions of growth. However, the stability is not absolute, and further changes can be induced by other circumstances.

Careful analysis of the seedlings treated with fertilisers at different stages of growth showed that the DNA changes took place in *all* the cells of a meristem (the growing zone of a plant) *simultaneously*. Thereafter, the growth characteristics of the meristem tended to improve, suggesting that at least some of the changes are adaptive.

These environmentally induced DNA changes are specific to different stimuli, and can be repeatedly generated. They are therefore, *not* random changes. Chris Cullis found similar DNA changes induced by fertiliser treatments in other plants, such as maize, pea and broad bean.

Unfortunately, John Innes Institute was undergoing privatisation just at the time (the 1980s), and Chris was unable to obtain further support for this work, which really pulled the rug from genetic engineering. The propensity for the plants to reorganise their own genome in different environments goes against the grain of genetic engineering, which assumes that genes cannot be changed directly by the environment in a repeatable way.

Gene conversion and concerted evolution

Many genes in the genome exist in multiple copies, from a few to thousands or tens of thousands. These are referred to as *multigene families*. They arose from duplications or amplification events in the past, in which multiple copies of a single gene were generated in the genome.

A puzzling phenomenon came to light as multigene families from different species were analysed. The members of multigene families within a species are much more homogeneous than expected if each member gene were to accumulate mutations independently of all the rest, as it would be expected to do. Furthermore, each species is homogeneous for variants that are diagnostic of the species. In other words, variation is accumulating between different species just as expected,

but within a species, *all* members of each multigene family tend towards uniformity.

Actually, there are degrees of uniformity, as each multigene family can consist of subfamilies, the members of which are more similar to one another than to members of a different sub-family. But the puzzle remains the same. What could be responsible for this "concerted evolution" of sequences, many of them dispersed throughout the genome?

Gabriel Dover achieved certain notoriety among the more orthodox community by drawing attention to this phenomenon very early on, calling it by the colourful term, "molecular drive", on grounds that the process would drive evolution much more substantially and rapidly than natural selection. Isolated populations could be internally 'driven' by these molecular mechanisms to diverge rapidly from one another, to become new species. That again, was a heretical thing to say, and he once gave a protest seminar in a car park outside a main event on evolution to which he was not an invited speaker.

There are several mechanisms that can homogenise sequences among members of a multigene family. One of them is gene amplification, or expansion (see above), in which an enormous number of tandem (head to tail) repeats are generated all at once. But these mechanisms cannot explain how dispersed copies of the gene can remain uniform; and even tandemly repeated copies will tend to accumulate mutations independently. Another mechanism is through transposons that multiply themselves as they move, but the replicated copies, again, will still tend to diverge after the event.

In the end, geneticists decided that there must be some process(es) of *gene conversion*, i.e., changing a DNA sequence from one to another. In the case of multigene families, it appears that all the family members tend to be converted to a uniform sequence; those sequences that have diverged are simply eliminated by substitution.

When sequences belonging to the 100 or so ribosomal RNA genes from *Neurospora* (the bread mould) were analysed, they were found to belong to 4 subfamilies. Within each sub-family, the coding regions differed in sequence by less than 0.2%, whereas the flanking 'spacer' regions differed by 3.5 to 7%. Differences between the sequences of each sub-family are such that they preserve the same secondary (three-dimensional) structure of the ribosomal RNA necessary for its function, suggesting that gene conversion may depend on gene function. One way for this to work is via reverse transcription,

which appears to be very active in eukaryotic genomes for which no other obvious function exists. It could be that gene product ribosomal RNAs that have been efficiently transcribed and remained stable, escape being broken down by ribonucleases (RNAase). Thus, only genes that function well may be reverse transcribed into complementary DNA (cDNA), which is then inserted into the genome, in place of those sequences that do not work well, by a process of homologous recombination, or exchange of DNA sequences. *Homologous* recombination takes place between DNA sequences that are very similar, and are derived from a common ancestral sequence. I shall have more to say on this fascinating process in the next chapter.

Wandering genes

As mentioned earlier, transposons can jump between species that do not interbreed. The transfer of genetic material between unrelated species is referred to as *horizontal gene transfer*, to distinguish it from vertical gene transfer that happens in normal reproduction within species or between individuals of closely related species.

A transposon called *mariner*, first discovered in *Drosophila*, was found to have jumped into the genomes of primates including humans, where it caused a neurological wasting disease. Transposons are now increasingly used in genetic engineering especially of insects like mosquitoes that suck blood and inject parasites into human beings. If these transgenic mosquitoes with transposons are released, the transposons are effectively given wings and sharp mouthparts to invade the genomes of human beings and of other warm-blooded species. Joe Cummins and I have warned regulators in the United States of that, but the researchers themselves seem undeterred.

Transposons are not the only carriers, or *vectors*, of horizontal gene transfer. Viruses and plasmids - usually circular pieces of genetic material that replicate independently of the chromosome in bacteria, yeasts and other cells - can also act as vectors (see Chapter 8, Figure 8.1). Recent evidence indicates that any piece of DNA can end up inside the cells of all organisms, from bacteria to human beings, and has the potential of becoming incorporated into the genome.

The extent to which horizontal gene transfer has contributed to genome evolution is a subject of debate among geneticists. However, the full scope of actual and potential horizontal gene transfer has come to light since the early 1990s, and especially since genome sequences

became available. There have been thousands of papers published since 1993, giving direct or indirect evidence of *recent* horizontal gene transfers. One of the strongest evidence for horizontal gene transfer is that identical, or nearly identical gene sequences turn up in unrelated species, which can only be due to recent events, as the sequences will diverge independently after they have transferred to different species. Transfers have occurred between very different bacteria, between fungi, between bacteria and protozoa, between bacteria and higher plants and animals, between fungi and plants, and between insects.

The current state of our understanding is pre-sented in Fig. 6.1, where the arrows indicate transfers for which direct or circumstantial evidence already exist. Genetic engineering biotechnology poses unique dangers in that it greatly facilitates horizontal gene transfer through vectors and constructs that are designed to jump into genomes and to overcome cellular defence mechanisms that break down or inactivate foreign DNA. Horizontal gene transfer will be dealt with in greater detail in Chapters 8 to 10.

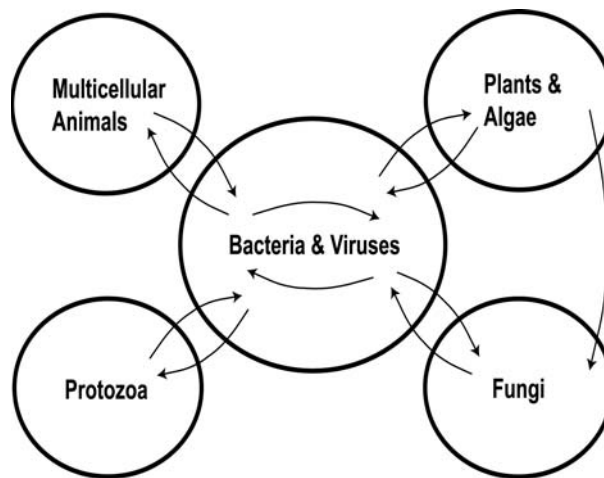


Figure 6.1. Horizontal gene transfer links the whole biosphere.

What it all means

One consequence of the fluid genome processes is that genomes within the same species or populations are infinitely variable, *reflecting the*

imprints of different individual experiences on the genome (see below). There is on average one base difference for every 500 between any two individuals in a population. The enormous variability of individual genomes is the basis of the DNA fingerprinting techniques used for forensic purposes. That's why the Human Genome Project to determine the sequence of all the genetic complement of human beings was subject to the criticism, "whose human genome is being sequenced?"

The answer, in the case of the public consortium, is a mixture of genomes from individuals of different ethnic groups, a kind of human-rainbow genome. It was considered a minor scandal when it turned out Craig Venter had merely sequenced his own genome.

The variability is especially in the many different kinds of repeated sequences scattered all over the genome, most of them in the 'junk DNA' that makes up 30 to 95% or more of the genomes of higher organisms. Some repeats code for abundant proteins such as actins or myosins, the proteins making up muscles, which exist in 'multigene' families. Many dispersed repeats are transposons or relict viruses - *provirus* - that have become inserted into the genome, and 'silenced' by chemical modification or other regulatory mechanisms. Numerous reverse transcripts are also present, including non-functional 'pseudogenes' that are reverse-transcribed from processed transcripts, and re-inserted into the genome. However, some of the repeats are actually important regulatory elements (see Chapter 11), especially those located next to genes.

Those who, like James Watson, claim to identify the 'blueprint' for making a human being from the genomic DNA sequence are deluding themselves and others. The genome looks more like an immense garbage dump of trashed DNA. But the 'trash' is not randomly distributed. There is an intricate dynamic order in the genome amid the chaos, which molecular biologists are just discovering, though far from understanding.

Genome fluidity should indeed have put the final seal on the coffin for genetic determinism, as there is actually no constant blueprint to read from, amid all the chopping and changing that goes on in the genome. But that's not all. The greatest neo-Darwinian taboo, that mystical Weismann's barrier supposedly keeping the 'germ plasm' immortal and unchanged, that too, has been breached and in many different ways, as we shall see in the next chapter.

Seven

Lamarck Resurrected

Breaching Weismann's barrier

Historians of science will one day reveal that 'Weismann's barrier', supposed to prevent acquired characteristics from being inherited, had been elevated to a mystical status based on scant empirical evidence. In the first instance, this 'barrier' can operate only when organisms have germ cells, which go to make the next generation, separate from somatic cells, which belong to the body and expire with the body.

Plants have no separation between germ cells and somatic cells, as any cell in a plant is capable of becoming a germ cell, so changes in DNA induced in a somatic cell can become inherited. That's why genetically engineered plants can be created simply by introducing foreign genes into plant cells in culture, which are stimulated by plant hormones to grow into whole plants.

In fact, the majority of animal *phyla* (phyla are major divisions of organisms, such as flat worms, round worms, insects, reptiles, and so on) also do not have separate germ cells and somatic cells. As any gardener will know, an earthworm broken into two becomes two earthworms. For animals that appear to have separate germ cells, there are two possibilities for the genome of the germ cells to change. The germ cells may also respond to the environment as somatic cells do; or there may be a feedback communication channel between the somatic cells and germ cells, so DNA changes acquired by somatic cells can become incorporated into the germ cells and passed on to the next generation.

The 'inheritance of acquired characteristics' is one of the greatest threats to the neo-Darwinian orthodoxy, and is hence also the greatest taboo. Few biologists will run the risk of being ex-communicated by the slightest hint that the process may be possible. It is often referred to as Lamarckian inheritance, after French scientist Jean Baptiste de Monet de Lamarck. Lamarck not only invented the subject of biology, he was also the first to propose the theory of evolution, which states that the diversity of living things on earth has arisen through natural causes in

long periods of geological time, rather than through special creation by God.

Charles Darwin's theory of natural selection is sometimes referred to as the 'special theory of evolution', as it offers a mechanism whereby evolution could have occurred. But Lamarck, too, suggested a mechanism of evolution more than 50 years before Darwin. He proposed that new species could arise through *changes in the relationship between the organism and its environment, in the pursuit of its basic needs*, in other words, its way of life, which thereby produced new modifications that became inherited after many generations.



Figure 7.1. A plaque commemorating Lamarck on a house next to the Museum of Natural History in Paris, where he lived and worked from 1795 until his death in 1829. It states unmistakably that Lamarck was "Professor of the Museum, author of the first theory of evolution".

Unfortunately, Larmack's theory was widely misrepresented as merely the inheritance of acquired characteristics, or caricatured as changes resulting from the 'wish-fulfilment' of the organism. Half a century later, Charles Darwin was to include a number of Lamarck's ideas in his own theory of evolution by natural selection, though without due attribution. For example, Darwin accepted the inheritance of acquired characteristics, much to the discomfiture of his followers. He also invoked 'the principle of use and disuse', the idea that use leads to reinforcement and strengthening of the characteristics, as attested to by the massive muscles of body builders; whereas disuse leads to loss, as for example, animals living in dark caves tend to be blind.

It is ironic that both Lamarckian principles should now be found to be operating at the molecular level.

Epigenetic inheritance

Epigenetic refers to features that arise in the course of development (see Chapter 2). For example, the state of gene expression in a cell, such as the pattern of methylation of its DNA - that determines which genes are silenced - is inherited by its daughter cells. In species which do not have separate germ cells and somatic cells, which include plants, fungi, bacteria, unicellular animals and many multicellular animals, patterns of methylation will be inherited in subsequent generations. In vertebrates, a de-methylation of genes that are tissue-specific in expression occurs as part of the 'reprogramming' of the genome in the egg. But sometimes, genes escape this reprogramming, leading to germline inheritance of acquired gene expression states.

There are many other forms of such epigenetic inheritance that do not necessarily involve changes in DNA base sequences, among them, the ill-defined *cytoplasmic* states - states of the cell outside the nucleus - that give rise to *cytoplasmic* inheritance not directly associated with DNA or RNA. They are known as *dauer-modifications* (lasting modifications), and tend to diminish in successive generations.

In my laboratory, we documented a case of cytoplasmic inheritance in the fruitfly in the 1980s. We found that briefly exposing embryos to ether (an organic solvent) during early development causes the adult fly to develop specific aberrations that appear to be partially inherited through the female. But the effect disappears after several generations if the embryos are no longer exposed to ether.

This effect may be similar to the changes in cells exposed to

carcinogens (cancer-causing agents), X-rays and metabolic stress. Cell biologist Harry Rubin in the University of California Berkeley has proposed that epigenetic changes, rather than mutations in putative cancer genes, may be the primary cause of cancer.

However, it is becoming increasingly evident that epigenetic changes cannot be neatly separated from genetic changes. For example, metabolic and other stress can mobilise jumping genes, which can change genomes quite substantially, and genes that are silenced tend to accumulate mutations and may become deleted.

It is more accurate to see genes (and genomes) as part and parcel of the entire physiological system that responds to the environment in myriad non-random, repeatable ways.

A major challenge to the 'random mutation' postulate of neo-Darwinism comes from hard experimental evidence.

Adaptive mutations to order

When *E. coli* bacteria are smeared onto petri-dishes containing high concentrations of a metabolite they cannot use, together with very low concentrations of another that keeps them alive but non-growing, strange things happen.

The bacteria grow until the nutrient they can use runs out. Then, the starving bacteria begin to mutate at high rates, apparently specifically in those genes that subsequently enable them to use the remaining metabolite to grow and multiply. The mutants do not exist before the bacteria are exposed to the metabolite, only *afterwards*; i.e., the mutants do not appear unless the metabolite is present.

Microbial geneticist Campbell and his team used a mutant strain of *E. coli* in which the gene coding for the enzyme that breaks down lactose, β -galactosidase, had been deleted from the genome, so the bacteria could not metabolise the sugar. The bacteria were smeared onto a petri-dish with high concentrations of lactose and some minimal nutrient that kept the bacteria alive. When the minimal nutrient was exhausted, mutant colonies began to appear which were able to metabolise lactose. The enzyme responsible in the mutant colonies was not the β -galactosidase that had been deleted from the genome, but another enzyme altogether, encoded by a gene called *ebg* (evolved β -galactosidase) mapping to the opposite side of the bacterial genome. This gene had acquired mutations that gave it lactose-metabolising activity. That result, though surprising, could be explained away in terms of the

artificial selection of a fortuitous variation that happened to confer lactose-metabolising ability on a bacterium that rapidly outgrew all the rest.

Two other geneticists, Hall and Hartl, later repeated the experiment, and isolated *thirty-four* independent lactose-utilizing mutant strains. All of the mutants contained enzyme activity identical to *ebg*. Moreover, in 31 of the strains, the synthesis of the newly evolved enzyme was regulated by lactose, i.e., there has been a mutation in another gene that interacts with lactose to regulate *ebg*. The chance of two random mutations arising in two different genes in the manner suggested by orthodox neo-Darwinian theory is 10^{-18} . One would have to grow about a hundred million litres of bacterial culture - the size of a football stadium - to get *one* of these double-mutants arising by chance.

Many subsequent experiments have confirmed that such mutations arise at frequencies many orders of magnitude above the 'spontaneous', 'random' mutation rates of about 10^{-9} , i.e., one in a billion, or smaller.

These findings led John Cairns and his colleagues in Harvard University, Cambridge, Massachusetts, to suggest that, "bacteria in stationary phase, have some way of producing (or selectively retaining) only the most appropriate mutations". The phenomenon has since been referred to as "directed mutations" or "adaptive mutations", much to the annoyance of the orthodox establishment.

The phenomenon exists, not only in bacteria but also in yeast and fruit flies, and I would be prepared to bet, in any other cells that geneticists care to look at (see below). But it doesn't seem to involve just one mechanism. DNA replication appears to be necessary, but beyond that, a variety of mechanisms may be involved, including hyper-mutations, genetic recombination, and single base-pair deletions.

It is clear that adaptive mutations in starving bacteria are due to a spectrum of genetic changes distinct from the 'spontaneous', random mutations that occur at extremely low frequencies. Despite that, neo-Darwinists are still trying hard to salvage the 'randomness' hypothesis by insisting that random errors are generated, which are subsequently selected.

But, as all the cells in the population are genetically uniform, and all the cells are potentially capable of generating those variations, there can be no question of selection in the usual neo-Darwinian sense. Instead it is a physiological response of starving cells undergoing

genetic changes that enable them to overcome the starvation.

A similar phenomenon has been discovered in human individuals where disease-associated defective genes spontaneously repair in somatic cells to become normally functional again.

Somatic cells in individuals with non-functional alleles (forms of a gene) have been discovered to revert spontaneously to functional, wild-type (normal) alleles. The best-known examples are in heterozygous individuals, i.e., those with two different alleles of the gene, one mutant and non-functional, the other normal and functional. Here, recombination within the gene during ordinary cell division, in which the two alleles exchange parts, could end up restoring function to the mutant allele. Gene conversion is also known, in which the functional normal allele converts the non-functional mutant allele.

New mechanisms of corrections have been found for homozygous patients (those with two identical, non-functional alleles) suffering from Fanconi anaemia. Abnormal alleles that arose from *frameshift* mutations, those due to insertion or deletion of basepair(s) that shift the reading frame of the triplet code, were restored to the correct reading frame. In one case, two separate single base-pairs were deleted from the mutant gene, and in the second, 5 contiguous base-pairs were inserted. In a third patient, the base T was mutated to G in position 1749 of the gene, and that mutation was compensated by a C to T substitution in position 1748. Although the protein resulting in all three cases differed in amino-acid sequence from the normal wild-type, they functioned just as well.

Whether those mutations are 'directed' in response to specific environmental or physiological stimuli is yet unknown. But it is an area that deserves further investigation.

Another indication that mutations may be largely non-random is that all organisms possess a wide range of DNA repair and proof-reading functions to remove accidental change to DNA sequences and correct errors resulting from physiological insults. University of Chicago microbial geneticist James Shapiro has pointed out that even bacteria may have "little tolerance for purely random variability".

What this means is that the fidelity of DNA replication is not guaranteed by the base-pairing mechanism. Under normal circumstances, the cell actively maintains its genetic material constant, which is why truly random errors are introduced when DNA replication reactions are carried out in the test-tube during genetic engineering.

However, the cells also have numerous biochemical mechanisms that deliberately change and reorganise DNA, in what amounts to “natural genetic engineering”, to enable them to respond to environmental challenges. The ability to activate those mechanisms under stress can significantly accelerate evolutionary change in crisis without threatening genetic stability under ordinary circumstances.

Shapiro has suggested that further studies of mobile genetic elements and DNA rearrangements would ultimately uncover truly directed mutations. Just as signal-transduction systems of the cell can direct the transcriptional complexes to specific locations in the genome to express specific genes, so mutational complexes (*mutatosomes*) could be similarly directed to mutate specific genes. We shall explore his suggestions further in Chapter 11.

Inheritance of environmentally induced changes in genes and genomes

In the previous chapter, I have given examples of *non-random*, repeatable changes that can be induced in the genes and genomes of cells and organisms exposed to toxic drugs, insecticides and herbicides, and in plants exposed to various fertilisers.

As plants and the majority of animal phyla do not have separate germ cells and somatic cells, those modifications will be inherited in subsequent generations. Even in animals with apparently separate germ cells, the germ cells may also respond directly to the same stimuli, or they may have communication channels with somatic cells. One such channel is via reverse transcription.

Feedback from somatic cells to germ cells

You have already come across the intriguing phenomenon of gene conversion in the previous chapter. There seem to be mechanisms that effectively report back to the germline, to change the genes or to stabilize them depending on the experiences of somatic cells, and according to the principle of use and disuse.

Reverse transcription, as you have seen, is extremely common in the genome. This has led to many speculations that it may be a normal physiological regulatory mechanism that can maintain the stability of organisms and also enable them to respond to environmental changes. Can the same mechanism prepare the next generation to cope with

environmental changes?

Reversed transcripts are abundantly re-inserted into the genome. Many are complementary DNA (cDNA) copies of the mRNA of genes, and largely non-functional; so they are referred to as 'pseudogenes'. But functional 'pseudogenes' have been identified. For example, one of the rat insulin genes was found to be part of a functional retro-transposon. Moreover, reverse transcriptase is now known to be in all genomes, being encoded by retrotransposons and retroviral sequences.

Australian-born immunologist Ted Steele proposed, back in 1979, that acquired immunological tolerance to foreign antigens - in which the body no longer produces an immune response against the foreign substance - was inherited via the *male* line, based on some experiments that he and his colleagues had carried out. The implication was that specific changes in germline DNA had resulted from exposing the organism to foreign antigens.

That was an explosive claim, especially in those days. With characteristic proselytising zeal, Ted did not just publish the results of his experiments in the journal *Nature*, after wrangling with the then Editor, John Maddox; he went on a world lecturing tour to publicize the results. Some of us cheered him on from the sidelines, but the orthodox community felt so threatened that they did their best to discredit him. Maddox called for his book describing his results to be burnt. We still don't know whether he may have been right about the inheritance of immune tolerance. But he came back twenty years later with what may be an example of feedback from somatic cells to germ cells via reverse transcription.

The immunoglobulin, or antibody protein, is shaped like a tuning fork and is made up of two heavy chains and two light chains. Both heavy and light chains have a variable region that binds to the foreign antigen (protein or other chemical) at the fork-end, which is crucial in eventually neutralising the antigen. The immunoglobulin protein is the result of several different genes assembled by chromosome rearrangements in the white blood cells that make the antibodies.

The germline genome consists of a number of different *V* genes, *D* genes and *J* genes, separated by some 300 000 base-pairs in the chromosome from the *C* genes encoding the constant region (Fig. 7.2).

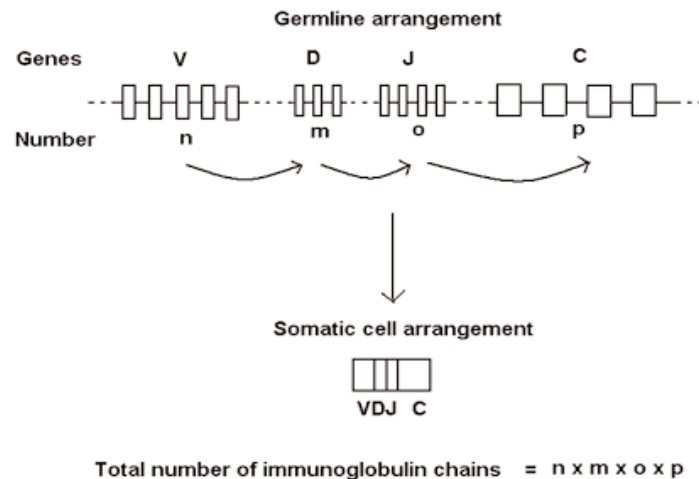


Figure 7.2. Several families of genes contribute to the initial diversity of antibodies.

To form an immunoglobulin coding sequence, one of the *V* genes must join to one *D* and one *J* gene, which together encode the variable *V* region of the immunoglobulin protein chain. That is further joined onto a *C* gene encoding the constant region. An initial diversity of immunoglobulins, consisting of millions of different immunoglobulin polypeptide chains, is thus generated by different combinations of the *V*, *D*, *J* and *C* genes; and one is expressed in each antibody-producing B cell.

The cells making immunoglobulins that bind well to the foreign antigen are stimulated to multiply to form clones (identical descendants of a single cell). These early antibodies are generally of low binding affinities, but in the later phase of the immune response, high affinity antibodies begin to appear.

It turns out that the high affinity antibodies are produced by hypermutations in the variable region of the molecule, *and only in the variable region*. The usual rates of random mutations are about one in a billion, but hyper-mutation rates are one in ten thousand to one in a thousand.

The mutations accumulate with time during the immune response

and are confined to the V region. By comparing the pattern of variation in the immunoglobulin DNA in somatic cells and in the germ cells, Steele and his colleagues found evidence suggesting that somatic V genes have been incorporated into the germline by homologous recombination. That is, a piece of DNA corresponding to the mutated V gene in a somatic cell has substituted for the V gene in the germ cell.

Hyper-mutation is most likely to be generated by rounds of transcription and reverse transcription, which are both known to have high error rates. (That is why retroviruses like the AIDS virus are notoriously difficult to get rid of.) During the later phase of the immune response, a B 'memory cell' divides a number of times into many daughter cells, which then undergo hyper-mutation by cycles of transcription and reverse transcription. The reverse transcript is reinserted into the cell's genome. When an antibody of sufficiently high binding affinity results, that B cell is stimulated to proliferate clonally, and the clone becomes the dominant cell population, producing large amounts of the particular high affinity antibody.

The dominant B cell population generates lots of reverse transcripts that may then be carried to the germ cells by retroviruses, according to Ted and his colleagues. If they are right, then that may be one of the functions of reverse transcriptase associated with retrotransposons.

Findings from 'gene therapy' experiments in recent years indicate that sperm cells are very good at picking up DNA, so reverse transcripts may well be taken up directly and undergo homologous recombination with the germline V genes. It is a case of gene conversion in which a sequence tested to work well for the body converts a germline gene, giving the next generation a little head start in life.

In the human cloning debate, it is worth remembering that although it is theoretically possible to produce a human clone without a sperm, ordinary reproduction does allow the embryo to pool genetic resources from both parents, giving the child double the benefits and shielding it from harmful genes, most of which have to be present in a double dose (one from either parent).

Could transposons be involved in the hyper-mutation of the V genes and in recombination with the germline genome in the manner suggested by microbial geneticist James Shapiro? Could other genes work in the same way? These are intriguing open questions that we shall explore in Chapter 11.

Gene conversion a physiological process?

Geneticists have recently made another big discovery suggesting that gene conversion by recombination may be a normal physiological process that goes on all the time.

It seems that substantial amounts of degraded genomic DNA are present in the bloodstream and in fluids bathing the cells, originating from cells in the body that have died. This circulating DNA binds to receptor proteins on the surface of living cells and is taken up and transported into the cell nucleus. Within the nucleus, homologous recombination takes place between DNA sequences that are very similar or nearly identical. So mutations that have occurred in certain genes in the cell's genome can be replaced with the correct non-mutated copy, or some other gene sequences can become converted to a copy present in the external DNA fragments.

This process would normally help to ensure that all the cells in the body have the same genome. Without this process, the cells in the body would mutate independently and their genomes would diverge more and more as the individual ages and the cells accumulate mutations independently of one another.

Such a process could account for 'concerted evolution' - the homogeneity of certain repetitive DNA sequences *within* species and their divergence between species (see previous chapter).

But there is a down side. When cancer cells die, or are killed, as by radiation or chemotherapy, their DNA is broken up into fragments which enter the general circulation. These DNA fragments may also be taken up by healthy cells, and could be responsible for the cancer spreading (a process referred to as *metastasis*). This raises serious questions over the safety of cancer therapies involving radiation and drugs that kill cancer cells, which may end up spreading the cancer itself. There is a school of thought to which I subscribe, that says cancer is above all, a *systemic* disease involving a fault in the intercommunication between the cancer cells and the rest of the body. The best way to cure cancer, therefore, is not to kill the cancer cells so much as to restore the intercommunication system that normally stop all cells in the body from multiplying out of control.

But why doesn't DNA from other plants and animals that we eat enter our cells and convert our genes? The quick answer here is that foreign DNA from other species does not have sufficient similarity (homology) to our DNA, and hence cannot enter our genome by homol-

ogous recombination. Homologous recombination increases the frequency of horizontal gene transfer a thousand fold or more. This may be the reason why most species are not cannibalistic, and why the practice of feeding cows rendered remains of its own species may have contributed to 'mad cow disease'.

But what about transgenic DNA in GM food, couldn't that be taken up by our cells and wouldn't that be hazardous? We shall address these important questions in more detail in connection with horizontal gene transfer in the next chapter.

Eight

Horizontal Gene Transfer - The Scourge of Genetic Engineering

What is horizontal gene transfer?

Horizontal gene transfer is the transfer of genetic material to unrelated species by processes other than normal reproduction. In the normal process of reproduction, which occurs only within a species or between closely related species, genes are transferred *vertically* with the germ cells (egg and sperm) that pass from parents to offspring.

Bacteria have long been known to exchange genes across species barriers in nature. There are three ways in which this is accomplished: *conjugation*, transduction, and transformation. In conjugation, DNA is passed between cells that are in contact with one another through long, thin tubular structures called *pili* (singular, *pilus*); in *transduction*, genetic material is carried from one cell to another by infectious viruses; and in *transformation*, the genetic material is taken up directly by the cell from the environment.

For horizontal gene transfer to be successful, the foreign DNA must become integrated into the cell's genome or become stably maintained in the recipient cell in some other form such as a plasmid in bacterial and yeast cells, or an *episome*, a replicated piece of (usually viral) DNA in the nucleus of animal cells, but outside the chromosomes.

Horizontal gene transfer and recombination between viruses take place when different viruses happen to infect and multiply in the same cell. Viral genomes often become integrated into the host cell's genome as provirus, to become replicated with the host cell genome. When proviruses are activated to develop into infectious viruses, they may take neighbouring host genes and DNA sequences along with them as they infect another cell. Viruses can also package loose pieces of genetic material along with its own genome that's tucked inside a protein coat (see below).

Horizontal gene transfer and recombination have played a major role in creating new viruses and bacteria associated with outbreaks of infectious diseases, and in spreading antibiotic and drug resistance among the disease agents, making the infections very difficult to treat.

For a long time, no one suspected that horizontal gene transfer could extend to higher plants and animals. Within the past decade, however, it has become clear that the scope for horizontal gene transfer is essentially the entire biosphere, with bacteria and viruses serving both as intermediaries for gene trafficking and as reservoirs for multiplying and recombining genes (see Chapter 6, Figure 6.1), and, I shall argue, it has become much more frequent with the help of genetic engineering.

Horizontal gene transfer in higher organisms

There are many potential routes for horizontal transfer of genes to plant and animal cells. Transduction is expected to be a main route, as there are many viruses that infect plants and animals, and plenty of opportunity for the viruses to pick up host genes and carry them from one host to another.

Transformation is potentially very important for cells of higher animals including those of human beings, and more than a decade of gene therapy research has left us in no doubt about that. A great variety of naked genetic material has been found to be readily taken up by all kinds of cells, simply by being applied in drops to the eye, rubbed into the skin, injected, inhaled, or swallowed. In many cases, the foreign gene constructs end up integrated into the genome, including those constructs that are not designed for integration. Some of us have been warning for years of the dangers caused by the integration of such foreign gene constructs into the cell's genome, among which is cancer, the uncontrollable proliferation of certain cells (see below).

Direct transformation may not be as important for plant cells, which generally have thick protective cell walls. But soil bacteria belonging to the genus *Agrobacterium* are able to transfer the T (tumour) segment of its tumour-inducing (Ti) plasmid into plant cells in a process that strongly resembles conjugation. This *T-DNA* is widely exploited as a gene transfer vehicle in plant genetic engineering, and has raised serious safety concerns (see Chapter 10).

Foreign genetic material can also be introduced into plant and animal cells by insects and arthropods with sharp mouthparts. In addition,

bacterial pathogens that enter plant and animal cells may be vectors for horizontal gene transfer. *Thus, there are almost no barriers to prevent the entry of foreign genetic material into the cells of probably any and every species on earth.* The most important barriers to horizontal gene transfer operate *after* the foreign genetic material has entered the cell.

Most foreign DNA entering a cell, such as that present in ordinary food, will get broken down to generate energy and building blocks for growth and repair. There are many enzymes that break down foreign genetic material. In the event that foreign genetic material is incorporated into the genome, chemical modification can still put it out of action, or the foreign DNA could become eliminated.

But under certain ecological conditions that are still poorly understood, the foreign genetic material escapes being broken down, and becomes integrated into the cell's genome. For example, heat shock and pollutants such as heavy metals enhance horizontal gene transfer; and antibiotics can increase the frequency of horizontal gene transfer between 10- to 10 000-fold.

Some genetic material can resist being broken down, especially that of genetic parasites such as viruses, plasmids and transposons. Viruses, plasmids and transposons not only undergo horizontal transfer most readily, but can also act as *vectors* (carriers) for transferring other genes (Fig. 8.1), and are employed as such in genetic engineering.

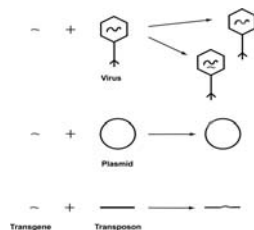


Figure 8.1. Vectors for horizontal gene transfer.

These genetic parasites have special signals and overall structures that protect them from the enzymes that degrade DNA (DNAase). A virus has its genetic material wrapped in a protein coat, which is shed on entering a cell. It can then either hijack the cell to make many more copies of itself, or its genetic material can jump directly into the cell's genome to become a provirus (Chapter 6). Plasmids can be indefinitely maintained in the cell separately from the cell's genome. Transposons - jumping genes - are blocks of genetic material that have the ability to jump in and out of genomes, with or without multiplying themselves in the process. Transposons can jump into plasmids to be propagated there, away from the chromosome, and to be transferred with the plasmids. Genes hitchhiking in viruses, plasmids and transposons, or spliced intentionally into them, have a greater probability of being successfully transferred into cells and to become integrated in genomes, which is why these genetic parasites are vectors for horizontal gene transfer.

Naturally existing genetic parasites are limited in the range of host-species they will infect, so, for example, pig viruses will infect pigs but not human beings, and cauliflower virus will not attack tomatoes. It is the protein coat of the virus that determines host specificity, which is why naked viral genomes (the genetic material stripped of the coat) are generally found to have a wider host range than the intact virus (see below).

Dormant proviruses in the genome are particularly dangerous, as when they meet cells of other species, they often turn into infectious viruses that attack the cells of the foreign species. This is one of the major risks of *xenotransplantation*, the transplant of cells, tissues and organs from animals such as pigs into human beings. Similarly, the signals for propagating different plasmids and transposons are usually specific to a limited range of host species, although there are exceptions. Some transposons are known to be very promiscuous, and can infect a wide range of species. These transposons are particularly popular with genetic engineers who use them as 'universal' vectors for gene transfer.

Genetic engineering has created a huge variety of artificial constructs and vectors by recombining the genetic material of the most promiscuous and dangerous bacteria, viruses, plasmids and transposons, which are designed to cross all species barriers and to invade essentially all genomes (see Box 8.1). In other words, artificial genetic

engineering greatly enhances horizontal gene transfer, the very process that creates new bacteria and viruses that cause infectious diseases, and spreads antibiotic and drug resistance among the pathogens.

What do you get when you cross a spider with a goat?

Genetic engineering is a collection of laboratory techniques used to isolate and recombine the genetic material of any species and to multiply the constructs in convenient cultures of bacteria and viruses in the laboratory. These techniques allow genetic material to be transferred between any and every species, especially those that would never interbreed in nature. That's how we have lost a favourite children's joke of what do you get when you make impossible crosses, as between a spider with a goat. For indeed, the spider's silk gene has become integrated into the goat's genome, in an attempt to make the poor female goat produce spider's silk in her milk. And silk worms have just been genetically modified to produce human collagen instead of silk.

In order to overcome natural species barriers limiting gene transfer and maintenance, genetic engineers have made a huge variety of artificial vectors, as mentioned above, by combining parts of the most promiscuous and infectious natural viruses, plasmids, and transposons from different sources. These artificial vectors generally have their disease-causing functions removed or disabled, but they have the capability to cross wide species barriers, so the same vector may now transfer, say, human genes spliced into the vector, to the genomes of bacteria in the first instance, and of all other mammals, reptiles, fish, insects, whatever. Genetic engineering is thus, facilitated and potentially unlimited horizontal gene transfer and recombination, with all the attendant risks.

Box 8.1

Artificial vectors enhance horizontal gene transfer

- They are derived from natural genetic parasites that mediate horizontal gene transfer most effectively.
- They have very mixed origins, containing genetic material from the viral pathogens, plasmids, and transposons of multiple species across kingdoms, and hence possess homologies (sequence similarities) to all of them. This facilitates widespread horizontal gene transfer and recombination. Homologous recombination is about a million times more frequent than non-homologous recombination under natural circumstances.
- They routinely contain antibiotic resistance marker genes (see Box 8.2), which enhance their successful horizontal transfer in the presence of antibiotics, either intentionally applied or present as pollutants in the environment. Antibiotics are known to enhance horizontal gene transfer from 10- to 10 000-fold.
- They often have 'origins of replication' and 'transfer sequences', signals that facilitate horizontal gene transfer and maintenance in cells to which they are transferred.
- Artificial vectors are well known to be structurally unstable, that is, they have weak links and tend to fragment, or break and join up incorrectly or with other DNA; this increases the propensity for horizontal gene transfer and recombination.
- They are designed to invade genomes and to overcome mechanisms that break down or disable foreign DNA, i.e., for enhanced horizontal transfer.

Box 8.2

What are antibiotic resistance marker genes?

Antibiotics are chemicals that kill bacteria and cells. There are many types of antibiotics. An antibiotic resistance gene codes for a protein that makes the bacteria or cell resistant to treatment with specific kinds of antibiotics. An antibiotic-resistant marker gene is one that accompanies foreign genes the genetic engineer wants to transfer into cells. It is placed next to the foreign genes in the

construct, so that those cells that have taken up the construct will also be resistant to the antibiotic. This provides a convenient way to select for transformed cells simply by using the antibiotic(s) to kill off the rest of the cells (see Chapter 11, Figure 11.1).

Although different classes of vectors are distinguishable on the basis of the mainframe genetic material, practically all of them are mosaics of genetic material originating from the genetic parasites of many different species of bacteria, animals and plants. Simply by creating such a vast variety of promiscuous gene transfer vectors, genetic engineering biotechnology has effectively opened up highways for horizontal gene transfer and recombination where previously the process was tightly regulated, with restricted access through narrow, tortuous footpaths. These gene transfer highways connect species in every domain and kingdom with microbial populations via the universal mixing vessel in genetic engineering, *E. coli*.

There is currently no legislation in any country to prevent the escape and release of artificial vectors and other artificial naked DNA constructs into the environment.

Terje Traavik, Scientific Director of the Norwegian Institute of Gene Ecology in Tromsø University, Norway, was the first to warn his government and the international community of the dangers of naked DNA constructs that are increasingly made by the biotech industry and discharged into the environment.

What alerted him was when, during a routine experiment, he and his colleagues injected the naked DNA genome of the human polyoma virus into rabbits as a control, for the intact virus was not infectious to rabbits at all. To his great surprise, the naked DNA viral genome produced a full-blown infection in the injected rabbits!

The simple manipulation of naked viral genomes during genetic engineering is already a danger.

Dangers from horizontal transfer of transgenic DNA

As artificial vectors and the artificial gene constructs they carry are predominantly made from genetic material of viruses and bacteria that cause diseases, they will therefore readily recombine with the naturally existing strains. In the process, newer, more dangerous strains could

be created. This is just common sense.

Has the growth of commercial-scale genetic engineering biotechnology contributed to the resurgence of drug and antibiotic infectious diseases within the past 30 years? A number of colleagues and I got together to review a combination of deductive, inductive and circumstantial evidence, suggesting that the answer may well be 'yes'; and we called for a major public enquiry. Of course, we did not, and are not saying that genetic engineering was solely to blame; the abuse and over-use of antibiotics, ecological destruction, deterioration of public sanitation, malnutrition, poverty, social disintegration, increase in travel, wars, etc., have all played their part. But no one has considered genetic engineering biotechnology.

Our review was published in 1998 in the journal *Microbial Ecology in Health and Disease*, edited by a distinguished medical microbial ecologist of Karolinska Institute, Stockholm, Sweden. At the time, there was already little remaining doubt among medical microbial geneticists that horizontal gene transfer and recombination have been responsible for creating new pathogens and spreading drug and antibiotic resistance among the pathogens. What we said was that the frequency and scope of horizontal gene transfer may have increased since genetic engineering began. We sent the review to a number of national and international regulatory bodies, including the World Health Organisation (WHO) and the UK Health and Safety Executive (HSE).

The WHO at first responded with categorical denial, but later sent a one-sentence reply, saying that the issue raised was receiving attention; and after that, complete silence. The UK HSE appeared much more responsive, and after a series of correspondence with me, eventually commissioned its own review from Timms-Wilson and colleagues in the Molecular Microbial Ecology Laboratory, Institute of Virology and Environmental Microbiology in Oxford. Perhaps predictably, the review simply considered all the incriminating evidence, and came to the following conclusion:

"This report has shown that, despite the recognised role of horizontal transfer in the evolution of pathogens, it is unlikely that a benign strain will be converted to a pathogenic strain by acquisition of genes horizontally transferred from a GMM [genetically modified micro-organisms]." What followed was a series of circumlocutions around "selective pressures", "specific consideration of the genes chosen for manipulation", with suitable apologies on there being only "a superficial under-

standing of the complexity and functional biodiversity of soils and the wider environment" and that improvements in knowledge is "essential for accurate predictions and evaluations of real versus perceived risk". At the end, it recommended that assessment of risks will have to be considered on a "case by case basis", a bureaucratic cliché in biosafety regulation which effectively allows business to go on as usual.

We were actually not the first to suspect that genetic engineering may inadvertently create new, infectious viruses and bacteria. The pioneers of genetic engineering called for a moratorium in the mid-1970s - the famous Asilomar declaration - precisely because they were worried about this obvious prospect. Unfortunately, the scientists succumbed to commercial pressures, and the moratorium was short-lived. The issue raised has never been resolved. If anything, subsequent discoveries have greatly increased its relevance, especially the persistence of DNA in all environments including the human gut long after the organisms are dead; and the ease with which all cells including those of human beings take up foreign DNA.

Accidental creation of killer virus in genetic engineering

In January 2001, the suspected came to light. Researchers in Australia, in the course of an apparently innocent genetic engineering experiment, accidentally transformed the harmless mouse-pox virus into a lethal pathogen that killed all its victims. The researchers also showed one of the ways in which this could be done: by putting into the virus a gene coding for a protein that suppresses the immune system, Interleukin-4. Genes similar to that are in pharm crops being field tested in Canada, as Joe Cummins has discovered. There is plenty of opportunity for benign viruses in the environment to turn into similar killer-viruses simply by picking up the gene for the immune suppressor protein from the pharm crop.

Horizontal gene transfer and recombination is one major route to generating pathogens (the other being mutation), and as said earlier, genetic engineering is nothing if not facilitated horizontal gene transfer and recombination. It is like winning a lottery - from the point of view of the virulent pathogen - the more tickets you buy, the more chances of winning.

That episode indeed brings home the hazards inherent to genetic engineering, which is in many ways worse than the intentional creation of biological warfare agents. The same tools and materials are

involved. Biological agents will normally be made in tightly contained facilities with the strictest precaution, and researchers will generally know exactly what they are making, so in case of accidents, proper remedial measures can be taken promptly. In the case of genetic engineering, however, no such safety precautions are taken, and unknown pathogens may well be created accidentally, which, more often than not, will completely escape notice.

That's not all. Geneticists have been intentionally creating hybrid superviruses in the laboratory during routine research on dangerous pathogens and in making vaccines. There is one called SHIV, a hybrid between human and monkey AIDS viruses, that kills in weeks, and is routinely used as a 'challenge' virus to test AIDS vaccines. Many AIDS vaccines themselves - made from the glycoprotein gene (*gp120*) of the human immune deficiency virus (HIV) - are so hazardous that a group of virologists headed by Veljko Veljkovic of the Institute of Nuclear Sciences, Belgrade, Yugoslavia, has called for clinical trials to be stopped.

The *gp120* is similar to the variable (V) region of the human immunoglobulin antibodies that binds to foreign antigens, and can therefore interfere with the immune response. The *gp120* gene also has special 'recombination hotspots', sites in the DNA especially prone to recombination. These sites are similar to those present in the human immunoglobulins as well as those in many viruses and bacteria, and can hence readily engage in horizontal gene transfer and recombination to create dangerous new pathogens.

The biotech company ProdiGene, recently prosecuted for contaminating food crops with 'pharm' crops engineered to produce vaccines and drugs (see Chapter 4), has put *gp120* into maize plants. Veljko Veljkovic and I referred to it as effectively the release of a slow bioweapon in a commentary published in the electronic journal *AIDScience*.

Most of the deadly pathogens from genetic engineering will be created in the environment when no one is looking. That's because large quantities of transgenic DNA, with viral and bacterial genes, in a profusion of new, unnatural combinations with genes of other organisms, are routinely discharged into the environment from 'contained use' facilities as transgenic wastes, still under the old, thoroughly discredited assumption that DNA will readily break down in the environment. Transgenic DNA is also deliberately released into the environ-

ment in GM crops that generate pollen, dust and debris; and among the GM crops are those engineered with dangerous genetic material to produce drugs and vaccines.

Microbial geneticists can culture and identify less than one percent of all the bacteria in the environment. The possibilities for creating new pathogens by horizontal gene transfer and recombination of transgenic DNA are endless.

Trangenic DNA and cancer

I mentioned cancer arising from the integration of foreign gene constructs into the cell's genome from gene therapy.

Gene therapy is simply the 'genetic modification' of human cells, using constructs very similar to those used in the genetic modification of animals and plants. The worrying thing about the cancer cases in France is that they have developed among the few (9 so far) unequivocal successes after 14 years of clinical trials and research in gene therapy in Europe and the United States. And the patients were treated with a procedure that was designed to minimise the risks of cancer.

Instead of exposing human subjects directly to the foreign gene (transgenic) constructs, bone marrow cells were taken out of the patient and transformed with a vector made from a retrovirus carrying the transgenic construct. The transformed cells were then selected and put back into the patient. The unanswered question, a question not even raised in the enquiry that followed, is whether the transgenic construct moved again, after the cells were transplanted back into the patient. Molecular characterisation was not done until after the first tragedy occurred.

In the first patient, the retroviral vector carrying the foreign genes had jumped into a gene, *LM02*, whose over-expression led to the uncontrolled proliferation of the white-blood cell containing this aberration, which is responsible for the leukaemia. It turns out that the retroviral vector has integrated into the same gene in the second patient to develop leukaemia. Since then, a similar insertion has been identified in a third child who is not, as yet, showing signs of leukaemia. Members of the United States National Institutes of Health (NIH) Recombinant DNA Advisory Committee (RAC) met on 10 February this year. They recommended that this particular kind of gene therapy should be used only if patients fail to respond to all other treatments, such as conventional bone marrow transplant from tissue-type matched donors. They

still failed to acknowledge that other vectors carry the same risks, let alone that all transgenic DNA carries the same risks.

The hazards of horizontal gene transfer from genetic engineering are summarized in Box 8.3.

Box 8.3

Potential hazards of horizontal gene transfer from genetic engineering

- Generation of new cross-species viruses that cause disease
- Generation of new bacteria that cause disease
- Spread of drug- and antibiotic-resistance genes among the viral and bacterial pathogens, making infections untreatable
- Random insertion into genomes of cells, resulting in harmful effects including cancer
- Reactivation and recombination with dormant viruses (present in all genomes) to generate infectious viruses
- Spread of dangerous new genes and gene constructs that have never existed
- Multiplication of ecological impacts due to all of the above.

Transgenic DNA not the same as natural DNA

Proponents of genetic engineering are fond of reassuring the public that DNA is DNA, no matter how it is made. They say, we have been eating lots of DNA in our food without being turned into cabbages or cows. So, why should we worry about transgenic DNA?

I have already given some of the reasons why we should worry about transgenic DNA (see Box 8.3). It is worth noting that in addition to being made predominantly from the genes of viruses and bacteria that cause diseases, they are brought together in combinations that have never existed in billions of years of evolution.

To see how different transgenic DNA is from natural DNA, you need to know its basic structure.

Genes are never transferred alone. They are transferred in unit constructs known as *gene-expression cassettes*. Each gene has to be accompanied by a special piece of regulatory genetic material, the *pro-*

moter, which signals the cell to turn the gene on, that is, to bind transcription factors and RNA polymerase that transcribes the DNA gene sequence into RNA (see Chapter 5, Figure 5.2). At the end of the gene, there has to be another signal, a *terminator*, to stop the transcription and so that the transcript can be further processed and translated into protein. The simplest expression cassette looks like this (Fig.8.2).

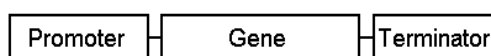


Figure 8.2. A gene-expression cassette.

Typically, each bit of the construct (promoter, gene, and terminator) comes from a different source. The gene itself may also be a composite of bits from different sources. Several cassettes are frequently linked in series or 'stacked' in the final construct. At least one of the expression cassettes will be that of an antibiotic resistance marker gene, to enable cells that have taken up the foreign construct to be selected with antibiotics, which often remains in the transgenic organism.

I have drawn the construct with lines linking the blocks of genetic material of different origins that are joined together, to indicate the potentially weak links. Such artificial constructs are known to be structurally unstable compared to the naturally existing DNA, tending to break and rearrange, so much so that the instability of artificially constructed vectors is a text book topic (see Chapter 11, Box 11.2). Structural instability will enhance horizontal gene transfer and recombination. Moreover, certain kinds of transgenic constructs are extra-unstable, and hence particularly prone to horizontal gene transfer and recombination with all the attendant risks.

The CaMV 35S promoter

The cauliflower mosaic virus (CaMV) infects plants of the cabbage family. One of its promoters, the 35S promoter, has been widely used in GM crops since the beginning of plant genetic engineering, before some of its worrying features came to light. The most serious of these is that it appears to have a 'recombination hotspot', where it tends to recombine with other DNA; although definitive evidence for that did not appear until much later.

Since the early 1990s, major doubts have arisen over the safety of viral genes that were being incorporated into GM crops to make crops resistant to viral attack. Many of the viral genes tended to recombine with other viruses to generate new and at times super-infectious viruses. Joe Cummins, Emeritus Professor of the University of Western Ontario, Canada, was almost a lone voice in objecting to the CaMV 35S promoter since 1994.

In 1999, definitive evidence for the recombination hotspot in the CaMV 35S promoter came from work published independently by two research groups. Interestingly, we turned up one of the papers during a computer search of the scientific literature for 'transgenic instability'.

It prompted us to write a review on the safety implications of this and other findings concerning the CaMV 35S promoter, in order to start a proper debate among scientists, or so I thought. We pointed out that the CaMV 35S promoter recombination hotspot is flanked by multiple motifs known to be involved in recombination, which are similar to other recombination hotspots, including the borders of the *Agrobacterium T* DNA vector most frequently used in making transgenic plants (see Chapter 10). The suspected mechanism of recombination - double-stranded DNA breaks followed by repair - requires little or no DNA sequence homologies, and recombination between viral transgenes and infecting viruses has been amply demonstrated. In addition, the promoter functions efficiently in all plants, green algae, yeast and *E. coli*. It has a modular structure, with parts common to, and interchangeable with, promoters of many other plant and animal viruses.

These findings suggested to us that transgenic constructs with the CaMV 35S promoter might be especially unstable and prone to horizontal gene transfer and recombination, with all the attendant hazards: gene mutations due to random insertion, cancer, reactivation of dormant viruses and generation of new viruses. These considerations were especially relevant in the light of Ewen and Pusztai's report in *The Lancet* published at around the same time, that certain transgenic potatoes containing the CaMV 35S promoter may be unsafe for young rats. They suggested that a significant part of the effects might be due to "the construct or the genetic transformation (or both)".

Consequently, we called for all GM crops containing the CaMV 35S promoter to be immediately withdrawn.

When our paper was finally accepted for publication, the Journal, *Microbial Ecology in Health and Disease*, put out a press release on its

website, labelling it a 'hot topic'. Within a day, someone by the name of Klaus Ammann organised at least nine critiques that rebounded around the Internet, ranging from the abusive and condescending to the relatively moderate. I later learned that Klaus Ammann is a key player in establishing (or undermining) biosafety standards on the international scene.

Ammann is Director of the Botanical Garden, University of Bern, Switzerland. Together with John Beringer, Julian Kinderlerer, Alan McHughen and Mark Tepfer (author of one of the critiques), Amman formed the International Society for Biosafety Research, whose journal, *Environmental Biosafety Research*, he edits. He also sits on steering committees for the following groups: The European Science Foundation group, Assessment of Impacts of Genetically Modified Plants (AIGM), Gensuisse, a GM promotion group funded by the pharmaceutical industry body Interpharma, and Internutrition, a promotion group for GM foods which has in its working group representatives from the food industry such as Nestlé, as well as the companies Monsanto, Hoffmann LaRoche, DuPont and Syngenta.

Amman is also co-editor of the Bio-Scope Frankfurt-Bern website, supported by GM industry group Europabio. With CS Prakash and others, Ammann signed an "Open Letter to the United Nations Commission on Sustainable Development" (April 2000). The letter warned the Commission against "needless over-regulation" of GM, and "the very real threat that an overly-strict adherence to precautionary regulation could pose to both the environment and to the well being of human populations around the world." That letter was obviously aimed against the Cartagena Biosafety Protocol that was finally agreed in January 2000 after a long and difficult negotiation going back at least to 1995. But the proposal for an international biosafety protocol to regulate genetic engineering was first presented to the United Nations Convention on Biological Diversity by Malaysia around 1990. The precautionary principle is enshrined in the Cartagena Biosafety Protocol, which has now been signed by 45 countries (as of 4 March 2003), including the European Union; although attempts to undermine it have continued unabated.

Amman further edits a strongly pro-GM e-mail list and has since been accused of playing a leading role, together with CS Prakash, in mobilising the attacks on David Quist and Ignacio Chapela's paper on the transgenic contamination of Mexican maize landraces published in *Nature*.

As to the attacks mobilised against our CaMV paper, curiously, the most moderate critique was from Monsanto, while the most abusive was from Paul Christou, the leader of one of the research teams (in JIC) that had provided the molecular evidence for the recombination hotspot, but had failed to draw what seemed to us the obvious conclusions.

We answered all the criticisms in a paper that was circulated on the Internet, and subsequently published in the scientific journal. Our critics have failed to respond to this day.

Unfortunately, the most outrageous and abusive remarks were incorporated into an 'analysis' piece written by an editor of *Nature biotechnology* under "Business and regulatory news" published in January 2000. That 'analysis', concocted entirely of hearsay and opinions, contained such defamatory, libellous statements that the journal had to give us a right to reply when challenged. There was a long delay, when the *Nature biotechnology* office kept 'losing' our e-mails. Our reply was eventually published several months later, along with the editor's 'apology' that he had failed to cite our rebuttal, but was actually another attack on us. This time, *Nature biotechnology* refused to let us reply.

All of the substantive scientific criticisms eventually turned up in a paper published in the journal where our original paper appeared. Roger Hull from JIC was first author and Phil Dale, a member of ACNFP, the third author. Their main criticisms boiled down to the following.

First, people have been eating the virus in infected cabbages and cauliflower for many years without harm, so why should they worry about the CaMV 35S promoter? Second, plants are already loaded with pararetroviral sequences, not unlike CaMV, so why should there be any risks?

We rebutted the criticisms thoroughly in a paper that was longer than the original, which appeared in the same journal soon afterwards.

We pointed out, among other things, that people have not been eating the CaMV 35S promoter plucked from its natural genetic and evolutionary context and incorporated into transgenic DNA.

The fact that plants are "loaded" with pararetroviral sequences similar to CaMV and other potentially mobile elements can only make things worse. Pararetroviruses are viruses that use reverse transcriptase, but do not depend on integrating into the host genome for repli-

cation. Pararetroviruses include a family that contains the human pathogen, hepatitis B virus. We thought that was significant, as the CaMV 35S promoter could activate dormant viruses like hepatitis B, which was also known to have integrated into some human genomes, and appeared to be associated with the disease.

Most, if not all of the elements integrated into the genome would have been 'tamed' in the course of evolution and hence no longer mobile. But integration of transgenic constructs containing the 35S promoter may mobilize the elements. The elements may in turn provide helper-functions to destabilize the transgenic DNA, and serve as substrates for recombination to generate more exotic invasive elements.

Evidence has emerged since, that integration of foreign genes into the genome associated with the genetic modification of *Arabidopsis* can indeed activate transposons and proviral sequences, leading to destabilisation of the genome. So we were not wide off the mark.

Roger Hull was asked to reply to us by a local group that called itself, "Concerned Citizens of Wivenhoe".

Hull initially promised to do so, but after a long delay, wrote back to say that he had discussed the whole matter with the Director of JIC and decided not to reply, and instead, "to draw up a briefing document targeted towards decision and opinion makers".

In the course of debating with our critics, we found even more damning evidence. It turns out that although the CaMV infects only plants in the cabbage family, its 35S promoter is promiscuously active in species across the living world, not just bacteria, algae, fungi, and plants, but also animal and human cells, as we discovered in a scientific paper dating back to 1990. Plant geneticists who have incorporated the CaMV 35S promoter into practically all GM crops now grown commercially are apparently unaware of that, and are still not admitting to it in public.

ACRE has no excuse for omitting the information in its latest Report (see Chapter 4), as I have personally drawn attention to it many times, both in written submissions and in oral evidence presented at several open hearings. Behind the scenes, however, the CaMV 35S promoter has been quietly withdrawn. It no longer appears in most of the GM crops under development

Recent evidence in connection with the transgenic contamination of Mexican maize landraces suggests that our predictions concerning the CaMV 35S promoter may have been borne out (see Chapter 11).

Once more with feeling

I must admit that the latest ACRE Report was a blow, even though we had to put on a brave face in public. We had worked so hard, for so long. Everyone in ISIS feels utterly exhausted, physically and emotionally. We have taken on Thomas Hobb's Leviathan, a giant political establishment with people paid and rewarded not to do their job properly, while people like us are making personal sacrifices and often suffering abuse to try to do their job for them.

The Penang meeting of 1994 seems so long ago. Many excellent activists have come and gone, worn down both by the mechanical bureaucracy that can see no further than guarding the *status quo*, and the sheer corruption of the corporate presence infiltrating so thoroughly into the collective public consciousness that critical voices are mistaken for cries of madness.

I had seriously considered boycotting the GM Science Review altogether; as all the indications are that the evidence will once again be dismissed (see Chapter 4). But a letter came from Chief Scientist David King himself, urging me to take part in the Review. It was no more than a form letter sent to many scientists, but it persuaded me to put the case across just once more.

I decided to focus on the ACRE Report, to show exactly where it has made false assertions, bland denials, misrepresentations, obfuscations and omissions, all of which add up to a serious abuse of science. We submitted that to the GM Science Review.

We heard nothing for almost a month. The Secretariat of the GM Science Review finally wrote me a letter, saying that my submission was turned down because of the language I used to describe ACRE. The offending paragraphs included a verbatim citation from a letter, published in the journal *Nature biotechnology*, from an international consortium of 18 pro-GM scientists attacking David Schubert, cell biologist from the Salk Institute in La Jolla, California. Schubert is the latest scientist to fall victim after expressing concerns over the safety of genetic engineering.

Schubert argued that, "GM food is not a safe option, given our current lack of understanding of the consequences of recombinant technology". He called for "all GM plant products destined for human consumption to be tested for long-term toxicity and carcinogenicity before being brought to the market."

My paper was eventually posted on the GM Science Review

website after I edited out the 'offending' paragraphs and re-submitted it with an accompanying letter that stated the following:

"I have consulted widely with local groups and other scientists, and they are unanimous in their agreement over the validity of, and justification for my remarks. In particular, my opening paragraph cites statements that were actually made by pro-biotech scientists against a colleague who is trying to draw attention to evidence of hazards that they have ignored. Many independent scientists including myself have been trying to do the same, and have received similar treatments and worse. So it is merely a case of putting the mirror back on those who have criticised us. As to the language I used, it is far more moderate than what has been said in print and in the popular media by a number of scientists on the Science Review panel against some of us on many occasions.

"Those whom I have consulted are strongly divided as to whether I should edit ISIS' contribution and re-submit. I have given the matter due consideration, and decided to do so in the interest of getting the scientific arguments properly aired.

"But in the interest of fair play, I must transmit the request made by those I have consulted, that members of the Science Review Panel should stop criticising and vilifying independent scientists who challenge their pronouncement that 'there is no evidence of harm'."

I don't hold out much hope that they will stop attacking us, but at least, the wider public ought to know what we are up against.

Why transgenic DNA may be more prone to horizontal transfer

Transgenic DNA is different from natural DNA in many respects, all of which contribute to its increased propensity for horizontal transfer into genomes of unrelated organisms, where it may also recombine with new genes (see Box 8.4).

In the next chapter we shall look at the evidence of horizontal transfer of transgenic DNA in some detail.

Box 8.4

Transgenic DNA more likely to spread horizontally

- Transgenic DNA has been designed to jump into genomes.
- The unnatural gene constructs tend to be structurally unstable and hence prone to break and join up or recombine with other genes.
- The mechanisms that enable foreign gene constructs to jump into the genome enable them to jump out again and reinsert at another site or in another genome. For example, the enzyme integrase, which catalyzes the insertion of viral DNA into the host genome, also functions as a disintegrase, catalyzing the reverse reaction. These integrases belong to a superfamily of similar enzymes that are present in all genomes, from viruses and bacteria to higher plants and animals. Recombinases of transposons are similar.
- The borders of the most commonly used vector for transgenic plants, the *T-DNA* of *Agrobacterium*, are recombination hotspots (sites that tend to break and join). In addition, a recombination hotspot is also associated with the cauliflower mosaic virus (CaMV) promoter and many terminators, which means that the whole or parts of the integrated DNA will have an increased propensity for secondary horizontal gene transfer and recombination.
- Recent evidence indicates that foreign gene constructs tend to integrate at recombination hotspots in the genome, which again, would tend to increase the chances of transgenic DNA *disintegrating* and transferring horizontally (see Chapter 11).
- Transgenic DNA often has other genetic signals, such as *origins of replication* left over from the plasmid vector. These are also recombination hotspots, and in addition, can enable the transgenic DNA to be replicated independently as a plasmid that's readily transferred horizontally among bacteria.
- The metabolic stress on the host organism due to the continuous over-expression of the foreign genes linked to aggressive promoters such as the CaMV 35S promoter will also increase the instability of the transgenic DNA, thereby facilitating horizontal gene transfer.

- The transgenic DNA is typically a mosaic of DNA sequences from many different species and their genetic parasites; these homologies mean that it will be more prone to recombine with, and successfully transfer to, the genomes of many species as well as their genetic parasites. Homologous recombination typically occurs at one thousand to one million times the frequency of non-homologous recombination.

Nine

Horizontal Transfer of Transgenic DNA

Evidence that transgenic DNA is different

Doing science is rather like solving a murder mystery. You have to follow your hunches, which often get you into blind alleys; you have to watch out for clues, look at the evidence from experiments, interpret the experimental findings and connect them up into a coherent picture. And then, you have to present the case to your peers and the public, as though in a court of law.

More importantly, the faculty of judgement that tells you whether the evidence is good enough to convict a murder suspect also tells you whether the scientific evidence is good enough to support your case. You cannot just accept the evidence at face value, but have to look into how the evidence was produced. Was the particular experiment well planned and executed? Did the findings support the conclusions, or were they misinterpreted?

(Doing science is rather like doing art as well, and the same faculty of judgement that enables you to distinguish good science from bad also lets you tell good art from bad, but that takes us too far afield for now.)

It is often argued that transgenic DNA, once incorporated into the transgenic organism, will be just as stable as the organism's own DNA. But direct and indirect evidence contradict this supposition.

Transgenic lines are notoriously unstable and often do not breed true. There are no molecular data documenting the structural stability of the transgenic DNA in either GM crops or GM animals in successive generations, both in terms of its site of insertion in the host genome and its arrangement of genes (genetic map). Instead, secondary mobility and rearrangements often occur, along with silencing of the transgenes, so that they are no longer expressed. I shall say more on transgenic instability at the end of Chapter 11.

As said above, scientific evidence is no different from ordinary evidence that might be presented in a law court for a suspected murder. More often than not, no one, apart from the murder victim, has actually witnessed the deed, and indirect evidence from different sources and of different kinds have to be synthesized. The same goes for the scientific evidence on the hazards of genetic engineering. Very few people will die immediately on eating GM foods. With the exception of acute toxins and allergens, the hazards are insidious, and will only appear in the longer term, provided there is systematic health or environmental monitoring. So all we can do is to look for somewhat indirect evidence of hazards.

In the previous chapter, I have presented many reasons why transgenic DNA is different from natural DNA and why it may be more likely to undergo horizontal gene transfer, a process that's responsible for potentially the greatest hazards from genetic engineering. Although my arguments were based on experimental findings, the experiments were not designed to address those questions directly. In this chapter, we shall look at some of the more direct experimental evidence indicating that transgenic DNA is different from natural DNA, and that it can indeed transfer horizontally.

Evidence that transgenic DNA is different

There has been only one experiment ever carried out to test the hypothesis that transgenes are the same (or not) as mutants induced by conventional means (mutagenesis), such as exposure to X-rays and chemical mutagens, which cause changes in the base sequence of DNA.

Joy Bergelson and colleagues in the Department of Ecology and Evolution, University of Chicago, United States, obtained a mutant for herbicide-tolerance by conventional mutagenesis in a laboratory strain of *Arabidopsis*, and created transgenic lines by introducing the mutant gene, spliced into a vector, into host plant cells.

The researchers then compared the rate at which transgenic and non-transgenic mutant plants spread the herbicide-tolerance trait to normal, wild type plants grown nearby. They found some surprising results.

The transgenes from the transgenic plants were up to 30 times more likely to escape and spread - presumed to be via pollination - to the wild-type plants than the same gene obtained by mutagenesis.

The results are very intriguing, and difficult to explain in terms of ordinary cross- pollination. Was it because introducing the transgene by means of a vector led to all kinds of unexpected effects? Did the transgenic plants produce more pollen, or more viable pollen? Was the pollen from transgenic plants more attractive to bees?

Another possibility for the increased spread of transgenes is horizontal gene transfer, via insects visiting the plants for pollen and nectar, or simply feeding on the sap or other parts of successive transgenic and wild type plants. I had posed this question to Joy Bergelson. She said they had no evidence for horizontal gene transfer, but could not rule it out. But they have not gone on to investigate that possibility.

Regardless of the manner in which the transgenes had spread, the experiment demonstrated that transgenic DNA did not behave in the same way as non-transgenic DNA.

When is scientific evidence considered sufficient?

If you have only ever heard from the proponents of genetic engineering and scientists advising our government, you might think that I have been speaking only of theoretical possibilities of horizontal gene transfer, not based on empirical evidence. Not so.

Horizontal transfer of transgenes and antibiotic resistant marker genes from genetically engineered crop plants into soil bacteria and fungi had been documented in the laboratory by the mid 1990s. Transfer of transgenes to fungi was achieved simply by growing the fungi with the GM plant, and transfer to bacteria achieved by applying total DNA from the GM plant to cultures of bacteria.

By the late 1990s, successful transfers of a kanamycin-resistance marker gene to the soil bacterium *Acinetobacter* were obtained with total DNA extracted from homogenized leaves in a range of transgenic plants: *Solanum tuberosum* (potato), *Nicotiana tabacum* (tobacco), *Beta vulgaris* (sugar beet), *Brassica napus* (oil-seed rape), and *Lycopersicon esculentum* (tomato). It was estimated that about 2 500 copies of the kanamycin-resistance genes (from the same number of plant cells) were sufficient to successfully transform one bacterium, despite the fact that there was a 6×10^6 -fold excess of plant DNA present. A single plant with, say, 2.5×10^{12} cells would be sufficient to transform 10^9 bacteria. Positive results of horizontal gene transfer in this system were obtained even with just a smidgen (100 microlitres) of ground up plant leaf added to the bacteria.

But from the beginning, obfuscation and misrepresentation reigned supreme (see Box 9.1).

Box 9.1

Obfuscating and misrepresenting positive findings of horizontal gene transfer

Despite the misleading title in one of the publications, which states that horizontal gene transfer in their experiment "occurs, if at all, at an extremely low-frequency", a high frequency of gene transfer of 5.8×10^{-2} per recipient bacterium was demonstrated under optimum conditions. The authors then proceeded to calculate a theoretical gene transfer frequency of 2.0×10^{-17} (close to zero) under extrapolated "natural conditions," by assuming that different factors acted independently, and by inventing the 'natural conditions', which are largely unknown and unpredictable, and, by the authors' own admission, synergistic effects from combinations of factors cannot be ruled out.

This paper, by Schluter, Futterer and Potrykus, was subsequently widely cited as showing that horizontal gene transfer does not happen.

In March 1999, I was invited along with Michael Antoniou, biomedical geneticist in Guy's Hospital who has also been very active in warning the public of the dangers of GM foods, to debate the issue with pro-biotech geneticists advising our UK government, in the office of the Minister for the Environment, Michael Meacher. There were five pro-biotech geneticists present, including the following: Mark Bailey, microbial molecular biologist who now sits on ACRE, Don Grierson, Nottingham University plant geneticist and creator of a GM tomato that was briefly on the supermarket shelves in Britain before it disappeared without a trace, and Phil Dale, JIC plant geneticist who now sits on ACNFP.

We discussed all the hazards, including the findings of Ewen and Pusztai, which had just appeared in print, and the dangers of the CaMV 35S promoter. The science advisors dismissed everything. On horizontal gene transfer, they said, "Just because horizontal gene transfer happens in the laboratory does not mean it can happen in nature". Not so long ago, they were denying that horizontal gene transfer could happen at all.

Towards the end our debate, I recall Michael Meacher asking what seemed to me the important question, "When will scientific evidence be considered sufficient to indicate that the risk is unacceptable?" The science advisors had no answer.

I drafted a special briefing paper for Michael Meacher immediately afterwards on the *Special safety concerns of transgenic agriculture*, in which I replied to Meacher's question explicitly as follows.

"When is scientific evidence considered sufficient to indicate that the risk is unacceptable? Risk is technically the extent of damage multiplied by the probability that the damage will occur. People take risks for a number of reasons: because they have to, or because there is overwhelming moral imperative for doing so, or because the likely benefits are compelling despite the potential damage. Not one of these reasons applies in the case of transgenic agriculture. On the contrary, existing scientific evidence pointing to the serious damages to health and the ecological environment that are likely to be incurred should compel us to call an immediate halt to the enterprise. That is in accordance with the generally accepted precautionary principle.

"Instead, scientists on the relevant advisory committees appear to have been operating on the inverse precautionary principle, according to which all processes and products must be approved unless proven absolutely unsafe. Arguments such as "no-one has been shown to have died from eating genetically engineered food yet" or "just because horizontal gene transfer happens in the laboratory does not mean it will happen in nature" go against the practice of good, sound science and are frankly irresponsible. It is like saying we have to wait for 8 000 babies to be born with truncated limbs before admitting there is sufficient evidence that thalidomide is harmful.

"The most rational, responsible course of action is to impose a five year moratorium at the very least, in order to create space for desperately needed research, and more importantly, for an open wide-ranging debate on the future of agriculture and food security for all."

Perhaps I should explain what the 'precautionary principle' is. There are many versions of it, presented in a comprehensive and informative book edited by Carol Raffensperger and Joel Tickner, a must read if you want to know why our governments are failing to stop anything, toxic discharges, nuclear wastes or GM crops, from damaging health and the environment. It is because our governments are not acting in accordance with the precautionary principle.

Adopting the precautionary principle will change our whole approach to environmental policies and to regulation. In general terms, this principle calls for protective, preventative actions against harm, even when scientific evidence is uncertain. More importantly, it shifts the burden of proof of safety to the perpetrators, instead of demanding regulators and civil society to provide scientific proof of harm.

The World Trade Organisation (WTO) operated against the precautionary principle when it judged Europe's ban on US growth hormone injected beef illegal. That is how the WTO undermines every single effort by citizens and governments all over the world to protect health and the environment.

In Europe, Sweden has the best environmental law based on the strongest version of the precautionary principle. In contrast, the UK, with a long tradition of "scientific corporatism and elitism" prefers to adopt the "long pipes and tall chimneys" approach to make optimal use of the "waste assimilative capacity of the environment".

When pressed to adopt the precautionary principle, the UK government came up with what must be the weakest version:

"Where there are significant risks of damage to the environment, [we] will be prepared to take precautionary action to limit the use of potentially dangerous materials or the spread of potentially dangerous pollutants, even where scientific knowledge is not conclusive, *if the balance of likely costs and benefits justifies it*. The precautionary principle applies particularly where there are *good grounds* for judging either that action taken *promptly at comparatively low cost* may avoid more costly damage later, or that irreversible effects may follow if action is delayed." (italics mine)

This is a frank admission that scientific evidence must bow to the profit motive, which is why there has been such great effort devoted to hyping the still potential benefits of GM. It will be useful to state the principle in another form:

"The precautionary principle states that where there is *reasonable suspicion* of serious irreversible harm, scientific uncertainty or lack of scientific consensus must not be used to postpone preventative action."

Let us stay with the law court analogy. The precautionary principle is about setting the burden of proof.

In the case of the murder suspect, the consequence of a wrongful conviction is a ruined life. That is considered so inhumane that society has agreed to set the burden of proof so as to bias against wrongful

conviction. A person is innocent until proven guilty *beyond reasonable doubt*.

Similarly, as pointed out by Peter Saunders, protecting the environment is so important that the burden of proof is set to prevent damage. But it doesn't mean we stop doing anything new, as critics of the precautionary principle have charged. We still need to look at the scientific evidence, but we base our judgement on a *reasonable suspicion* of serious irreversible harm, not on absolute proof.

In the same way, to require the perpetrators to demonstrate safety, does not mean we are demanding that something should be "100% safe" before it can be accepted, as critics have charged, only that it should be safe *beyond reasonable doubt*.

It is clear that the precautionary principle is not separate from science. It is part and parcel of good science, which requires us to interpret scientific evidence appropriately, to allow for uncertainty. That's because uncertainty is the hallmark of any *active knowledge* system, which is what science is, as opposed to religious fundamentalism.

The valid use of scientific evidence is to set precaution, and not to set permissive standards for scientists and corporations to continue to treat life and our life-support system as one vast laboratory.

There was, in 1999, already sufficient evidence to support *reasonable suspicion* that genetic modification is inherently harmful, for preventative action to be taken.

My paper was widely circulated through the Third World Network as well as on the ISIS e-mail list, and presented at a very important conference on 'biodiversity law' organised by the amazingly radical judiciary of Brazil, which took its independence most seriously. The paper was published in the conference proceedings later that year, and has probably played a role in maintaining Brazil's rejection of GM crops against tremendous pressures from Monsanto. Monsanto formed an unholy alliance with the previous federal government to appeal repeatedly against a ban on commercial growing of GM crops that had been imposed as the result of legal action brought by a coalition of civil society organisations and Brazil's then minister for the environment.

Today, the weight of evidence is sufficient, *beyond reasonable doubt*, I believe, to put a halt to transgenic agriculture altogether; and Michael Meacher has all but said so in public, in an exclusive interview with *The Ecologist* magazine, which has been widely reported in the press.

Meacher said GM technology is not necessary to solve world hunger and could prove dangerous over the longer term. He expressed dissatisfaction with the fact that “we have to rely on the biotech companies themselves to tell us if they discover any other problems such as health risks”.

He also expressed concern over Lord Sainsbury's role as the government's science minister on the cabinet committee in charge of overseeing the GM trials. “As far as I know, the only way he seeks to avoid this conflict of interest is by absenting himself when decisions are taken,” said Meacher.

Michael Meacher has put his job on the line by so openly declaring his position, but he is more than justified in view of all the evidence that has come to light since I took part in the debate in his office in 1999.

I am going to concentrate on the evidence of horizontal transfer of transgenic DNA in the rest of this chapter, and I invite you to consider the evidence carefully, as though in a court of law.

Evidence on horizontal transfer of transgenic DNA

In 1999, researchers in Germany had already reported the first, and still only, field-monitoring experiment in the world, that provided *prima facie* evidence that transgenic DNA had transferred from the GM sugar beet plant debris to bacteria in the soil (see Box 9.2). I circulated a detailed review of this evidence, and duly submitted it to our government's science advisors. They dismissed that evidence too, and worse, cited it as evidence that horizontal gene transfer did not occur.

Box 9.2

Prima facie evidence of horizontal gene transfer in the field

Researchers Frank Gebhard and Kornelia Smalla at the Institute of Plant Virology, Microbiology and Biosafety in Braunschweig, Germany, began a series of experiments in 1993 to monitor field releases of transgenic rizomania-resistant sugar beet (*Beta vulgaris*) containing the marker gene for kanamycin resistance. They looked at the persistence of transgenic DNA, and its transfer into soil bacteria. It was the first, *and still the only*, such experiment to be carried out after tens of thousands of field releases and tens of millions of hectares have been planted with transgenic crops. It is therefore

important to examine the results in some detail.

Transgenic DNA was found to persist in the soil for at least 2 years after the transgenic crop was planted, confirming previous findings that DNA is not readily broken down in the environment. Although the researchers did not comment on it, the data showed that the proportion of kanamycin-resistant bacteria in the soil increased significantly between 1.5 and 2 years. Could this be due to horizontal transfer of the antibiotic resistance marker gene in the transgenic DNA liberated from the transgenic plant debris as the debris degraded? This question was not raised.

A total of 4 000 kanamycin resistant bacterial colonies isolated were individually screened with a 'dot blot' technique to identify sequences that bind to, or 'hybridize with' the transgenic DNA-specific probes. This technique is more direct, but much less sensitive than polymerase chain reaction (PCR), which amplifies specific sequences aiding their detection. The researchers reported that, "A few isolates giving weak hybridization signalswere detected". These were checked with the PCR technique, but none gave PCR products, and hence the results were dismissed as 'false positives'.

There were obvious limitations to this experiment. First, 4 000 is a small number of isolates, and most of them were probably from bacteria already carrying pre-existing kanamycin resistance. Second, the failure to obtain PCR products does not mean transgenic DNA is absent. The PCR method has its own limitations. It will not detect fragments of the transgenic DNA not covered by the stretch of the DNA being amplified, nor will it give a positive result if the stretch of DNA being amplified happens to be rearranged. In order to rule out those possibilities, it would have been necessary to do more extensive molecular analyses.

Two of seven samples of total bacterial DNA isolated directly from the soil yielded positive results for transgenic DNA at 18 months after the GM crop was harvested. This suggested that horizontal gene transfer had taken place, although the specific bacteria that had taken up the transgenic DNA could not be isolated as colonies. That was not surprising, as every microbiologist knows, less than 1% of all the bacteria in the soil can be cultured and identified, but the researchers themselves had failed to point that out. They also failed to connect the positive identification of transgenic DNA in soil bacteria at 18 months with the simultaneous appearance of excess kanamycin resistant

bacteria in the soil that I have noted above. The researchers were careful, however, not to rule out the possibility that transgenic DNA might have been adsorbed to the surface of bacteria, rather than having been transferred into the bacteria.

The researchers also carried out microcosm experiments to which total transgenic sugar beet DNA was added to non-sterile soil with its natural complement of microorganisms. The intensity of the signal for transgenic DNA decreased during the first days and then *increased*. The most obvious interpretation of the result is that the transgenic DNA was taken up by bacteria and became amplified as it replicated along with the bacteria. The researchers failed to make this interpretation, or to raise this possibility.

In parallel, the bacteria from soil samples were plated and the total bacterial lawn was allowed to grow for 4 days, and then DNA was extracted. Several positive signals were found, which, according to the researchers, "might indicate uptake of transgenic DNA by competent bacteria."

The authors were cautious not to claim conclusive results simply because the specific bacteria carrying the transgenic DNA sequences were not isolated, and possibly also because they were afraid of being seen to be suggesting too explicitly that horizontal gene transfer had occurred. Their results do, however, provide *prima facie* evidence of horizontal gene transfer. At the very least, follow-up research should have been undertaken, but it was not.

Gebhard and Smalla's honest, but perhaps overly 'cautious' apologies over the limitations of the methods they used to detect horizontal gene transfer unfortunately gave licence to pro-GM scientists, including those advising the UK and other governments, to dismiss their results, or worse, to cite it as 'evidence' that horizontal gene transfer did not occur under field monitoring, which is a complete travesty.

DNA not only persists in the external environment, both in the soil and in water; it is not broken down sufficiently quickly in the digestive system to prevent transgenic DNA transferring to micro-organisms resident in the gut of animals.

Such transfer could start in the mouth. Researchers in the University of Aberdeen, led by Derry Mercer in Harry Flint's group,

reported in 1999 that a genetically engineered plasmid had a 6 to 25% chance of surviving intact after 60 minutes of exposure to human saliva. Moreover, the partially degraded plasmid DNA was capable of transforming *Streptococcus gordonii*, one of the bacteria that normally live in the human mouth and pharynx. The frequency of transformation dropped exponentially with time, but it was still significant after 10 minutes. Human saliva actually contains factors that promote transformation in bacteria resident in the mouth.

This research was done in the test-tube, and the authors clearly stated that, “further investigations are needed to establish whether transformation of oral bacteria can occur at significant frequencies *in vivo*.” However, no such studies have been carried out since, which is difficult to understand, *as the original research was commissioned by the UK government, as part of the Novel Foods Programme*.

Another group in Leeds University, however, got a grant from the then newly established Food Standards Agency to investigate the possibility of horizontal gene transfer in the stomachs of ruminants, where food remains for long periods of time (see Box 9.3).

The researchers found that transgenic DNA was rapidly broken down in the fluids from the rumen and the silage, but that nevertheless, horizontal transfer could take place before the transgenic DNA was completely degraded.

They also found that transgenic DNA was very slow to break down in saliva, and therefore, the mouth could be a major site for horizontal gene transfer. This confirmed the results obtained by the research group in the University of Aberdeen. But once again, no follow-up work was done in live animals. Was it a case of avoiding doing the obvious experiments for fear of finding positive results that would be more difficult to dismiss?

Box 9.3

Gene transfer in ruminants?

An insect-resistant maize line contained three bacterial genes: the *cry1A(b)* specifically toxic to lepidopterans, the *bar* gene conferring tolerance to glufosinate, and a *bla* gene encoding TEM-1 β -lactamase (ampicillin resistance). The *bla* gene originated from the cloning vector pUC18, derived from a bacterial plasmid, and is not expressed in maize, but has bacterial regulatory sequences that would allow it to

become functional were it to be transferred back into bacteria. There are at least two copies of *cry1A(b)* and *bla* genes integrated into the DNA of the maize line.

A research group in Leeds University, headed by John Heritage, who sits on the ACNFP, was commissioned by the FSA to investigate the survival of DNA from the transgenic maize and the transfer of the antibiotic resistance *bla* gene to bacteria in the presence of saliva, rumen fluid and silage effluent, which are relevant to horizontal gene transfer in the oral cavity, the rumen, and in silage, *but the scope of the research project did not include studies in live animals.*

E. coli was the test micro-organism for horizontal gene transfer, although it would have been more informative to use bacteria that are normally resident in the mouth, rumen and silage respectively. Degradation of DNA was followed by gel electrophoresis as well as by polymerase chain reaction (PCR). Both pUC18 plasmid and transgenic maize DNA were used in the experiments.

On gel electrophoresis, plasmid DNA and maize DNA appeared to be degraded rapidly by rumen fluid or silage effluent within one minute, but both were incompletely degraded after one hour in saliva.

On PCR, a much more sensitive analysis, large fragments of the *bla* gene (>350bp, bp stands for base-pairs, the usual units for measuring the size of DNA) were still found in rumen fluid after 30 minutes for the plasmid and one minute for maize DNA. Even larger fragments (>350 and >684 bp) from plasmid and maize DNA were found after 30 minutes incubation in silage effluent, and after 24 hours and 2 hours respectively in saliva.

PCR analysis also showed that fragments of the *cry1A(b)* (>1 914bp) in maize DNA could be found after one minute in rumen fluid, 5 minutes in silage effluent, and 60 minutes in saliva.

Plasmid DNA exposed to saliva for 24 hours was still capable of transforming *E. coli* to ampicillin resistance, but at low efficiency: 20 cfu (colony forming units) per millilitre in saliva compared with 1 600 cfu per millilitre after 24 hours in sterile water. Previous exposure to rumen fluid for just 30 seconds reduced transformation by 5-fold. No transformed bacteria were obtained after the plasmid DNA was exposed to silage effluent or rumen fluid for longer than one minute.

However, when *E. coli* bacteria and plasmid were simultaneously added to silage effluent or rumen fluid, 4 750 cfu per millilitre trans-

formants were recovered after 4.5 hours in rumen fluid and 11 cfu per ml were recovered after 3 hours in silage effluent.

Thus, the experiments suggest that horizontal gene transfer can occur before the DNA is completely broken down, even when the breakdown is relatively rapid, as in the rumen or in silage. DNA breakdown is extremely slow in saliva, and hence the oral cavity may be a very important site for horizontal gene transfer.

As the experiments were only done under *in vitro* conditions, the researchers stated that "further studies are needed" to assess the significance of their results. Three years later, and no follow up studies on live animals have been done. Is it a case of avoiding doing the obvious experiments for fear of actually finding a positive result?

There's more to the scope of horizontal gene transfer as revealed in the existing scientific literature. A research group led by Walter Dörfler in the Department of Genetics, University of Cologne in Germany, has carried out a series of experiments on the fate of foreign DNA in food, beginning in the early 1990s.

They fed mice DNA, either isolated from the bacteria virus M13, or as the cloned gene for the green fluorescent protein inserted into a plasmid. They found that a small, albeit significant percentage of the viral and plasmid DNA not only escaped complete degradation in the gut, but could pass through the wall of the intestine into the blood stream, to get into some white blood cells, spleen and liver cells, and become incorporated into the mouse cell genome. When fed to pregnant mice, the foreign DNA could be found in some cells of the foetuses and the newborn animals, indicating that it had gone through the placenta.

This work underlines the hazards of all kinds of naked DNA, including viral genomes, created by the genetic engineering industry, that Terje Traavik and others have drawn attention to.

In a paper published in 1998, Dörfler and Schubbert stated, "The consequences of foreign DNA uptake for mutagenesis [generating mutations] and oncogenesis [causing cancer] have not yet been investigated". The relevance of this remark is striking with regard to the cancer cases identified among the recipients of gene therapy (see previous chapter). It makes the point that exposures to transgenic DNA carry the same risks, regardless of whether it is from gene therapy or from GM

foods.

In a report published in 2001, the team compared the fate of ordinary soybean DNA from soybean leaves with that of transgenic plasmid DNA. They confirmed their earlier finding. Transgenic plasmid DNA invaded the cells of many tissues (see Box 9.4).

But like most of the research projects that I have reviewed in this chapter, they too, seemed to have stopped short of attempting to obtain clearer, definitive results, which they could have done by feeding mice transgenic soya, and monitoring for the fate of both the transgenic DNA and the plant's own DNA. That would have gone some way to settle the issue I have repeatedly raised in this book: there are many reasons to believe transgenic DNA may be more invasive of cells and genomes than natural DNA (see Chapter 8, Box 8.1 and Box 8.4).

Box 9.4

Fate of transgenic DNA in food, another wasted opportunity?

Researchers in the Department of Genetics, University of Cologne, in Köln, Germany, have been investigating the fate of ingested DNA since the early 1990s. In their latest experiments Hohlweg and Döerfler compared the fate of ordinary soybean leaf DNA with that of transgenic plasmid DNA containing green fluorescent protein (GFP) linked to viral promoters from the human cytomegalovirus (HCMV), the Simian virus (SV40) or Rous Sarcoma virus (RSV).

The ordinary soybean DNA was tracked by probing for the presence of the plant-specific, ribulose-1,5-biphosphate carboxylase (Rubisco) gene.

They found that the plant gene, or fragments of it, could be recovered in the small intestine of mice from 2 hours and up to 49 hours after feeding. In the caecum (first part of the large intestine), it could be detected for up to 121 hours after ingestion. Plant-associated DNA was much more stable in the intestinal tract than naked DNA. The researchers also looked for the DNA in different tissues of the mice fed soybean leaves. Out of a total of 37 mice, Rubisco DNA was found in three organs in two animals.

When the plasmid with transgenic GFP-DNA were fed to mice, however, a much higher proportion of the DNA was found in many tissues: liver, spleen, kidney, blood, and the intestinal wall, starting from just 3 to 8 hours after feeding. The numbers of mice in different feed-

ing experiments that showed transgenic DNA in their tissues varied from a low of 3 out of 16, to 5 out of 8; the total summed over 8 experiments was 28 out of 87, or nearly one-third.

The researchers noted that far greater amounts of transgenic DNA were fed, typically 50 micrograms, equivalent to 10^{13} copies compared with about 6×10^8 copies of the Rubisco gene, which could have accounted for the greater numbers of mice with transgenic DNA in their tissues. But, according to their findings, plant-associated DNA was much more stable and resistant to breakdown, remaining in the gut far longer, which might have increased the chance of the DNA being passed through the gut wall and into the bloodstream and other tissues.

When injected, into muscle, the green fluorescent protein DNA fragments could be amplified by PCR for up to 17 months afterwards. The GFP DNA was also identified in organs remote from the site of injection up to 24 hours after injection, and from intestinal contents 6 hours after injection. This indicates that transgenic DNA *can* circulate around the body, regardless of its site of application.

It is difficult to understand why these seasoned researchers didn't do the obvious experiment that would have avoided the ambiguity of the present results: feed transgenic plant material to the mice, and probe for both the plant gene and transgenic DNA.

In fact, this obvious experiment, which would have settled the crucial question of whether transgenic DNA is more prone to horizontal gene transfer than ordinary DNA, appears to have been avoided altogether so far.

The UK government eventually commissioned research to look for horizontal gene transfer into bacteria in the gut of human volunteers *and found positive results*. But once again, the findings were dismissed and downplayed, all the more scandalous as the experiment was already designed to *bias against* detecting horizontal gene transfer in the first place. More in the next chapter on the bad science that's routinely accepted and perpetrated, it seems, by our regulatory bodies and vested interests.

Ten

Sense & Nonsense in Horizontal Gene Transfer

Transfer of transgenic DNA in human gut

Transgenic DNA transferring to bacteria in the human gut is not at all unexpected. We already know that DNA persists in the gut, and that bacteria can readily take up foreign DNA, as reviewed in the previous Chapter. Why had our regulators waited so long to commission the research? *And when they did, the experiment appeared to have been designed to stack the odds heavily against finding a positive result* (see Box 10.1). *Despite that, they still found a positive result; which the regulators immediately dismissed and obfuscated.*

The research in question is the final part of the UK Food Standards Agency (FSA) project on evaluating the risks of GMOs in human foods.

Box 10.1

Stacking the odds against finding horizontal gene transfer

The research was done by a team in Newcastle University led by Netherwood, and the report was released in July 2002 on the FSA website.

A *single* meal containing GM soya was fed to human subjects. It consisted of 150 grams of commercial soya meal (El Corte Ingles, batch number and GM content unspecified) mixed in the burgers, and 100 grams of soya protein supplement (Holland and Barrett, batch number and GM content again unspecified) mixed in 'milk' shakes. No data were presented on the amount of DNA present in the commercial samples, nor on the state of the DNA, whether it was degraded and to what degree. Needless to say, the transgenic DNA inserts in the GM soya were not characterised at all.

The method of detecting transgenic DNA was highly flawed. It depended on amplifying a small part - 180bp - of the entire transgenic DNA insert that was at least ten or twenty times as long. So, any other fragment of the insert would not be detected, nor would a fragment that did not overlap the whole 180bp amplified, or that had been rearranged. The chance of obtaining a positive result is 5% at best, and likely to be much, much less. *Thus, a negative finding with this detection method most probably would not indicate the absence of transgenic DNA.*

More revealing still, the researchers checked for transgenic DNA only in the gut contents, but failed to check if the DNA had passed through the gut into the bloodstream and blood cells. This omission is inexcusable, as a series of experiments in mice dating back to the mid 1990s had already documented that viral and plasmid DNA could pass through the gut and placenta into the bloodstream, to be taken up by some cells in the blood, liver and spleen and cells of the foetus and the newborn (see previous chapter).

In the first trial, the GM meal was fed to seven people who have had an operation that removed part of their lower bowel, and were wearing a colostomy bag. The digested food from the colostomy bag was analysed, and transgenic DNA was detected in all seven subjects. As much as 3.7% of the transgenic DNA was recovered in one subject. *Bearing in mind the limitation of the method used for detecting transgenic DNA, the values are most probably gross underestimates. More worrying is the distinct possibility that much of the DNA may have disappeared into the bloodstream undetected.*

In the second trial, the meal was fed to 12 healthy volunteers with intact bowels. No transgenic DNA was detected in the faeces, which the researchers claimed indicated that the nucleic acid did not survive passage through the complete intestine. *But this claim, too, is most likely to be false, due to the limitation of the transgenic DNA detection method, and the possibility that the DNA may have passed out of the gut and into the bloodstream.*

Microbes in the digested food that had passed through the small intestine were cultured through 6 passages in broth containing glyphosate herbicide. Bacteria grew to a density of 10^8 /ml in each passage. In samples taken from 3 subjects at 180, 240 and 300 minutes after eating the meal, the transgene was found in all the sub-cul-

tures. *This is a further gross underestimate. The method depends on the bacteria having taken up an intact gene coding for glyphosate tolerance, and would not have detected bacteria that have taken up fragments of that gene, or other parts of the transgenic DNA containing other genes or gene fragments.*

One subject was found to have glyphosate-tolerant bacteria before the GM meal, though at very low concentrations, possibly from having consumed GM soya independently prior to the trial. The bacteria harbouring the transgene could not be isolated from the subject, so the researchers concluded that, "although present, the bacterium represented a very small proportion of the indigenous intestinal microflora". *This is false reassurance, as only a tiny fraction of all the bacteria resident in the intestine can be cultured by present methods. And as bacteria are capable of multiplying, even rare gene transfer events cannot be ignored.*

The researchers were disingenuous when they expressed surprise at the relatively large proportion of transgenic soya DNA that had survived passage through the small bowel. But in earlier research already published, the same group had found that DNA in food or mixed up with food was much slower to degrade than naked DNA.

Despite the severe limitations placed on detecting transgenic DNA, and an experimental design biased towards *negative* results, irrefutable positive evidence was nevertheless obtained. It means that the transfer of transgenic DNA in the human gut could be much more extensive than the data indicate, which makes it all the more astonishing for the FSA to claim that "the findings had been assessed by several Government experts who had ruled that humans were not at risk".

In a statement on its website, the FSA said that the study had concluded it is "extremely unlikely" that GM genes can end up in the gut of people who eat them. I have shown why this statement is highly misleading and very likely to be false.

Our government's scientific advisers have been guilty of persistent denial in the face of mounting evidence that horizontal gene transfer can happen and has happened. They are guilty of commissioning and accepting bad scientific research that misleads the public, of downplaying positive evidence, and of taking the absence of evidence

as evidence of absence.

I first pointed out the dangers of horizontal gene transfer to MAFF in a series of correspondence beginning in 1996. Their scientific advisers said there was no evidence it could happen. When it became clear that it could readily happen in the laboratory, the scientific advisors said, "Just because it happens in the laboratory does not mean it will happen in the field".

When positive findings turned up in the only field monitoring experiment in the world that has ever been performed, the science advisors swept that under the carpet. ACRE's latest report gave the following misleading statement: "The authors of these publications did not demonstrate, nor did they claim to provide evidence of horizontal gene transfer under laboratory conditions or in field studies." This is almost a complete travesty.

As I have clearly shown (Chapter 9, Box 9.2), the results offered *prima facie* evidence of horizontal gene transfer. The authors themselves had indeed considered that possibility, first and foremost. ACRE is guilty of misrepresentation, and failing to interpret the experimental results correctly by *excluding obvious* explanations. At the very least, ACRE should have called for further investigations

This latest finding of transgenic DNA transferred to human gut bacteria is the last piece of damning evidence that horizontal transfer of transgenic DNA can indeed happen, has already been happening, and cannot be controlled if GM crops continue to be released to the environment.

I sent my review of this research report to the FSA, and asked them to reply to my scientific criticisms. The FSA evaded my questions. I submitted my critique to an open meeting organised by the ACNFP, and still got no response. I tabled a written paper on my criticisms at the open meeting on 13 November 2002. I was granted three minutes to present my case, and still, the panel of experts evaded my questions.

Finally, ACNFP and ACRE issued their joint report just before Christmas, repeating their mantra that "there is no evidence of harm".

In the previous chapter, I have given the reasons why transgenic DNA, as opposed to natural DNA, is optimised for horizontal gene transfer. I have also pointed to the risks of horizontal transfer of transgenic DNA, in creating new viruses and bacteria that cause diseases, in spreading drug and antibiotic resistance among the pathogens, and in triggering cancer by jumping into genomes of mammalian cells.

I also mentioned new pharm crops being developed, containing drugs and vaccines that could poison our water and soil, affecting all organisms in our food web. The ecological impacts are unthinkable. What more do we need for an immediate comprehensive ban on GM crops?

And why are other scientists not speaking up loud and clear; especially those knowledgeable about horizontal gene transfer and who have no vested commercial interests?

In the entire episode surrounding our published papers on the CaMV 35S promoter, we received positive comments from just two scientists, compared to more than a dozen others who wrote hostile critiques, or worse, circulated irrelevant and abusive personal attacks. One of the scientists who supported us is Veljko Veljkovic, virologist in Belgrade, Yugoslavia, who has since made public his own concerns about the CaMV 35S promoter and recombination sequences present in the *gp120* gene from the AIDS virus, HIV. The other is a senior virologist, who expressed support for our “general aim” and “enjoyed” reading our reply to the comments of Hull, Covey and Dale, which he thought, were “mostly trivial, and occasionally irrelevant”.

But those were exceptional individuals within the scientific community. The problem for most scientists is that they are either stuck within the old paradigm, or do not have the courage of their conviction to challenge the orthodoxy. For they risk being ostracised, and worse.

Averting sense for nonsense

What do most scientists do when faced with findings that threaten to topple the ruling paradigm? They describe the findings at great lengths in technical language that not even scientists in other disciplines can comprehend; they fail to interpret the findings altogether or interpret them incorrectly, avoid discussing the practical implications, and above all, dismiss incriminating evidence suggesting that what their colleagues are doing could be dangerous. At the same time, they try desperately to paper over the cracks of the crumbling edifice of the old paradigm, and engage in rampant speculations.

Horizontal gene transfer goes counter not only to modern genetics, but also to the dominant theory of evolution. In the case of evolution, both the general theory of evolution - that organisms descended with modification from earlier common ancestors - and the special neo-Darwinian theory - that organisms evolve by the natural selection of

random genetic mutations - are under threat.

While Charles Darwin's *Origin of Species* took Victorian Britain by storm, Gregor Mendel's work, published the same year, languished in obscurity until it was rediscovered to inaugurate the search for the material basis of heredity. As everyone knows, that search culminated in the triumphant discovery of the DNA double-helix.

The significance of DNA for neo-Darwinism is encapsulated in Francis Crick's 'Central Dogma' of molecular biology: that 'genetic information' flows in one direction only from DNA to RNA to protein, and never in reverse. The Central Dogma lent support to the most hard line genetic determinist assumptions of neo-Darwinism (see Chapter 2, Box 2.1).

The first assumption, that the organism's characteristics are more or less hardwired in the genes, had been severely criticised ever since modern genetics began. Many biologists, including especially Richard Goldschmidt and later, Conrad Waddington, pointed to the large influence that the environment has on development, emphasizing that the effects of genes and the environment cannot be disentangled.

But perhaps nothing rivals the discoveries made within the past twenty five years, on the fluidity of genes and genomes in both structure and function, which really disposes of genetic determinism (see Chapters 5 to 7).

The end of genetic determinism also spells the end of the neo-Darwinian theory of evolution by the natural selection of random variations. The latter is based on a two-fold separation: between the genes passed on to the next generation and the environment, so that acquired characters cannot be inherited, and between variation and selection, such that the 'selective pressure' or 'selective force' cannot directly influence the process of variation, hence the dogmatic insistence on *random* variation.

You have come across many examples (Chapters 5 to 7) of variations in the genome and gene mutations that are *not* random, and more will be presented later on (Chapter 11). Consequently, it is difficult to separate the process of selection from that of variation, as envisaged by the doyen of neo-Darwinism, Ernst Mayr (see Chapter 2). The organism and its genes and genome are ever in flux, always changing and responding to the environment. The static and constant genes and genomes are a mechanistic fiction, rather like the unmoved mover that was supposed to set the universe going at the beginning of time.

Horizontal gene transfer - a key process of the fluid genome - makes neo-Darwinism untenable in perhaps the most explicit fashion. It tells us not only that genes and genomes are unstable, but also that the 'characteristics' encoded or influenced by the genes belonging to one species can be acquired by another species as the genes themselves are transferred, *to become inherited in the second species*. This momentous significance was not even hinted at, in a recently published volume entitled *Horizontal Gene Transfer*, replete with detailed examples of horizontal gene transfer between species within and across every living kingdom on earth.

One chapter - the shortest in the volume - promised to spell out the relevance of horizontal gene transfer to the release of genetically modified organisms. Michael Syvanen from the Department of Medical Microbiology and Immunology, University of California Davis, one of the two co-editors, recalls how, when GM crops were first approved, the US Food and Drug Administration responded to fears of the spread of antibiotic resistance genes by denying this could take place. He himself, however, was just then trying to convince his colleagues that horizontal gene transfer between life's kingdoms was a natural process.

"If these speculations were true, it would make more sense *to defend the transgenic crop industry* (italics mine) by arguing that gene transfer is a natural phenomena [sic] than by arguing that it does not occur." He continued in a paragraph full of mistakes, though the gist of what he was trying to say was unmistakable,

"As is well-documented throughout this book, we can infer numerous cases cross-species gene transfers from the whole genome sequencing studies. However, this does not necessarily mean that the frequency of transfer is high enough for gene transfer on the farm may to occur [sic]."

Actually, an important chapter in the same volume written by Russell Doolittle of the Center for Molecular Genetics, University of California San Diego, warned against over-estimating horizontal gene transfer in our evolutionary past.

When the human genome was first announced, the sequencers claimed that more than 200 genes in the human genome were transferred from bacteria. Doolittle dismissed every one of them in his chapter.

Sure enough, a paper published in *Nature* in 2001 by another research team announced in its title, "Phylogenetic analyses do not

support horizontal gene transfer from bacteria to vertebrates". Those 'bacterial' genes were found in the genomes of 'lowly' animals, such as the nematode and slime mould. So those genes could not have been transferred from bacteria into the human genome, but were inherited from very distant invertebrate (animals without backbones) ancestors of our species, before vertebrates (animals with backbones) were evolved. Another problem was simply the contamination of human DNA with bacterial DNA during sequencing procedures.

Doolittle pointed out that horizontal gene transfer is certainly not so very frequent in our evolutionary past that it changes the general theory of evolution, i.e., descent with modification, with each species keeping genetically distinct, and only very occasionally exchanging genes horizontally.

But genetic engineering is something else. It sets out to overcome the genetic barriers between species and to *enhance* horizontal gene transfer, which is why we suspect that horizontal gene transfer may have increased since commercial genetic engineering began (see Chapter 8). One main concern is the transfer of antibiotic resistance marker genes to bacteria that cause diseases.

Syvanen claimed to have screened for antibiotic resistance marker genes in the faeces of mice fed transgenic plant material, and found none. But those results were unpublished and non-peer reviewed, and presumably, should be dismissed in the same way that he disposed of the study reported by Hans Katz, a professor in the University of Jena, Germany, in 2000.

Katz carried out a piece of rather clever research. He set up a contraption in the beehive to collect transgenic pollen from incoming bees, and fed the pollen grains to the bee larvae in a controlled experiment. He found that the antibiotic resistance marker gene in the transgenic pollen had transferred to the yeast and bacteria living in the gut of bee larvae.

In retrospect, this result is just what one would expect to find. But the journal *Nature* rejected the paper, which was submitted elsewhere after it was released to the press. But the paper subsequently disappeared without a trace. Was it another case of suppression of adverse results? Or worse, was it self-censorship for the sake of a quiet life?

Syvanen then cited a report claiming there was no gene transfer to bacteria growing on transgenic vegetables, not mentioning that the report actually documented high frequencies of gene transfer in the lab-

oratory, *which were reduced down theoretically to almost zero by arbitrary assumptions as to what they would be under hypothetical 'natural conditions'* (see Box 9.1, previous chapter).

Syvanen wrote that experiments demonstrating gene transfer with purified DNA from transgenic plants is due to "high frequency recombination", thus creating the misleading impression that we need not worry. He did not explain that such high frequency recombination arises from sequence homology (similarity in DNA base sequence), and as transgenic DNA is highly mosaic, it would have homologies to a *wide variety* of bacteria and viruses, and is hence optimised, in that respect alone, for horizontal gene transfer (see previous chapter).

I have been saying that for years, and am now no longer alone. Kaare Nielsen, a Norwegian geneticist whose specialty is horizontal gene transfer, recently wrote, "Transgenes often contain DNA sequence homology to prokaryotes thereby increasing their likelihood of integration in bacteria significantly. Many studies have shown that DNA homology is the main barrier to HGT [horizontal gene transfer] of chromosomal DNA (such as transgenes) in bacteria."

Syvanen also dismissed findings that transgenic DNA fed to mice ended up transferred to the mice's cells, and failed to cite the key experiment that provided *prima facie* evidence of horizontal gene transfer during field monitoring (Chapter 9, Box 9.2).

All the more striking then, when after dismissing every single piece of evidence on horizontal gene transfer in our time, Syvanen and other contributors to the volume then indulge in rampant speculations on how horizontal gene transfer in our distant evolutionary past could explain away one of the major difficulties with neo-Darwinian theory: parallel or convergent evolution, i.e., similar or parallel changes occurring in lineages of organisms that are not related to one another.

A classic example of convergent evolution is the similarity between our eyes and the eyes of the squid or the octopus. They are similar in construction and action, and are just like the camera we use to capture visual images.



Figure 10.1. Convergent evolution. Our eyes and those of the squid have evolved independently, but they are very similar in construction.

But the human species belongs to the major division of living things called *vertebrates* (animals with backbones), whose ancestor had split off from the *invertebrates* (animals without backbones), to which squids and octopus belong, long before their respective eyes have evolved. So their eyes must have evolved independently. There are many other examples of convergent or parallel evolution, and at every stage of evolution.

The evolution of marsupial mammals in Australia, which carry their immature embryos in pouches, shows astonishing parallels to the evolution of placental mammals in other parts of the world, which carry their embryos in the womb and nourish them by means of a placenta. There are Australian marsupials that look and behave like wolves, cats, squirrels, anteaters, moles and rodents.

The difficulty in trying to explain such convergent or parallel evolution by the natural selection of random genetic mutations is that one must suppose the same set of genes being selected in the correct sequence to produce the very similar morphologies and habits in organisms with different collections of genes, living in widely disparate geographic areas and during entirely separate geological epochs.

"We simply have not identified those genes responsible for the morphologies that characterise the major geological periods," Syvanen admitted. Nonetheless, he was not deterred from presenting "a series

of more speculative pieces that describe the explanatory power of a theory of horizontal gene transfer as it may affect a general theory of evolution.”

So, with one bold stroke of speculation, Syvanen and others could dispose of this major difficulty for neo-Darwinian theory. Why not have the unrelated species exchange the relevant genes by horizontal gene transfer? They suggested that viruses and bacteria could help, and so could insects. Of course, those species would have had to inhabit the same geographic area at the same time for horizontal gene transfer to work.

If those agents had operated in the past, why would they not operate similarly today?

I was among those critics of neo-Darwinism who have followed in the footsteps of the great Scottish scholar biologist, D'Arcy Thompson, evolutionist Richard Goldschmidt in the United States, and later, English mathematician/code breaker Alan Turing, and have argued how morphology can only be understood in terms of 'principles of organisation', or in more modern language, dynamical processes that *generate* patterns and forms.

How else can I not but see those mindless acts of speculation as evidence of the intellectual decline symptomatic of the degenerate research programme that's neo-Darwinian biology?

“The only possible explanation for parallel evolution,” it was stated in one chapter, “is similarity in selective force operating on the genetically isolated populations.” The idea that there might be other more appropriate explanations had apparently never crossed their minds.

Let me describe a much more reasonable and probable scenario based on empirical evidence presented in their volume; and there are several others left as exercises for the many authors themselves.

Gayle Ferguson and Jack Heinemann of the Department of Plant and Microbial Sciences, University of Canterbury, Christchurch, New Zealand, in their chapter “Recent history of trans-kingdom conjugation”, described evidence that should raise serious concerns over the release of GM crops, although the authors themselves failed to do so.

Agrobacterium tumefaciens, the soil bacterium that causes crown gall disease, has been developed as a major gene transfer vector for making transgenic plants. Foreign genes are typically spliced into the *T-DNA* - part of a plasmid called Ti (tumour-inducing) - that ends up

integrated into the genome of the plant cell that subsequently develops into a tumour; that much was known, at least since 1980.

But further investigations revealed that the process whereby *Agrobacterium* injects *T-DNA* into plant cells strongly resembles conjugation, i.e., mating between bacterial cells.

Conjugation, mediated by certain bacterial plasmids, requires a sequence called the origin of transfer (*oriT*) on the DNA that's transferred. All the other functions (called *tra*) can be supplied from unlinked sources, referred to as 'trans-acting functions'. Thus, 'disabled' plasmids, with no trans-acting functions, can nevertheless be transferred by 'helper' plasmids that carry genes coding for the trans-acting functions. And that's the basis of a complicated vector system devised, involving *Agrobacterium T-DNA*, which has been used for creating numerous transgenic plants.

But it soon transpired that the left and right borders of the *T-DNA* are similar to *oriT*, and can be replaced by it. Furthermore, the disarmed *T-DNA*, lacking the trans-acting functions (virulence genes that contribute to disease), can be helped by similar genes belonging to many other pathogenic bacteria. It seems that the trans-kingdom gene transfer of *Agrobacterium* and the conjugative systems of bacteria are both involved in transporting macromolecules, not just DNA but also protein.

That means transgenic plants created by the *T-DNA* vector system have a ready route for horizontal gene escape, via *Agrobacterium*, helped by the ordinary conjugative mechanisms of many other bacteria that cause diseases, which are present in the environment.

In fact, the possibility that *Agrobacterium* can serve as a vehicle for horizontal gene escape was first raised in 1997. McNicole and his colleagues reported in a study sponsored by the UK Government, that it was extremely difficult to get rid of the *Agrobacterium* in the vector system after transformation. Treatment with an armoury of antibiotics and repeated subculture over 13 months failed to get rid of the bacterium. Furthermore, 12.5% of the *Agrobacterium* remaining still contained the binary vector (*T-DNA* and helper plasmid), and *were hence fully capable of transforming other plants*.

Several other observations make gene escape via *Agrobacterium* even more likely. *Agrobacterium* not only transfers genes into plant cells; there is possibility for *retrotransfer* of DNA from the plant cell to *Agrobacterium*, according to a chapter written by co-editor, Clarence Kado, also from the University of California, Davis.

Research published elsewhere showed that high rates of gene transfer are associated with the plant root system and the germinating seed, where conjugation is most likely. There, *Agrobacterium* could multiply and transfer transgenic DNA to other bacteria, as well as to the next crop to be planted. These possibilities have yet to be investigated empirically.

Finally, *Agrobacterium* attaches to and genetically transforms several human cell lines. In stably transformed HeLa cells (a human cell line derived originally from a cancer patient), the integration of *T-DNA* occurred at the right border, exactly as would happen when it is transferred into a plant cell genome. This suggests that *Agrobacterium* transforms human cells by a mechanism similar to that which it uses for transforming plants cells.

When will these scientists and scientists advising our government regulators dare to tell the truth about horizontal gene transfer to themselves and to the public?

Experiments that pro-GM scientists dare not carry out

I have, of course, communicated my critique of that important publication on horizontal gene transfer to ACRE and ACNFP together with a series of obvious experiments that the Food Standards Agency should commission in a paper tabled at an open meeting organised by ACNFP. These are described in a slightly revised form in Box 10.2.

Box 10.2

Missing experiments on the safety of GM food and crops

The following are some definitive experiments that would inform on the safety of GM food and crops. They seem to have been intentionally avoided so far.

1. Feeding experiments similar to those carried out by Dr. Arpad Pusztai's team, using well-characterized transgenic soya and/or maize meal feed, with appropriate, unbiased monitoring for transgenic DNA in the faeces, blood and blood cells, and post-mortem histological examinations that include tracking transfer of transgenic DNA into the genome of cells. As an added control, non-transgenic DNA from the same GM feed sample should also be monitored.
2. Feeding trials on human volunteers using well-characterized transgenic soya and/or maize meal feed, with appropriate, unbiased mon-

monitoring for transgenic DNA and horizontal gene transfer in the mouth and in the faeces, blood and blood cells. As an added control, non-transgenic DNA from the same GM feed sample should also be monitored.

3. Investigation on the stability of transgenic plants in successive generations of growth, especially those containing the CaMV 35S promoter, using appropriate quantitative molecular techniques (see next chapter for evidence of widespread instability of transgenic lines).

4. Full molecular characterisation of all transgenic lines to establish uniformity and genetic stability of the transgenic DNA insert(s), and comparison with the original data supplied by the biotech company to gain approval for field trials or for commercial release.

5. Tests on all transgenic plants created by the *Agrobacterium T-DNA* vector system for the persistence of the bacteria and the vectors. The soil in which the transgenic plants have been grown should be monitored for gene escape to soil bacteria. The potential for horizontal gene transfer to the next crop via the germinating seed and root system should be carefully monitored.

Eleven

Natural vs Artificial Genetic Engineering

Is genomic fluidity a licence for genetic engineering?

How are we to understand artificial genetic engineering in view of the natural fluidity of the genome? There are indeed many pro-GM scientists who now take genomic fluidity as a licence for the arbitrary modification of genomes, in the firm belief that 'natural selection' will cure all in the end.

They have forgotten that such 'selection' is highly unnatural, and could result in killing a lot of innocent organisms including human beings who happen to be poisoned by toxins or allergens, or die of infectious diseases from the deadly viruses and bacteria created, or develop cancer from having transgenic DNA integrated into the genomes of their cells.

Not so long ago, the same pro-GM scientists were denying that the genome is fluid or that horizontal gene transfer could occur.

In this chapter, I will present emerging evidence that the fluidity of the genome due to 'natural genetic engineering' is quite precisely regulated by the organism as a whole, whereas artificial genetic engineering is uncontrollable, unreliable and unpredictable, hence the high failure rates. Of course, the evidence is very incomplete, because almost no one is looking in the right way, or at the relevant processes. Their heads may have been turned by commercial exploitation for far too long.

Natural genetic engineering is non-random

Despite the plethora of natural mechanisms that can alter genes and genomes, the changes are far from random or arbitrary. There are many examples of repeated, predictable changes that occur under given environmental conditions, as for example, in plants treated with different fertilisers, a phenomenon extensively analysed by molecular

biologist Chris Cullis (see Chapter 6). This also applies to the process of directed mutations, where the same mutations consistently turn up in bacteria exposed to a given nutrient that they at first are unable to metabolise. Intriguingly, in rats exposed to the same foreign antigens, repeatable mutations are frequently generated in the immunoglobulin antibodies.

There is a large degree of non-randomness and repeatability in molecular events, which are reminiscent of the extensive parallelism in morphological evolution mentioned in the previous chapter. The molecular events are far more mysterious, however.

Molecular geneticist James Shapiro has remarked, “[Bacteria] may have little tolerance for purely random variability.” Bacteria, like higher organisms, have a wide range of repair and proof-reading enzymes to remove accidental changes to DNA sequences and correct errors resulting from physiological insults. But the same cells also have numerous biochemical mechanisms for changing and re-organising DNA whenever the need arises, suggesting that such changes are the consequences of a kind of “natural genetic engineering”, said Shapiro, which enables the cells and organisms to overcome a range of environmental challenges. Shapiro speculated that, just as signals received by the cell can direct the cell to transcribe specific genes, so the appropriate signals might also direct specific genes to mutate, possibly by transposons jumping across the genome to mutate the right genes.

Shapiro has offered a new perspective to help make sense of the way genomes are organised and how that relates to the fluidity of genomes and gene function. One important feature of genomes, most obvious in bacteria, is that they are organised hierarchically into functional units.

Genes coding for proteins are composed of subunits or 'domains' that are shuffled to make different proteins that carry out specific functions, even new functions that have never existed. Several genes are collected into regulatory units called operons (see Chapter 5, Figure 5.2). Several operons are grouped to form a higher unit, the *regulon*, that's made up of all the operons required for complex functions. For example, chemotaxis (movement towards certain chemical substances, such as food), motility and flagellar biosynthesis, three complex functions that have to be coordinated, are encoded by 15 operons in five regions of the *E. coli* genome. In higher organisms, the complexity of co-regulated suites of genes can be far greater, especially for

the cell cycle, cellular differentiation and development.

In many organisms, reorganisation of the genome is part of the normal life cycle, as when the egg cell matures in vertebrates, or in plants that have juvenile and adult phases. Many cells within the body also undergo reorganisation and rearrangements of the genome, as in the cells that make immunoglobulin antibodies. Without that natural genetic engineering in our white blood cells, we would perish from severe immune deficiency.

One of the keys to integrating the complex of functions involving the expression of many genes is by means of the repeated sequences scattered throughout the genome, that through binding common transcription factors, can result in the *coordinated* expression of many genes that don't have to be linked.

This ingenious scheme was predicted by Britten and Davidson in 1969, and explained in an Open University text - probably the first and only - that has dealt with the fluid genome. That excellent, wide-ranging Genetics Course sat uneasily within the Open University Biology curriculum, and my colleagues in the department made sure it was discontinued to make room for more conventional courses as soon as possible.

Britten and Davidson's scheme has been amply confirmed in recent years by the shared motifs found between the regulatory elements of genes that are expressed in concert. This enables cells to express a different repertoire of genes whenever and wherever appropriate (see Chapter 5).

That may be one of the reasons why genomes are full of repetitive sequences. Repetitive sequences serve a range of functions, as Shapiro pointed out (see Box 11.1).

Box 11.1

Some functions of repetitive DNA sequences

1. Coordinated expression of unlinked genes
2. DNA replication origins
3. Chromosome ends, or telomeres, for stability
4. Distribution of chromosomes during cell division
5. Pairing of chromosomes in cell division
6. Chromosome organisation and timing of gene expression

As can be seen, repetitive DNA sequences play key roles in both the 'architecture' and the functional organisation of the genome. They are responsible for maintaining the stability of the genome as a whole, its replication and its correct separation after replication to the daughter cells. Repetitive sequences may also determine the timing at which different combinations of genes are expressed.

But this organisation is dynamic, and can be changed as the need arises.

Indeed, many repeated sequences are transposons that generally have regulatory elements for gene expression at their ends, and transposons may be the natural molecular genetic engineers of the genome.

Jumping genes as molecular genetic engineers

Could regulated movement of transposons be involved in linking the expression of previously disparate genes? That's what Shapiro has suggested.

"The modular organisation of genomes as hierarchical systems requires a capacity for cut-and-splice changes (i.e., natural genetic engineering) that transposable elements can provide to cells."

All genomes contain significant numbers of transposons. Some species of bacteria have as much as 10% of the genome consisting of transposon-like insertion sequence (IS) elements. Mammalian genomes contain large amounts of repetitive DNA and retrotransposons. About 50% of the human genome consists of proviral sequences (viral sequences integrated into the genome) and transposable elements, many with reverse transcriptase. Plant genomes too, are rich in such sequences. Over 50% of the maize genome is composed of retroviral-like transposons, and in some species, the fraction can be as high as 95%.

Interestingly, each mammalian order (a unit in classification) has its own set of short interspersed nuclear elements (SINEs), which are transposon-like.

Some transposons have well-defined functions. One example is in fruitflies, where retrotransposons are involved in regenerating the *telomeres*, repeated sequences at the ends of chromosomes, which are important for keeping chromosomes intact (see Box 11.1).

From analysing genomes, a growing number of cases are docu-

mented in vertebrates in which regulatory elements for genes can be traced to vestiges of inserted transposons. All transposons carry transcriptional regulatory elements, such as promoters and enhancers, which boost the expression of genes.

Thus, transposons can create new genome architectures, new functions and complex of functions. The dispersal of multiple copies of a single regulatory element to many locations in the genome can affect the regulation of individual genes, including the activation of silent genes. In addition, transposons can move part of one gene and fuse it to another, to create a new gene. Transposons can rearrange segments of chromosomes, or move them wholesale to another chromosome. They can duplicate small and large chunks of chromosome. They are to all intent and purposes natural genetic engineers at the molecular level, compared to which, the human genetic engineers are unspeakably crude, awkward and ineffectual.

But what directs transposons to the right locations in the genome to activate the right genes? That's almost a complete mystery.

The movements (transposition) of many transposons are regulated, that much is known. In bacteria, molecular genetic analysis has revealed sites in transposons that interact with proteins that regulate the cell cycle, and with transcription factors and other factors that influence gene expression and transposition.

In the first example of adaptive mutation that was described, a transposon-like mechanism was found to be responsible for joining the front end of one gene (*araB*) to the tail end of another (*lacZ*) to generate a hybrid *araB-lacZ* coding sequence. That was enough to enable the bacterium to metabolise lactose. In other examples of adaptive mutation, the activation of a specific silent gene was achieved by the insertion of a transposable IS element.

In yeast and higher organisms, the activity of retrotransposons is also regulated. The retrotransposons move in response to specific environmental stimuli, and to insert at specific sites.

In *Drosophila*, for example, transposons such as the *P* factor and certain retrotransposons tend to be stable in their normal host, but can transpose at high frequencies when introduced into an egg cell from a naïve population lacking the transposons.

P factor activity is further limited to the germline by regulated alternative RNA splicing (see Chapter 5, Figure 5.1). In the germ cells, all four exons encoding the active transposase enzyme are spliced

together after transcription, while in somatic cells, only the first three exons are correctly spliced, leading to the production of a truncated inactive version of the transposase protein.

Methylation - chemical modification of the DNA - is used in bacteria, fungi, plants and mammals to regulate the activity of various repeated sequences, including transposons.

In summary, there is evidence that the 'cut and splice' action of transposable elements in the genome responds appropriately to biological signals via regulatory networks that determine both the timing and extent of the genetic change they cause, and are always appropriate to the organism as a whole.

But exactly how transposons are directed to the correct locations in the genome is not known.

Artificial genetic engineering vs natural genetic engineering

Artificial genetic engineering is disrupting just this exquisite regulatory network of natural genetic engineering by making use of aggressive promoters and vectors made from viruses and transposons to get foreign genes into the genome. It is anything but precise. It introduces into organisms new genes and combinations of genes that have never existed, and deliberately sets out to cross all species barriers and to break down the integrity of the organism.

Genes are typically constructed in 'gene-expression cassettes' (see Chapter 8, Fig. 8.2) each consisting of a very strong promoter, usually from a virus, linked to the gene, in order to make the gene over-express continuously at perhaps 1 000 times the rate of any host gene. This effectively prevents integration of the transgene into the regulatory system of the organism as a whole, and subjects the system to physiological stress. Imagine what it is like to have a bar of heavy metal music played, over and over again, on the most powerful amplifier while a live concert of Mozart is going on.

One of the major problems of 'gene therapy', or genetic modification of human cells, is to get the inserted gene not only to express itself, but more importantly, to be *properly regulated* by the body as a whole, so that it is not expressed where and when it is not needed, and only when and where appropriate, at the correct level. Cancer results when this regulation breaks down, and when certain genes are over-expressed that results in the cell multiplying out of control.

Several gene-expression cassettes are often linked in tandem,

including the antibiotic resistance marker. The gene-constructs are either introduced directly into plant cells by physical methods that damage cells, such as a gun that shoots in gold particles coated with the genetic material or a strong electric field that creates holes in the cell membrane, in the case of animal cells. Or, more usually, the constructs are spliced into artificial vectors, made up of bits of different viruses, transposons and plasmids (Fig.11.1). As said in Chapter 8, while natural viruses, transposons and plasmids are generally restricted in their movements by genetic barriers between species, the artificial vectors and gene-constructs are *designed* to cross all species barriers and to overcome normal mechanisms that break down, inactivate or inhibit foreign DNA, and are therefore more likely to be transferred horizontally.

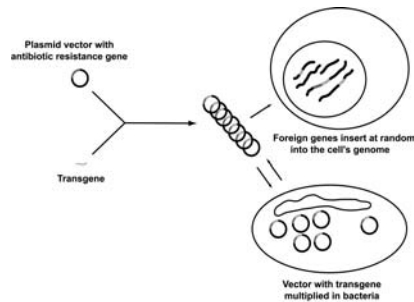


Figure 11.1 How to make a transgenic organism.

Artificial genetic engineering is uncontrollable and unpredictable

The insertion of foreign genes into the genome is controlled neither by the organism nor by the genetic engineer. Where in the genome the inserts end up, and in what form, are almost completely random and indeterminate. There may be hotspots for foreign gene integration, as recent experiments suggest (see below), but the important thing is that gene integration cannot be targeted to specific sites in the genome. The insert can end up almost anywhere in the genome and rearrangements and deletions of the transgenic construct frequently occur. This results in correspondingly random, unpredictable genetic effects.

A transgenic plant is regenerated essentially from a *single* transformed cell, from which a transgenic line is eventually derived. In the case of animals, a single egg or embryo is transformed, which is bred to produce a transgenic line. There are now newer ways for creating transgenic animals from embryonic stem cells that may eventually make transgenic animals more similar to the way transgenic plants are produced.

It is therefore impossible to predict the properties of the resulting transgenic lines; unintended changes abound, including possible toxins and allergens, from the genes introduced, from random gene integration, and from interactions between the foreign genes and host genes. The foreign insert can inactivate host genes, or the strong promoters in the insert could activate host genes inappropriately. The effects can extend to parts of the host genome far from the site at which the foreign genes have integrated.

Consequently, each transgenic line is different from every other, even if the same plant cells, transgenic constructs and vectors are used. I have argued why it is important to have 'event-specific' characterisation, in order to identify where in the genome the transgenic insert or inserts have landed and in what form, and to show that the insert(s) remain stable for at least five generations in the field. But our regulators have failed to require such data from the companies to this day.

In 2001, Belgian scientists carried out such 'event-specific' characterisation of Monsanto's Roundup Ready soya. The transgenic insert was considerably scrambled, and an extra piece of DNA of unknown origin, more than 500 base pairs long, was also present. This was very different from the characterisation given in Monsanto's original submission for commercial approval.

The new European Directive for Deliberate Release does require such 'event-specific' characterisation. But as the implementation is up to the individual country, all the signs are that UK's ACRE will water it down.

Proponents often claim that undesirable, unstable plants can be selected out in early generations. Not so. Transgene instability can become worse in later generations of growth under field conditions.

Transgenic instability, the best-kept open secret

There are two major causes of transgenic instability. The first has to do with the defence mechanisms protecting the integrity of the organism that 'silence' or inactivate foreign genes integrated into the genome, so that they are no longer expressed. Gene silencing was first discovered in connection with integrated transgenes in the early 1990s, and is now known to be part of the organism's defence against viral infections. Gene silencing is a very active research area.

A 1994 review by Finnegan and McElroy gave a devastating description of transgene instability due to gene silencing:

"While there are some examples of plants which show stable expression of a transgene these may prove to be the exceptions to the rule. In an informal survey of over 30 companies involved in the commercialisation of transgenic crop plants,....almost all of the respondents indicated that they had observed some level of transgene inactivation. Many respondents indicated that most cases of transgene inactivation never reach the literature".

The second major cause of instability has to do with the *structural* instability of the transgenic constructs themselves, their tendency to fragment, to break along weak joints and to recombine incorrectly, and often with other DNA that happens to be around. That's perhaps the more serious from the safety point of view, as it enhances horizontal gene transfer and recombination.

There is yet another source of instability more recently discovered. There appears to be certain 'receptive hotspots' for transgenic integration in both the plant and the human genomes. These receptive hotspots may also be 'recombination hotspots', prone to breaking and rejoining. That, too, would make inserted transgenes more likely to come loose again, to recombine, or to invade other genomes.

I have pointed all that out to national and international regulatory bodies time and again, only to be met with persistent evasion and

denial, basically because to admit transgenic instability would essentially undermine the entire industry. The instability of transgenic lines is perhaps the best-kept open secret of the industry. Everyone knows about it, and agrees to have nothing said on the matter. UK's ACRE Report at the end of 2002 (see Chapter 4) effectively gave the green light to commercial growing of GM crops in Britain. On transgenic instability, it stated, "There is also no evidence in the literature to support the idea that transgenic DNA is inherently less stable than native DNA." This statement is simply false.

There is such a vast literature on the instability of transgenic plants that ACRE's denial is embarrassing. We at ISIS have reviewed many papers in the scientific literature, among which are reviews on the subject of transgenic instability; some of the work coming from John Innes Centre (JIC), which has a prominent presence on ACNFP, ACRE's regulatory twin.

Some quotations on the instability of transgenic constructs and transgenic plants are presented in Box 11.2. The first is from a standard textbook on genetic engineering, and the last, from JIC's own Annual Report, documenting the instability of constructs with the CaMV 35S promoter, written by the JIC group that discovered the CaMV 35S recombination hotspot, led by Paul Christou.

Box 11.2

Quotations on structural instability of transgenic constructs and transgenes

Instability of artificial vectors

Old RW and Primrose SB. *Principles of Gene Manipulation*, 5th ed., Blackwell Science, Oxford, 1994 (p.164):

"Structural instability of plasmids may arise by deletion, insertion or rearrangement of DNA. Some of the earliest reports of deletions were in chimaeric plasmids which can replicate in both *E. coli* and *B. subtilis*...Spontaneous deletions have now been observed in a wide range of plasmid, virus and chromosomal DNAs. A common feature of these deletions is the involvement of homologous recombination between short direct repeats....Artificial plasmids with multiple tandem promoters are particularly prone to deletion formation....However, deletion formation can occur between two sites with no homology.....Transcription dramatically affected deletions in an orientation-dependent way such that 95% of deletion endpoints

were localized downstream from the inserted promoter when it faced the major plasmid transcripts..."

Prazeres DMF *et al.* Large-scale production of pharmaceutical-grade plasmid DNA for gene therapy: problems and bottlenecks. *TIBTECH* 1999, 17, 169-74 (p.171):

"...Structural instability, which leads to problems such as the formation of multimers, is more difficult to eliminate....These genetic-stability problems of plasmids usually increase with the size of the DNA fragment inserted."

Instability of transgenes

Pawlowski WP and Somjers DA. Transgene inheritance in plants genetically engineered by microprojectile bombardment. *Molecular Biotechnology* 1996, 6, 17-31:

"Molecular analyses of transgenes introduced by direct gene transfer....often reveal that a high frequency of transgenic events exhibit extensive rearrangements of transgenic DNA sequences....rearrangements of transgenic sequences are either deletions or ligations [i.e. joining] of introduced DNA...." (p.19)

"....Loss of transgene sequences during plant regeneration and reproduction has been described in a number of cases. All or only a portion of the transgene sequences may be eliminated..." (p.25)

Kumpatla SP *et al.* Genome intruder scanning and modulation systems and transgene silencing. *Trends in Plant Sciences* 1998, 3, 97-104 (p.97):

"The widespread occurrence of transgene inactivation in plants...suggest that all genomes contain defence systems that are capable of monitoring and manipulating intrusive DNA. Such DNA might be recognized by its structure, its sequence composition relative to that of its genomic environment and possibly by its disruption of normal biochemical function. Although methylation, especially of repeated sequences is widely associated with gene inactivation, other attributes, including chromatin modification, may be involved. Elimination of inactivated intrusive sequences.....may also contribute to genomic defence mechanisms..."

Kohli A *et al.* Engineering pest-and disease-resistance in cereal crops: fundamental and applied aspects of transgene expression. John Innes Centre and Sainsbury Laboratory Annual Report

1998/1999 (p.22-23):

"Many factors can influence the behaviour of foreign DNA when it integrates into the plant genome. Such factors include the position of integration, possible rearrangement of the exogenous DNA by recombination, and the activation of endogenous plant defence systems that have evolved to suppress the activity of 'invading' DNA.

"Fig. 17. Transgene rearrangements often occur at regions rich in DNA secondary structure, such as the CaMV 35S promoter, which can form the cruciform structure shown above. This allows recombination to occur, as shown by the green arrow."

Paul Christou became the fiercest critic of our papers on the CaMV 35S promoter, possibly because we had drawn the obvious conclusion from his research that he has refused to draw.

Two years later, in the 2001 JIC Annual Report, Christou's research team admitted to the need to avoid the CaMV 35S promoter as well as other bits of DNA containing recombination hotspots.

"Analysis of junctions between genomic and transforming DNA, and between individual plasmid molecules at integration sites, demonstrates the predominance of microhomology-mediated illegitimate recombination events involving regions with secondary structure. One such region occurs in the CaMV 35S promoter, widely used to drive transgene expression in plants. The plasmid backbone provides other such regions, including the origin of replication*The influence of transgene rearrangements on expression and silencing has been understated in the past...*" (italics mine)

This 40-strong research team was mysteriously 'disbanded' in April 2002, in the midst of the controversy over transgenic contamination of Mexican maize landraces.

The most controversial finding in the report from Berkeley biologists Quist and Chapela was not the transgenic contamination itself. It was molecular data suggesting that the transgenic DNA containing the CaMV 35S promoter may be "fragmenting and promiscuously scattering throughout the genome" of the landraces. Such observations would be consistent with our expectations given the recombination hotspot in the promoter.

In the same 2001 Annual Report from JIC, another team docu-

mented instability arising *in later generations* of transgenic barley lines, thus giving the lie to the claim that unstable plants are selected out in early generations.

“Data from the 1998 trial showed that transgenic barley lines performed as well as non-transformed control plants and controls from tissue culture-derived parents for several agronomic traits, including yield. For other traits, a significant difference was seen between transgenic and control lines. The transgenic lines were significantly shorter and also slightly later flowering.... When we examined the next generation of the same transgenic line in the field during 1999, there was evidence that the transgenic plants were more variable compared to the controls than those in the 1998 field trial. This could be because somaclonal variation, resulting from the tissue culture and transformation procedures, and was more obvious in later generations. *These results show that transgenic lines need to be examined over a number of generations under field conditions to obtain the necessary data on transgene stability and agronomic performance.*”(italics mine)

The findings on transgenic barley have been replicated in different forms in every species of transgenic plants investigated with the appropriate molecular techniques.

Despite that, the only criterion on which ACRE and ACNFP assert transgenic lines are stable is transgene expression, and even here, they have simply accepted the company's word, with no numerical data or independent verification.

This practice of accepting only the company's data is the standard procedure. Derek Burke, ex-chair of ACNFP, admitted to that at a recent public 'science debate' (*Are GM foods safe to eat?*); the reasons being, “it is too expensive” to do otherwise, and it “has to be paid for”. That debate was supposed to be the first of a series to feed into the national GM Science Review. But it was no debate at all. There were no strong challengers to GM on the panel, and no serious challenge was permitted from the floor. The chair cut me off when I tried to tell the audience about the critical scientific evidence that has been left out. The audience was stuffed with people from the rapidly growing GM debate industry. The usual defenders were there in force, and I even met an academic whose job it was to 'assess' the debate.

Transgenic instability in the field

By the biotech companies' own admission (see above), most cases of transgene inactivation or instability never reach the literature. But farmers and ordinary citizens have certainly witnessed these failures in the field that have ruined many livelihoods.

You will recall the massive failures of GM cotton that have hit the major cotton growing areas in India in 2002 (Chapter 1, Box 1.1). That was not an isolated case.

The same had struck Indonesia the year before, as I discovered while on a lecture tour there. Monsanto planted 500 hectares of GM cotton within 9 districts of Sulawesi, in open 'field trials'. The company had invited journalists to one of the sites where it claimed the GM cotton out-performed the indigenous variety planted side-by-side. It turned out that the GM cotton failed to out-perform the indigenous variety in *all* but one of the 9 districts. Worse yet, the GM cotton succumbed to drought and the brown hopper. Vivid photographs appeared in the major national newspaper *Kompas* (8 Feb. 2001) showing the browned-out GM cotton field next to the lush green field of indigenous cotton, which was resistant to both drought and the brown hopper.

Another earlier failure of Bt cotton in Texas, United States, was reported in *Nature biotechnology* under the news headline, "Farmers say Monsanto's engineered cotton drops bolls." In the first year of planting, and in up to 50% of the acreage, the Bt cotton failed to provide protection against the cotton bollworm (*Helicoverpa zea*). Numerous farmers had problems with germination, uneven growth and lower yield. A class action was filed against Monsanto, which was settled out of court in 2000, as reported by Michael Hansen of the Consumers Union in the United States.

With such a record, the world was taken by surprise when a paper appeared in the top US journal, *Science* in February 2003, which reported glowing successes of Bt cotton in India, claiming that "the technology substantially reduces pest damage and increases yields" by some 70 to 80%. It was hyped all over the world.

The paper turned out to have been written by two non-scientists, Matin Qaim from the Center for Development Research in the University of Bonn, Germany, and David Zilberman from the Department of Agricultural and Resource Economics, University of California, Berkeley. They were reporting not actual field data, but field trials. The trials were carried out, not by them, but by the company

Mahyco, a subsidiary of Monsanto, in 2001. Furthermore, the authors thanked “researchers of Mahyco for making the field-trial records available” to them.

Not only was the company Mahyco-Monsanto entrusted with carrying out its own field trials, the same practice as in Britain, the results were never made public despite pressure from many Indian civil society organisations. And based on those results, the Indian government approved the commercial plantings that have led to this year's widespread disaster.

Devinder Sharma, Indian policy researcher and food analyst, who runs the highly successful e-mail list AgBioIndia, was the first to expose the shoddy science behind the *Science* paper. It was so bad that a pro-biotech scientist, Santhu Shantharam, formerly with the United States Department of Agriculture now working for Syngenta, wrote a stinging attack on *Science* magazine for publishing the paper, and posted it on the pro-GM website, AgBioView. He stated, “This paper really questions the current standards of peer review in a prestigious journal like *Science* that has a century old reputation for high scientific standards.”

Commenting on the incident, Devinder wrote, “We hope Dr Shantharam's reaction will inspire the public sector scientists to also stand up and be counted. After all, what is at stake is the future of good science.”

Closer to home, the Munlochy GM Vigil, a local protest group near Inverness in Scotland, witnessed a GM oilseed rape crop - supposed to be glufosinate-tolerant - almost completely shrivel up as the herbicide was applied, and captured the evidence on photograph. The company Aventis has never come up with an explanation.

These GM oilseed rape varieties are really bad news, for they are engineered to be male sterile, so farmers have to purchase new seed every year. They are versions of the terminator crops engineered to be sterile, solely to protect patented seeds and traits. Monsanto publicly announced it would not develop these crops several years ago. Behind the scenes, however, such crops have been field tested and commercially grown since 1989. Joe Cummins and I explained it all in a paper submitted to the GM Science Review entitled, “Chronicle of an ecological disaster foretold”, and it has created quite a stir. We first drew attention to these crops two years ago, but it took Brian John, a scientist belonging to the local group in Pembrokeshire that's campaigning for a GM-free Wales to recognize the implications, and to alert the media

accordingly.

The same problems of transgenic instability beset the genetic modification of animals. Transgenic animals simply do not breed true, which is why cloning from adult cells was attempted in the first place. Gene therapy, the genetic modification of human beings has had little or no success either, and so far, the hazards of cancer are threatening to outweigh the benefits to the handful of successes.

So why are transgenic organisms unstable? The reason is because the transgenes are transgressing the natural integrity of the organism, which reasserts itself in a very concrete way. It will be instructive to look at gene therapy.

Transgene products are treated as foreign by the host organism

In a review on gene therapy published in 2000, Helge Grosshans of Heidelberg University, Germany, summed up the situation: "Gene therapy follows a simple principle: causal therapy instead of symptomatic treatment....By now, however, it has become evident that particularly in those cases where the idea of "causal therapy" appears most appropriate, i.e., monogenic diseases, success is minimal. This is due, among other factors, to the cell being a very complex and dynamic system. A change in the genetic make-up that causes a cellular defect also brings about a number of compensating mechanisms. Mere addition of the "health" gene does not automatically re-create the original situation, because the compensatory mechanisms will not necessarily be turned off again."

Also, "A newly synthesised normal protein will appear abnormal to an immune system that has never been exposed to it."

In other words, the cell and ultimately the entire organism, works as a whole, so practically every part of it will have changed when even a single gene is mutated. Consequently, restoring that gene is unlikely to put things right, and may even result in the gene product being targeted by the body's immune defence.

The same reactions have now been discovered against gene-drugs that involve replacing or augmenting hormones or proteins found wanting in the body, as Andrew Pollack first reported in the *New York Times* (Box 11.3). These findings underline the importance of seeing organisms as integrated wholes, and not as a collection of separate, replaceable parts.

Box 11.3

Hazards of gene drugs generic, quality control impossible

Many biotechnology drugs are promoted on grounds that they are "the body's own solutions". But an outbreak of serious illnesses linked to the anaemia drug Eprex shows that some patients react as if the protein is foreign, and their immune systems try to destroy it.

Eprex, made by Johnson & Johnson and sold only outside the United States, is widely believed to be responsible for 141 cases of pure red cell aplasia, in which the body is unable to produce red blood cells, making some patients dependent on transfusions to survive.

It is now clear that virtually all gene drugs provoke immune responses in some patients, though usually just tiny fractions. But these add up as the number of protein drugs increases.

Patients can experience allergic reactions or even potentially fatal shocks. In most cases, the body makes antibodies that attack the protein, rendering it less effective.

Up to one-third of haemophiliacs develop immune reactions to the blood-clotting protein Factor VIII. In desperation, some undergo year-long treatments at a cost of \$1 million to try to restore the drug's usefulness.

Between 5 to 40 percent of multiple sclerosis patients develop resistance to beta interferon, according to Dr. Richard A. Rudick, director of the Edward J. and Louise E. Mellen Center for Multiple Sclerosis Treatment and Research at the Cleveland Clinic. Although there is debate about whether the presence of the antibodies is important, Dr. Rudick said, many such patients are unlikely to benefit from the drug.

What makes the illnesses associated with Eprex more alarming is that the antibodies produced by the patients not only make the drug ineffective, they also attack the patient's own protein. So the patients can no longer produce any red blood cells, ending up with worse anaemia than they would have had without the drug.

The reasons for the immune responses are not clear. Putting the human gene for the protein into bacteria or animal cells could result in proteins subtly different from the natural protein, in the sugars attached to the protein, or in the way they are folded. In other cases, the drug may have an impurity.

The factory in Puerto Rico, which makes Eprex, is under

criminal investigation. The same protein, sold in the United States as Procrit by Johnson & Johnson and as Epogen by Amgen, is linked to just a handful of cases of red cell aplasia. Amgen, which developed the drug, makes both brands in a factory in Colorado. However, the antibodies developed cross reacts with all commercial brands of the hormone.

Another problem, experts say, is that biotech drugs made by living cells are not as predictable as those made chemically. (This recalls the epidemic linked to a batch of genetically engineered tryptophan in the United States in 1989, which resulted in many deaths.)

The 141 suspected cases of red cell aplasia must be seen against the background of the three million people who have been treated with Eprex or its American counterparts for more than 10 years. Many of those patients are able to work or lead more normal lives because the drug eliminates the fatigue of anaemia.

The drug is a form of erythropoietin, a protein hormone produced by the body to spur the production of red blood cells, which ferry oxygen around the body. It is normally produced by the kidney and is widely given to people with kidney disease and on dialysis machines.

Human insulin made by genetic engineering has supplanted insulin from animals.

The human version generally provokes far fewer immune reactions and lets diabetics control their insulin better. Nevertheless, a small percentage of diabetics say they cannot tolerate the human version and are trying to keep cow and pig insulin available.

Eprex has been the biotechnology industry's most successful product. Eprex and its American cousins exceed \$5 billion in combined yearly sales. But, ironically, this new problem is thought to help in the long run, by eliminating generic competition.

For drugs made chemically, generic manufacturers do not have to run expensive clinical trials. They just need to show that their drug is identical to the brand-name product. But the biotechnology industry and the FDA have said that, "it is too difficult to show that two biotech drugs are equivalent without full clinical trials".

This actually sums up the generic nature of hazards in gene drugs, and the impossibility of quality control, and should make regulators and the general public more wary.

It is on account of the organic fluidity of the genome and gene function *in the maintenance of the whole* that artificial genetic engineering is so disruptive and dangerous. It also reveals the futility of the attempts to exert mechanistic control over organic processes that are radically uncontrollable, because organisms work on entirely different principles.

The final chapters of this book will expose the futility of the new 'bio-informatics' and health genomics in the light of all we have learned about the fluid genome, and explore what it means to be living with the fluid genome.

Twelve

The 'Bio-informatics' & Health Genomics Bubble

The genomics hype continues

When Craig Venter pricked the genomics bubble by telling the world there were too few genes in the human genome to sustain the genetic determinist myth, investment companies were quick to see the implications, "Mapping the genome could be route to disaster", so said a newspaper headline. The human genome project could bankrupt the biotech and pharmaceutical industry, it warned. The "information overload" will cost much more than previously thought. Genomics, it said, "threatens to increase not only the associated research and development costs, but also the average cost per new drug."

The proponents redoubled their efforts to keep up the hype. The American Museum of Natural History put on a lavish exhibition, "The Genomic Revolution", funded by a corporate-friendly charity. It predicted that, "By the year 2020 it is highly possible that the average human life span will be increased by 50 percent; gene therapy will make most common surgery of today obsolete; and we will be able to genetically enhance our capacity for memory". Meanwhile, an art show promoting genetics called "Paradise Now" went on tour around the country.

Our genome scientists have distanced themselves from the crude propaganda. Instead, they said the human genome map is just "the end of the beginning", and much more money would be needed before the goods can be delivered in cancer cures, eradication of disease, genetic enhancement, gene therapy, personalised medicine and a prescription of lifestyle based on our genetic make-up.

The British government, in 1999 and 2000, committed some £2.5 billion to genomics over the next four or five years. The Wellcome Trust, the biggest UK charity long regarded as friendly to the pharmaceutical industry, gave £300 million to Cambridge's Sanger Centre for genomics research. The centre changed its name to the Wellcome Trust Sanger

Institute.

Genomics is a huge undertaking. 'Health genomics' alone includes information gathering, not just on the human genome, but the genomes of ultimately all animals, plants, fungi, bacteria and viruses that 'serve as disease models' or cause or transmit diseases of one form or another.

Complete genome sequences are already available or announced for yeast, *Arabidopsis*, fruit fly, nematode and mouse; and rat, zebrafish, chicken and dog are well on the way. Within the next five years, complete genome sequences will be available for all model organisms used in biomedical research.

In addition, the genomes of hundreds of viruses and bacteria associated with major infectious diseases are sequenced, with hundreds more to follow. Genomes of many protozoa parasites are available including those associated with malaria, sleeping sickness and Leishmaniasis. And genome sequences of insect vectors, such as the mosquito, *Anopheles gambiae*, and the *Aedes species*, have been announced.

Tens of thousands of genes will be identified in each genome. Then, one needs to find out which genes are transcribed under different conditions, in '*transcriptomics*', and what proteins are made and what each protein does, in '*proteomics*', and to complete the set, '*metabolomics*' will define all the metabolites generated by the enzyme proteins. The crystalline structure of each protein will have to be determined in big synchrotron X-ray facilities. That's not all.

There are hundreds of variants of every gene, and any two human individuals will differ by one in every 500 bases in their genomes. These single nucleotide polymorphisms (SNPs), millions of them, account for most of the genetic differences between individuals, and are the great white hope to 'personalised' medicines. A public-private consortium of the Wellcome Trust and 13 pharmaceutical companies is supporting mapping and analysis of SNPs in the Sanger Institute plus four other centres in the United States.

In addition, 'genome-wide' scans for patterns of genes expressed and proteins synthesized are made with gene chips and protein chips to find out what happens in different diseases under different conditions. Endless reams of data will be generated, demanding ever bigger and faster computers to compile and analyse them, and much more efficient and compact means of data storage.

The Wellcome Trust website, well into 2002, told us that pharmaceutical companies believe they will get many additional “druggable targets”. “Currently, the entire drug cabinet is focussed on just 483 targets”, so “enormous possibilities for new therapies exist”. The SNPs are even better. Pharmaceutical companies are said to be “eagerly awaiting this flood of new information about human variability” to accelerate the development of drugs based on individual differences

In reality, the prospects for genomics look bleaker than ever before, and creeping doubt had developed into full-blown scepticism by the end of 2001 (Box 12.1).

Box 12.1

Creeping doubt on genomics developing into full-blown scepticism

A row broke out in the European Parliament over the European Commission's proposal for its sixth framework for research funding, which covers public research funding within EU countries for the next 6 years. Members of the European Parliament (MEPs) and researchers criticised the proposal for focussing too much on genomics and biotechnology, and not enough on individual diseases like cancer, diabetes and Alzheimer's.

Plans to build a new genomics facility in Grenoble were in jeopardy due to lack of funding. The project was a joint venture between the European Synchrotron Radiation Facility, the Institut Laue-Langevin and the European Molecular Biology Laboratory, and was due to be financed in large part by an industrial consortium.

Russia cut its spending on genomics research through its 12 year old national Human Genome Program by 50%, and put the money into a general pot for basic research.

‘Bio-informatics’ the next financial bubble

The multibillion-dollar industry of genetic engineering could well turn out to be the biggest financial bubble perpetrated by the scientific establishment. And as industry is deserting the enterprise (see Chapter 1, Box 1.1), our governments are led further down the garden path. They are throwing even more public money away on the ‘new science’ of ‘bioinformatics’, to ‘make sense’ of the genome sequences, genes

and proteins spilling out of the pages of learned journals.

The vain hope is that by putting all the information in supercomputers, knowledge will automatically emerge out of this vast graveyard of data.

Governments including Iceland, UK and Estonia as well as private healthcare providers in the United States are supporting mega-projects to collect human DNA 'Biobanks' of the entire nation, or of substantial populations up to a million, on the stated belief that correlating gene sequences with disease and lifestyle data will "revolutionise" healthcare and deliver "personalized medicine". In fact, it will do nothing of the sort. Nothing useful could come out of such projects, and drug companies themselves have said such studies will not help drug development.

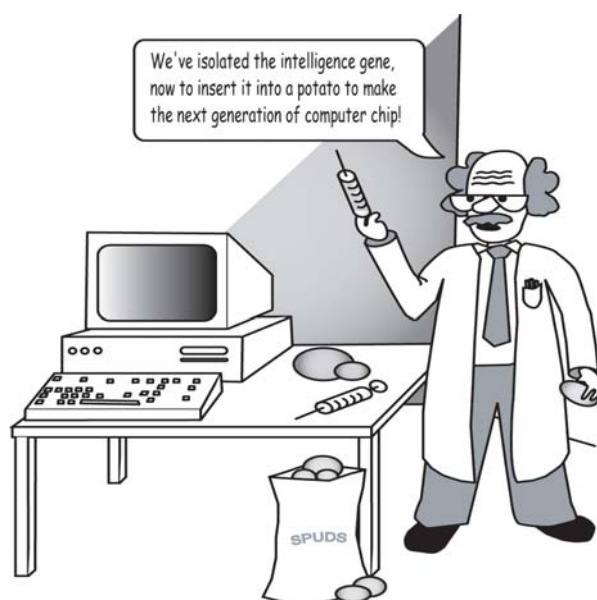


Figure 12.1. From genomics to bio-informatics.

Research released by the Tufts Center for the Study of Drug Development in 2001 showed that the 'biotech revolution' had, so far, greatly slowed drug development while increasing cost.

During the last two decades, the average time spent on clinical trials increased from 33 months to 68 months, while the average cost of

developing a new prescription drug had jumped to \$802 million from \$231 million a decade ago. This figure included the cost of human trials, pre-clinical studies, expenses for product failures, and the impact of long development times on investment costs.

Most of the top 20 multinational pharmaceutical companies are not generating in-house the new products needed to sustain the rates of growth they had enjoyed in the past."No serious industry onlooker could dispute this depressing picture", wrote David Horrobin, CEO of Laxdale Research, Stirling, Scotland in an Editorial in the December 2001 issue of *Nature biotechnology*.

Anthony Sinskey, a co-director of MIT's Program on the Pharmaceutical Industry, stressed that the millions of chemicals sitting on pharma's shelves need to be investigated. And that it is up to the 'array makers', 'microfluidics developers', and 'informatics providers' to help predict and measure outcomes and bring products to market.

But Horrobin pointed out that combinatorial chemistry has been around for over a decade, and yielded relatively few products considering the extraordinary size of the investment. "Could it be that there is something wrong with the technology in principle, and that the target choices and the target configurations are fundamentally flawed?" he asked.

'Pharmacogenetics' a non-starter

Allen Roses, scientist from the Genetics Directorate of Glaxo Wellcome, wrote an article in *Nature*, ostensibly to promote 'pharmacogenetics' - the study of how genetic differences influence individual response to drugs. But a different sobering message came across.

It is very expensive to validate the new drug targets, so, pharmaceutical companies prefer to make new variants of old drugs. "The best-validated targets are those that have already produced well-tolerated and effective medicines."

Roses distinguished 'discovery genomics' from 'discovery genetics'. The former uses databases of DNA sequence information to identify genes and families of genes for possible drug targets, but these are not known to be associated with any disease. The latter uses human populations, like the human DNA Biobank collection, to identify disease-related susceptibility genes. But susceptibility genes are not drug targets. They "will not be tractable targets or amenable to high throughput screening methods to identify active compounds".

Companies now 'mine' sequences from the human genome, to identify gene families that have sequences similar to one that is known. But this approach is full of pitfalls, because similar sequences often turn out to have entirely different functions, just as completely different sequences can have the same function.

The major challenge remains the validation of the drug target. "Each failure is very expensive in lost time and money".

The new methods promoted, such as 'genome-wide scans' are generating patterns so complicated and so variable that they defy efforts to interpret or to validate the methods themselves. They have not led to specific targets for drugs.

There are not even experiments demonstrating that the differences in tissue expression of a particular gene are related to the disease. Neither have these screening methods been applied to rare mutational diseases, i.e., 'single gene disorder', which comprise less than 2% of all human diseases, for "proof of principle", that they give any useful information. Yet these methods are now the major investments in several research programmes on diseases such as Alzheimer's disease.

Several genes are found to be responsible for the rare early-onset form of Alzheimer's disease, and a susceptibility gene has been identified for the common late-onset form. But routine genetic screening is not recommended by consensus groups for either form of the disease, on account of the poor predictive value of the gene tests. Nor have these genes yielded useful targets for drugs (Box 12.2).

So where does that leave pharmacogenetics? "Pharmacogenetic approaches will no doubt confirm what clinicians already know - disease diagnosis is not easy nor necessarily homogeneous and accurate." Doctors already classify diseases according to their differential responses to existing drugs.

The SNPs can speed up discovery of susceptibility genes, but as already mentioned, these are not automatically suitable drug targets, and problems of validation remain. Could SNPs not be used to select for patients that are more likely to respond positively to new drugs? Roses pointed out that regulatory authorities might justifiably look askance at this practice, and would be concerned that when the drug is sold, those patients who do not have the right SNPs might be sold the drug by mistake.

Most of all, Roses warned that gene tests based on susceptibility

genes and SNPs cannot be equated with those used for 'single gene' disorders; furthermore, "recommendations for patients and relatives with these 'complex' diseases are frequently miss-stated with authority by authors whose only experience is in mutational diseases".

In other words, gene tests based on susceptibility genes and SNPs are far less reliable or predictive. But even those based on single genes cannot give accurate prognoses for the individuals.

Box 12.2

Genetics and genomics of Alzheimer's disease

Less than 5% of people with Alzheimer's disease have a strong family history of the disease, with early-onset under 60 years of age. About a fifth to half of these cases are attributed to mutations in three genes coding for the *amyloid precursor protein* (APP), and *presenilin 1* and *2* (PS1 and PS2). The mutations for APP and PS1 are dominant, that is, the presence of one copy of the mutant gene will give the disease. (A person has two copies of each gene, one from mother and the other from father.) However, the severity of the disease is variable and unpredictable.

APP is the precursor of the beta-amyloid, which accumulates in the brains of Alzheimer's patients. The function of PS1 and PS2 are not well understood, they appear to be involved in signalling within and between cells, and influence the processing of APP into beta-amyloid.

The more common, late-onset Alzheimer's disease is not linked to APP, PS1 or PS2. A particular variant of the blood protein apolipoprotein E gene, *apoE4* has been identified as a 'susceptibility gene'. Case-control studies comparing the genotypes of people with and without Alzheimer's disease suggest that people with the disease are about 3 times more likely to carry a copy of the *apoE4* than people without the disease, and 8 to 12 times more likely to carry two copies. However, this claim has been contested.

At least one large study failed to confirm the link between *apoE4* and late-onset Alzheimer's disease. Another study found that the mutation is not predictive of Alzheimer's disease alone as it also occurs in other kinds of dementia. Within the same gene, certain markers may be associated with increased risks while others with decreased risk of disease. It is clear that association alone without

knowledge of biological relevance is of no use for the development of drugs.

Although about 34-65% of Alzheimer's patients have at least one copy of *apoE4*, so do 24-31% of unaffected adults. The association between *apoE4* and dementia is not seen in all populations either.

A possible target for drug intervention in Alzheimer's disease is the presenilins. It was thought that by inhibiting presenilin, less beta-amyloid would be produced. Unfortunately, presenilin has other vital functions besides processing APP. And no useful therapeutic treatment has resulted.

On account of the large uncertainty in linkage between the mutant genes identified and disease development and progression, routine gene testing is not recommended by consensus groups in the US and in Britain.

The Public Health Genetics Unit in Cambridge gives the following advice regarding early-onset and late-onset Alzheimer's disease on its webpage:

"If a mutation has been detected in an affected individual, accurate testing is possible for the same mutation in other family members, but a positive test does not necessarily give a good indication of the severity of disease."

"There is no reliable predictive genetic test for late-onset Alzheimer's. The *apoE4* allele can be detected easily and rapidly; however, a positive test result does not guarantee that disease will develop, nor does a negative result rule it out."

Time for realism

Two medical geneticists wrote in the *New England Journal of Medicine*, that the 'genetic mantle' "may prove to be like the emperor's new clothes."

Holzman and Marteau concluded, "In our rush to fit medicine with the genetic mantle, we are losing sight of other possibilities for improving the public health. Differences in social structure, lifestyle, and environment account for much larger proportions of disease... Those who make medical and science policies in the next decade would do well to see beyond the hype."

Sir David Weatherall, one of UK's most eminent geneticists, stated that "the remarkable complexity of the genotype-phenotype relationship has undoubtedly been underestimated.... It has led to many statements being made about the imminence of accurate predictive genetics that are simply not true". Nevertheless, he still believes that the information gained will eventually result in accurate predictive genetics.

Richard Strohman, Emeritus Professor of the University of California, Berkeley, has long expressed doubt that genomic information will have much impact on human life span and health.

Genes influencing general health and longevity are numerous, perhaps hundreds or even thousands, but their effects are confounded by interaction with the environment. It is simply impossible to pin them down. Scientists face a 'computational barrier' because the possible number of interactions that have to be taken into account far exceeds our ability to cope with them, even with the fastest, biggest computers.

The potential to longevity is equally present in a wide variety of genomes, with environmental factors dominating.

The improvement in hygiene, medical care and public health measures in the first half of the 20th century added nearly 30 years of life-expectancy to the populations in the rich countries, which had little to do with genetics. This must now be the priority for Third World countries. Instead of genomics, which will turn out to be the biggest white elephant that will surely bankrupt the nation, they should have access to decent living conditions, nutrition and hygiene.

Indeed, environmental variables - such as hygiene and living conditions, exposure to toxic substances, social class, poverty, diet, exercise, and other 'lifestyle' habits - can all significantly influence disease and disease progression across all ethnic groups and major genetic differences. For the vast majority of diseases, the environmental variables will swamp the effects of 'susceptibility genes'.

'Susceptibility', by definition, denotes weak linkage, and can never give firm predictions. Moreover, it describes the *population attribute* of certain gene variants, and says nothing at all about the susceptibility of *individual* human beings. One thing the human genome project has taught us is how genetically diverse we are. Each of us is genetically unique, except for identical twins at the beginning of development, before each can accumulate genetic mutations independently in the cells or the body.

Associations between a disease and a 'genetic marker' (of unknown function) can happen by chance and many have proved to be spurious. The effect of each gene depends ultimately on the 'genetic background' consisting of all the other genes in the genome, which is why predictive medicine based on susceptibility genes or SNPs is scientific nonsense at best. At worst, it can be an excuse for genetic discrimination and eugenics.

The importance of genetic background is seen even in diseases that can be attributed to the effects of single genes: the same gene mutation does not give the same disease in different human populations. The gene mutation that causes 80% of all cystic fibrosis in the white European population is associated with another condition altogether in the Yemen population. And in the Yemen population, conditions identified as 'bona fide' cystic fibrosis are due to mutations in another gene altogether.

And even in the same genetic background, the effect of the gene can depend drastically on hormonal and other physiological influences. The so-called breast cancer gene, *BRCA1*, said to predispose women who already have a family history of breast cancer to develop the condition, shows no association with breast cancer in men at all. Thus, it can be predicted that susceptibility genes, even if they can be firmly identified for one population, will have little or no relevance for another population.

'Susceptibility', one has to conclude, simply conceals the fact that much, much more important environmental variables are left out of consideration, as for example, chemical carcinogens, at least several hundreds confirmed and admitted by the United States National Cancer Institute. It is well known that the incidence of cancer increases with industrialisation and with the use of pesticides. Women in non-industrialised Asian countries have a much lower incidence of breast cancer than women in the industrialised North. However, when Asian women emigrate to Europe or the United States, their cancer rate jumps to that of native European women within one generation. And when DDT and other pesticides were phased out in Israel, breast cancer mortality in pre-menopausal women fell by 50%.

'Health genomics' projects and related studies are indeed worse than useless, for they may deliver us to the brave new world of privatised eugenics, that promises to rid us of all the 'bad genes' that make the individual 'susceptible' to diseases and disabilities of every kind,

and promulgate the 'good genes' that code for intelligence, beauty and every other desirable trait. All in accordance with the age-old folklore that has given rise to the genetic determinist myth in science.

The conspiracy theorist, on the other hand, will see one possible use of human DNA 'biobanks': it is every military dictator's dream to be able to keep everyone of its citizens under absolute surveillance.

Genomics enthusiasts corrupt science

In a stinging attack published in *The Guardian* newspaper, David Horrobin dismissed the triumphalism about molecular biology, genomics and the human genome project. "Popular science magazines and the general media have united in a chorus of praise concerning the supposedly dramatic effects this is going to have on human health." Instead, he accused genomics enthusiasts of having "corrupted scientific endeavour and destroyed real hope of progress".

From the 1930s to the 1960s, biomedical science looked like an integrated whole, with researchers working at every level of biological organisation - from molecules, to cells, organs, animals, and humans. That was the golden age, he said. "Anyone receiving the best medical care in 1965 was incomparably better off than anyone in 1930".

Solutions to health problems came from many different levels: social, cultural, educational and economic, all contributing to the control of malaria, for example. And in every field, "progress depended on a constant exchange of knowledge"

But since the 1960s, molecular biologists and genomics specialists have taken over biomedical science, and everything has been reduced to the genome. "Journals and grant-giving bodies came to be dominated by reductionists. Clinical and physiological studies lost out."

The first genetic disease to be defined in molecular terms was sickle cell anaemia, an abnormality of the haemoglobin in human red blood cells common among people originating from Africa. The exact mutations were worked out, as well as the three-dimensional structure of the protein. But what is the clinical impact of this molecular knowledge? "Precisely nothing", Horrobin stated. "The clinical picture of the disease cannot yet be understood in terms of the molecular biology".

He singled out two areas - stomach and duodenal ulceration and organ transplantation - where medical treatment has advanced dramatically, but in almost every other field, people are little better off than those receiving the best treatment in 1965.

What about cancer? "We have made dramatic progress in a very limited range of rare cancers: the leukaemias, lymphomas and testicular tumours," he said. "But even there many of the drugs used were introduced before 1965. We have simply learned to use them better."

He called the hype surrounding the malaria genome project and the claims that it will lead to a solution to malaria in five years "absurd". Malaria is enormously complex, and part of the complexity, as pointed out by veteran malaria expert Brian Taylor, is that there are more than 65 known species of mosquitoes that serve as vectors for malaria.

Horrobin reminded us of the "disgraceful" campaigns waged in the 70s and 80s by gene hunting scientists. "Give us the money, we'll find the gene and then your problems will be solved." The gene was found, and nothing has been delivered in 14 years of gene therapy, on account of complexities that every clinician knew about.

"The idea that genomics is going to make a major contribution to human health in the near future is laughable," said Horrobin. "But the tragedy is that the whole-organism biologists and clinicians who might have helped to unravel the complexity have almost all gone, destroyed by the reductionists."

A multi-billion industry built on a scientific myth

By now, you must be wondering: In view of all that was known about the fluid genome already in the 1980s, why did the human genome project - to sequence the human genome - ever get the go-ahead? Geneticists must have known that the genome sequence, the sequence of the regions containing genes, which in the human genome comprises less than 5% of the actual genome, would tell us next to nothing about what it means to be human, much less to give us a blueprint for making a human being. Nor, as it turns out, can the human genome sequence tell us anything really useful about something as basic as health and disease.

Was it a conspiracy of silence in the molecular genetics establishment? Barry Commoner wrote in *Harper's Magazine* (see Chapter 5),

"By the mid 1980s, therefore, long before the \$3 billion Human Genome Project was funded, and long before genetically modified crops began to appear in our fields, a series of protein-based processes [such as alternative splicing that gives multiple proteins from one gene] had already intruded on the DNA gene's exclusive genetic franchise ... that a DNA gene exclusively governs the molecular processes

that give rise to a particular inherited trait.

“The credibility of the Human Genome Project is not the only casualty of the scientific community's stubborn resistance to experimental results that contradict the central dogma [of molecular biology]. Nor is it the most significant casualty. The fact that one gene can give rise to multiple proteins also destroys the theoretical foundation of a multibillion-dollar industry”.

The processes of the fluid genome include much more than the alternative splicing of gene transcripts to give multiple proteins that Commoner was referring to. The genome itself is subject to turnover as it adjusts and responds to the environment, to repair and correct errors, or to expand, contract, to rearrange, to hyper-mutate, to insert or delete, or to convert sequences. In other words, the genome can keep itself dynamically constant under balanced ecological conditions, or it can undergo small or cataclysmic changes according to the extent of environmental challenge that the organism experiences. It is part and parcel of the physiological system of the organism that must constantly adjust and respond to the environment in order to sustain itself.

There really is no holding the genome still.

What use is it, therefore, to have a huge catalogue of genomes that are in constant flux, of which only 5% can be captured in silico (on the computer) as genome sequence, the remaining 95% being repetitive 'junk' DNA that's beyond the reach of our sequencing, cataloguing technologies? How can any predictions be made based on something in constant flux?

And in any case, it is those repetitive sequences amidst the 95% inaccessible 'junk' that hold the key to gene regulation, as we have seen in Chapter 11. They could tell us how each gene could possibly know when and where to turn on and by how much, or become silenced altogether. Perhaps they could even tell us how each cell knows which splice forms of the RNA transcript to use, whether to edit the RNA, to reverse transcribe, to convert, to hyper-mutate, to rearrange, which transposons to move when and where, *in coordination with the activities of all the other genes in the genome* as the organism acts on and responds to the environment in the course of living its life.

But even then, we may not grasp what makes the organism whole.

Wouldn't it make much more sense, and be much more cost-effective, to devote resources to improving the environment so that every-

one can have the best of health, regardless of their genetic make-up?

'Bio-informatics' is not just a phenomenal waste of public money, it blocks other more promising approaches to health, and worst of all, it commits generations of scientists to the most mind-numbing work, a sheer waste of scientific imagination.

A planetful of help desks

Jaron Lanier, a pioneer in virtual reality, is one among a number of information technology scientists who have not let the informatics revolution go to their heads, let alone the bio-informatics revolution. Even if computers can become fast enough, and there's every indication they won't be, at least, not unless different kinds of computers are made, like quantum computers (see next chapter), the software simply cannot keep up, especially if biotech companies are hoarding their data on proprietary databases, as they are doing at the moment. His vision of terror is this. While there are all sorts of useful databases and modelling packages developed, they all exist in isolated bubbles. So vast resources will be expended to get data from one bubble to another.

'Moore's Law' in information technology expresses the fact that the number of components on a computer chip, as well as its speed doubles every two years. Lanier writes,

"If Moore's Law is upheld for another 20 to 30 years, there will not only be a vast amount of computation going on planet earth but also the maintenance of that computation will consume the efforts of almost every living person."

We may well end up with a planetful of help desks!

What to do with the existing genome sequences and databases

Why not abandon the mausoleums of useless genomic information to employ scientists in more imaginative, less mind-numbing work?

In fact, this is happening already. A recent article in the *Boston Globe* reports on the continuing gloom in bio-informatics. Many companies have shut down their databases to concentrate on drug discovery, or gone bust altogether. Incyte Genomics Inc. closed its operations and reduced its work force by 37% in November 2002 to move deeper into drug discovery. Celera Genomics Group went through a similar reorganization in June. InforMax and NetGenics were lucky to find

merger partners in Invitrogen and LION Biosciences AG, respectively. But Doubletwhist Inc., a once hot start-up company, closed down last spring.

"We are in a data-rich environment, but the fact is we are information poor," says Peter Sorger, an associate professor of biology at MIT and co-chairman of the school's budding 'systems biology' initiative. "You look at biological systems with much more complexity than before. Bio-informatics has concentrated entirely on sequence information, but it's only a tiny piece of the puzzle."

But you might object that something useful and interesting may yet come out of all that data, one can never tell.

At least, we should rationalise the management of the data so they will not swallow us up in the way that Lanier has envisaged. Besides, data held in private databases, not freely available, would not serve science or scientists in any case.

The solution is simple: *Keep all genomic data on the public database, Genbank, where the data can be accessed by everyone, everywhere, for free.*

That way, no one needs to miss out, it is most efficient and it doesn't cost the nation or the earth.

Thirteen

Living with the Fluid Genome

Lessons from the fluid genome

The fluidity of the genome and gene function is *integral* to life itself, and reflects the constant intercommunicating of all the parts, from organ systems, tissues and cells down to genes, and proteins that are necessary to sustain life. The processes are part of a sophisticated regulatory system that can maintain stability and constancy of the whole, or generate hyper-variability for specific genes or parts of the genome as and when required.

I have drawn a diagram contrasting the DNA-centred Central Dogma with the new genetics (Figure 13.1).

The new genetics acknowledges that genes have a very complex ecology from which they receive layers of biological feedback over every scale of space-time (in the new physics, space and time are not separate). The ecology of genes consists of all other genes in the genome, the particular kind of cells in which the genes find themselves, and the physiology of the whole organism in its ecological environment. Genes are anything but selfish, they are sensitive, responsive and dynamic. The idea that one can patent genes or DNA sequences is absurd, because by themselves they can do nothing, what DNA sequences do depend ultimately on the entire ecological context, which can change the DNA sequences themselves.

In the same way, ecosystems, which include human beings, do not consist of individuals in constant competition of one against all and all against nature. Instead, ecosystems consist of organisms of diverse species, and are sustained as a whole by diverse, mutual, symbiotic relationships over all space-times from the very local to global, from the extremely slow to the ultra-fast.

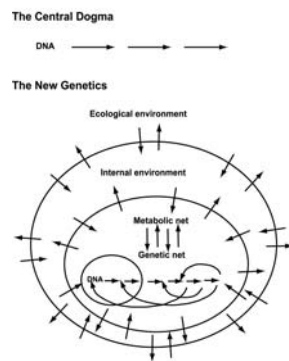


Figure 13.1. Genetics old and new.

The new genetics is holistic genetics. It says that changes in ecological conditions can propagate inwards to affect the organism, including its genes and genome. Conversely, a foreign gene introduced by genetic engineering into an organism may indeed have influences that propagate outwards to affect the ecosystem, if it isn't disarmed or eliminated by the defence mechanisms in the organism itself. The responsiveness of genes and genomes to the environment makes clear that the only way to keep genes and genomes constant and healthy is to have a balanced ecology. In a balanced ecology, genes may even be able to mend themselves, as new findings are suggesting (see Chapter 7). On the other hand, it is definitely futile to think that we can go on

ruining our ecosystem and stay healthy so long as we have 'good' genes.

Genes, unlike diamonds, are not forever. Estimated mutation rates turn out to be "surprisingly high", according to population geneticist James Crow of the University of Wisconsin, Madison, United States; about 100 new mutations per individual. Some genes mutate much faster than the average, for reasons yet unknown. The gene coding for the protein dystrophin on the sex determining X-chromosome mutates to give Duchenne muscular dystrophy at rates approaching one in ten thousand per generation.

But there are deeper implications to living with the fluid genome, as we shall explore in this final chapter.

Do genes make sense?

There is a profound malaise in the molecular genetics community beneath the aggressive hype of the bio-informatics revolution, and the next 'breakthrough'. Discovery of genes is now routine. There are new technical toys in the laboratory, such as gene chips that allow one to do 'genome wide' scans of all the genes all at once, and 'quantum dots' that label DNA and antibodies in pretty colours for quick efficient detection of any DNA or protein species. But beyond that, there have been few real scientific 'breakthroughs', in understanding cancer, aging, or any other human condition; and especially, in understanding how all the genes could become coordinated and regulated in the organism as a whole.

There is a proliferation of more and more complicated models of 'genetic control networks' that give anyone but the most dedicated a splitting headache just by glancing at them on the page. The question - which genes turn on which others under which circumstances to make which of a number of possible proteins - is almost certainly irrelevant, because the organism always functions as a whole and changes as a whole.

Biologists who study the brain have amply demonstrated that every encounter of apparently the same experience is distinct, in terms of the electrical activities that can be measured in the brain. Life never repeats because each experience changes the *whole* organism. There is no holding nature still and looking at it, as the English philosopher-mathematician Alfred North Whitehead said, and no holding the genome still, either.

Nanotechnology is the next white hope of corporate science, now teaming up with biotechnology, information technology and brain science in a "Brave New World Quartet".

One device fills me with particular terror - brain implants that allow us to control electronic devices based on recording the activities of individual brain cells. It is not so much that fine electrodes are inserted into hundreds of nerve cells in the brain. It is not even that such a device cannot be removed without the help of a surgeon. It is that one must train oneself to repeat the same thoughts precisely for any given task, so that the pattern of brain cell activities could remain the same or similar enough to be read by a robot. It is the thought that such a device will turn one into a true robot that's really terrifying. That is the stuff of the Brave New World.

Another reason why the molecular details are largely irrelevant is that in physics, it is well known that astronomical numbers of different microscopic molecular states map into the same macroscopic state defined by some macroscopic variables, such as temperature and pressure. To a large extent, it is irrelevant what each molecule is doing or where it is. In the same way, it will be largely irrelevant what individual proteins are doing at any given time.

Gene-chip scans now confirm what we've always known: that when even a single factor is changed in the cell's environment, the expression of thousands of genes are invariably altered, simply because the functions of ultimately all the genes are interconnected and entangled. It is somewhat like dropping the smallest pebble in a pond and watching the ripples grow and spread.

But still, the hope is that with the aid of the fastest computers in bio-informatics, we may one day make sense of all that molecular complexity. And better still, the organism could itself be understood as a supercomputer that can keep track of all the molecular interactions.

Is the organism a supercomputer?

James Shapiro indeed proposed a "computational" model of the cell as a sophisticated "information processing" system, in which the cell's 'signal-transduction networks' are supposed to "compute data from multiple inputs" to "make decisions about cellular behavior." But the computation metaphor all too readily breaks down.

To put organic function and computing in perspective, IBM has been building a new supercomputer, Blue Gene, projected to be

complete in two years time, which will be the most powerful supercomputer on earth. It will be equipped with something called SMASH (simple, many and self-healing), which will dramatically simplify the use of instructions carried out by each processor. Instead of a single microprocessor on a chip, Blue Gene's chips will hold 32 processors and about a million microprocessors, so it will perform one quadrillion (10 to the power 15) operations per second, or a 'petaflop'. Even then, it will take a year to simulate an average protein folding into shape.

Our body contains some 100 trillion (10 to the power 14) cells, each expressing a unique repertoire of the millions of proteins that could be made. Not only do the proteins fold simultaneously to perfection in split seconds, they also carry out millions of catalytic reactions at the rate of thousands to hundreds of thousands cycles per second. And all the diverse activities are orchestrated, from the molecular and microscopic through to the macroscopic domain, from trillionths of a second to hours, days and years.

What makes the organism whole?

The organic whole is quintessentially diverse and pluralistic. The organism is the antithesis to uniformity and homogeneity. It is truly remarkable that we should habitually think of ourselves as "I" in the singular considering the 100 trillion cells, each with its own repertoire of trillions of protein molecules processing at least ten times as many molecules of metabolites. So what makes such a vast conglomerate of diverse molecules and cells a whole?

That's the question mechanistic science seldom addresses. One way to think of the tens of thousands of genes and millions of proteins that must work together as a whole, is that they constitute a 'molecular democracy of distributed control'. That phrase was first introduced by Hungarian born Scottish biochemist/geneticist Henrik Kacser, who worked all his life in Edinburgh University. He invited us to look at the metabolic network as a whole, rather than in terms of specific control points. He initiated a way of studying metabolism, which demonstrated that 'control' invariably shifts between different enzymes, depending on circumstances. The entire network changes and readjusts as the circumstance changes.

The problem is that conventional biochemistry still tends to see enzymes and their substrates bumping into each other as they wander about haphazardly and randomly in a solution of water inside the cell.

But since the early part of the last century, evidence has been accumulating that the cell is intricately organised, perhaps down to the level of individual molecules; much of that evidence was reviewed by Joseph Needham, distinguished Cambridge biochemist later turned historian of Chinese science, in his book, *Order and Life*, published in 1936.

Not just enzymes and substrates, but even the water molecules are organised in a liquid crystalline continuum, as Needham had predicted in his remarkably prescient book, and as we discovered in my laboratory in 1992, when we half stumbled upon a novel, non-destructive imaging technique that enabled us to see the insides of live organisms in pure, vivid colours. It inspired me to write a book around what the colours mean, entitled, *The Rainbow and the Worm, The Physics of Organisms*.

So how do the right molecules meet up with one another? They may well do so by electromagnetic vibrations, that, according to quantum theory, is present in all molecules. Molecules also radiate electromagnetic waves when they are excited, on absorbing energy. Furthermore, each molecule has its own characteristic frequencies of vibrations, and those with the same frequencies will attract one another and vibrate in a correlated fashion, as is well known in chemistry.

This picture is beginning to look very different from the static, conventional lock-and-key model of how the enzyme and its substrate are 'fitted' to one another according to their complementary shapes.

In this new vision of the organism as a liquid crystalline continuum, electromagnetic energies of all kinds enable the molecules to intercommunicate and coordinate their activities. When intercommunication and coordination are perfect, we get to a state of *coherence* or wholeness.

But there is something special about the coherence of organisms that demands a quantum rather than classical description. For a quick summary of the difference between quantum and classical physics, see Box 13.1.

Box 13.1

Quantum versus classical physics

At the beginning of the 20th century, German physicist Max Planck discovered that energy, like matter, is discontinuous, and comes in discrete packets.

From this follows a quantum physics that's diametrically opposed to the classical, mechanical physics of René Descartes and Isaac Newton, which had dominated western scientific thought for more than three centuries up to then.

One of the fundamental breaks with the classical tradition is the need to abandon the linear, deterministic or 'causal' description of Newtonian physics for a quantum probabilistic description. The consequence is that it is impossible to predict or control the outcome of any single event. This is especially so at the level of atoms and molecules, where quantum fluctuations rule.

There are many other 'quantum paradoxes', such as electromagnetic radiation and matter being simultaneously wave-like and particle-like; the indeterminacy of the quantum state in which all potentialities are entangled until the moment it is 'observed'; and the possibility of instantaneous communication at a distance.

In my book, often referred to by friends as *The Rainbow Worm*, I presented a theory of *quantum* coherence to describe the organic whole, inspired by others who had made similar suggestions. Very few scientists believed quantum theory could have any relevance for biology when I first presented the idea in 1993, let alone quantum coherence. Since then, however, quantum coherence has been invoked to explain the synchronous firing of brain cells over large areas of the brain, which cannot be explained in conventional classical terms.

A more appropriate computation metaphor for how genes and proteins function in the organism could be quantum computing. More precisely, the organism, being quantum coherent, can carry out quantum computation to 'solve' infinitely complex molecular problems instantaneously by the quantum entanglement of infinitely many possible molecular states until it is 'observed', at which point, the 'solution' somehow drops out. Maybe that's how transposons are directed around the genome to carry out the natural genetic engineering required.

If you think that's merely hand waving, you are quite right, because no one *really* understands how it could be done, even though we know organisms are really doing it all the time.

Much effort is currently dedicated to researching quantum computing. But we may discover that when the perfect quantum computer

is eventually built, it will be useless for the instrumental purposes of human beings. Why? Simply because the perfect quantum computer cannot be controlled, just as organisms, as opposed to machines, cannot be controlled. That's why the mechanistic orientation of genetic engineering goes so much against the living grain.

From control to intercommunication

How molecules and cells can work together in the body is ultimately by intercommunication and reciprocity. In the ideal of quantum coherence, each gene and protein is as much in control as it is sensitive and responsive.

What does quantum coherence look like in the body? Perfectly fluid, dynamic and scintillating, which is why the genome is fluid. Think of the diverse activities in the body as 'quantum jazz', where every single player, however small, is improvising freely from moment to moment, and yet keeping in beat and in sync with the whole. There is no composer or conductor, the music emerges spontaneously as it is played, in endless variations that never exactly repeat.

Organisms are not controlled from the outside, that much is clear. They are not even controlled from the inside. There is no central controlling agency, no driver propelling or pulling the parts into action. There is no switching mechanism to turn any part on, no line-managers transmitting chains of command from top to bottom. There is no 'master-gene' turning development on, nor armies of 'control' genes that are not themselves also 'controlled'.

The ideal organic whole, quintessentially pluralistic and diverse, works by total participation and intercommunication, and is at once most coherent and most free, for the parts as for the whole.

In contrast, the dominant mechanistic culture reduces all to uniformity and homogeneity, it operates in rigid hierarchies of controller versus the controlled, and sees only inevitable opposition between the individual and the collective, the public and the private. This failure to see nature as organic, entangled wholes is the ultimate source of all wars and conflicts.

Taking the ideal organic whole seriously involves entirely new programmes of scientific research in sustainable agriculture, holistic health and sustainable energy use, with equal partnerships between scientists and traditional health practitioners and farmers in the field (see the ISIS-SGR-INES-TWN-Tebtebba discussion paper, *Towards a*

Convention on Knowledge, widely adopted by scientists and indigenous peoples' organisations).

I am often asked how I manage to keep on going in the face of what seems like insurmountable hurdles. The answer, very probably, is that I am not just the person you see, but an entangled, delocalised quantum entity, sustained by all who are dedicated to bringing about the paradigm change, wherever and whenever they exist, have existed, or have yet to exist.

To live with the fluid genome challenges us to shift from a mechanistic culture to a thoroughly organic one that would transform the very texture, quality and meaning of our lives; that would truly nurture individual freedom and diversity with universal love.

Glossary

Adaptive mutation, or Directed mutation The phenomenon whereby bacteria and yeast cells in stationary (non-growing) phase, have some way of producing (or selectively retaining) only the most appropriate mutations that enable them to make use of new substrates for growth.

Allele A particular variant of a **gene**.

Allergen A substance that causes the body to react hypersensitively to it.

Antibiotic resistance marker genes Genes coding for antibiotic resistance used in **genetic modification**. They allow the cells that have taken up the foreign **GM construct** to be selected with antibiotics, and frequently remain in the **genetically modified organism** and **transgenic line** created.

Autosome A **chromosome** other than the **sex-chromosome**, or sex-determining chromosome.

Bacteriophage Any virus that infect bacteria, also known as **phage**.

Cell line A supposedly genetically uniform population of cells derived from one individual cell. The genetic identity of all the cells is a fiction, as the genetic material is subject to many 'fluid genome' processes that constantly make cells genetically different from one another, and especially in culture. Both plant and animal cells are subject to large variations known collectively as **somaclonal variations**.

Chromosome A structural unit of genetic material consisting of a long molecule of DNA complexed with special proteins in **eukaryotes**, but not in **prokaryotes**.

Clone An identical copy of an individual organism, a cell, or a gene, or the totality of all the identical copies made from an individual organism, a cell, or a gene. In genetics, the clone implies identical genetic make-up to the original.

Conjugation The mating process in bacteria, which requires cell-to-cell contact being established, and in which genes are transferred between cells.

Directed mutation See **Adaptive mutation**.

DNA DeoxyriboNucleic Acid, the genetic material consisting of a long chain of individual units called nucleotides, each consisting of a base joined to a sugar and a phosphate group.

DNAse Deoxyribonuclease, an enzyme that breaks down **DNA**.

DNA sequence The sequence of bases in a stretch of **DNA**. There are 4 different bases, which are simply represented by the alphabets, A, T, C and G. An example of a DNA sequence is as follows: ATTTCCGC-TACGCGTTA... An **RNA sequence** is similar, except that the alphabet T is replaced by U.

DNA methylation A process in the cell which adds a methyl group, -CH₃, to the base cytosine or adenosine, resulting in gene-silencing, or failure of the gene to become expressed.

DNA polymerase An enzyme that makes **DNA**.

Dominant allele An **allele** which is expressed when only one copy is present in an individual, i.e., in heterozygous condition.

Ecosystem The totality of all plant and animal species that constitute an interdependent, interrelated community.

Enzyme A **protein** produced by living organisms that acts as a catalyst for a specific biochemical (metabolic) reaction.

Eukaryote The major class of living things, including all multicellular, higher organisms and some single-celled organisms, that have a **nucleus** in their cells, containing the chromosomes.

Frameshift mutation Mutation due to addition or deletion of base-pairs, so that the reading frame of the triplet code is shifted, giving rise to a completely different protein.

Gene A unit of heredity, usually a stretch of genetic material (**DNA** or **RNA**) with a defined function in the organism or cell, such as one for a protein. There are many genes within a genome. For example, the human genome is now found to contain about 30 000 genes, while the rice **genome** has about 50 000.

Gene amplification The process whereby **genes** or a sequence of **DNA** in the **genome** is greatly increased in number of copies.

Gene cloning The technique of making many copies of a **gene**, isolating the gene and identifying it.

Gene expression The synthesis of the gene-product or **protein** encoded by the **gene**.

Gene expression cassette A unit of **transgene** in genetic modification that allows the gene transferred to become expressed. It consists of a **promoter** and a **terminator** flanking the gene itself.

Gene silencing The process(es) whereby certain genes in the genome are prevented from being **expressed** by chemical modifications and other means.

Gene therapy Treating diseases by replacing the defective **gene**, either by incorporating a normal copy of the gene in the germ cells (egg or sperm) or in the embryo (germline gene replacement therapy), or by supplying copies of the normal **gene** to be taken up and incorporated into cells of the adult (somatic cell gene replacement therapy).

Genetic code The code establishing the correspondence between the sequence of bases in nucleic acids (**DNA** and the complementary **RNA**) and the sequence of amino acids in **proteins**.

Genetic determinism Determinism is the doctrine that all acts, choices and events are the inevitable consequence of antecedent sufficient causes. Genetic determinism is the doctrine that the organism is the inevitable consequence of its genetic make-up, or the sum of its genes.

Genetic engineering The manipulating of genetic material in the laboratory. It includes isolating, copying, and multiplying **genes**, recombining genes or **DNA** from different species, and transferring genes from one species to another, bypassing the reproductive process.

Genetic map A representation of the **genome** or any piece of **DNA** giving the order of **genes** or genetic markers and their sizes in numbers of base-pairs.

Genetic marker Any segment of **DNA** that can be identified, or whose **chromosomal** location is known, so that it can be used as a reference point to map or locate other genes. Any **gene** which has an identifiable **phenotype** that can be used to track the presence or absence of other genes on the same piece of **DNA** transferred into a cell.

Genetic modification A euphemism for the genetic engineering procedure (**transgenesis**) that makes **transgenic organisms**. This involves making artificial or modified genetic material (**GM** or **transgenic constructs**) that are inserted into the **genomes** of cells or embryos. The cell or embryo is regenerated to an organism, out of which a **GM line** or **transgenic line** is derived.

Genetically modified organism (GMO) An organism that has foreign **DNA** inserted into its **genome** by means of **genetic modification** in the laboratory.

Genome The totality of all the genetic material (deoxyribonucleic acid or **DNA**) in an organism, organised in a precise, though by no means

fixed or constant way. In the case of **viruses**, most of them will have ribonucleic acid or **RNA** as the genetic material.

Genotype The precise variant(s) of the gene(s) carried by an individual.

GM construct **Genetically modified** construct, an artificial combination of genetic material created in the laboratory.

GM line See **Transgenic line**.

Heterozygote An individual who has two different **alleles** of a **gene**.

Homozygote An individual who has two identical **alleles** of the **gene**.

Horizontal gene transfer The direct uptake of foreign genetic material by cells and incorporation into the cells' **genome**.

Insertion carcinogenesis Triggering cancer by the insertion of foreign **DNA** into **genes**, leading to abnormal gene expression.

Insertion mutagenesis Causing mutation by the insertion of foreign **DNA** into **genes**.

Insertion sequence element (IS) **Transposon**-like elements in bacteria.

Interrupted gene A **gene** whose coding sequence is interrupted at intervals by long stretches of non-coding sequences. The coding regions came to be known as *exons* and the non-coding regions as *introns*. This structure is now found to be characteristic of most eukaryotic genes. The number and size of introns vary greatly, and they are often much longer than the coding sequences. After **transcription**, the intron regions are removed, or *spliced* out from the **RNA transcript** before it is **translated** into protein.

Messenger RNA The **RNA** intermediate in **protein** synthesis containing a **transcribed** copy of the gene sequence that specifies the amino acid sequence of the **polypeptide** it encodes.

Metabolism The sum total of the chemical processes that take place in living organisms, resulting in growth, development, and all other forms of energy transformation.

Metabolite One particular chemical intermediate generated in metabolism.

Micro-organism An organism that can be seen only under a microscope, usually, an ordinary light microscope. It includes bacteria, mycoplasma, yeasts, single-celled algae and protozoa. Multicellular

organisms are normally not included, nor fungi apart from yeasts. **Viruses** are also not automatically included, as many scientists do not classify them as organisms.

Missense mutation Mutation resulting from a single base-pair substitution that changes one amino acid in the **polypeptide** translated.

Mitochondria Membrane-bound cellular organelles in which organic substrates derived from food are oxidized to provide energy for all kinds of vital activities. They carry their own complement of **DNA** and are replicated independently so that when the cell divides, each daughter cell will receive half of the mitochondria.

Mobile Genetic Element, also called **Transposon** or **Transposable Genetic Element** A sequence of **DNA** that can move (transpose) from one place to another in the **genome** of a cell.

Multigene families **Genes** that exist in multiple copies in the **genome**, from several copies to many thousands or hundreds of thousands of copies.

Mutagen A substance or agent that causes genetic **mutations**, or chemical alteration of the genetic material, **DNA**.

Mutation Any change in the base sequence of a gene.

Nonsense mutation **Mutation** that results in creating a stop codon in the **polypeptide** encoded by the gene, leading to premature termination of **translation** and a truncated **polypeptide** chain.

Nucleus A structure in the **eukaryote** cell bounded by a membrane, which contains the genetic material, in the form of **DNA** organized into **chromosomes**.

Oncogenes **Genes** associated with cancer.

Operon A multigene unit consisting of a regulator gene, a promoter-operator region and one or more other genes whose expression is regulated.

Paradigm A dominant model in science that shapes thought and practice in a pervasive way, which spills over into the general culture. For example, we can refer to the mechanistic (Newtonian) paradigm of linear, simplistic causation, or the Darwinian paradigm that emphasises competition and the survival of the fittest.

Pathogen Any agent that can cause disease.

Phage A bacterial virus

Phenotype The expressed characteristics, or an expressed character of an organism due to its genotype.

Plasmid A piece of parasitic genetic material found in a cell that can propagate using the cell's energetic and other resources.

Polymerase chain reaction, PCR Technique for amplifying a specific stretch of DNA present in trace amounts, using a heat-stable **DNA polymerase**.

Polypeptide or **Protein** A long chain of different amino-acids joined together by special chemical (peptide) bonds.

Polygenes The (hypothetical) many genes affecting a character, each having a small, additive effect on the character.

Prokaryote The class of living things, including all bacteria, that do not have a **nucleus** in their cell.

Promoter A piece of genetic material that acts as a **gene** switch, so that a gene can become **expressed** in the cell. It is the region at which the **RNA polymerase** binds to start **transcription**. Most promoters are located upstream of the **gene**, except that some eukaryotic genes have promoters internal to the gene.

Proto-oncogenes Cellular **genes** which, when mutated, or over-expressed become **oncogenes**.

Provirus A **virus** that has inserted its **genome** or a complementary copy of its **genome** into the host cell **genome**.

Recessive allele An **allele** which is not expressed unless two copies are present in the individual, i.e., in **homozygous** condition.

Recombination The formation of new combinations of **alleles** or new **genes** which occur when two homologous **DNA** or **chromosomes** break and exchange parts.

Recombination hotspot A frequent breakpoint in the **DNA** or **RNA**, which promotes recombination.

Reductionism The doctrine that complex systems can be completely understood in terms of its simplest parts. For example, an organism is to be completely understood in terms of its genes, a society in terms of its individuals, and so on.

Retrotransposon A mobile genetic element that depends on a **reverse transcription** step to move and to duplicate.

Reverse transcription The reverse of transcription - making a copy of

complementary **DNA** (cDNA) from an **RNA** sequence - catalyzed by the enzyme, reverse transcriptase.

Ribosome An organelle in the cell required for protein synthesis.

Ribosomal RNA RNA molecules which make up the **ribosome**.

RNA RiboNucleic Acid, similar to **DNA** except for the sugar in the nucleotide unit, which is ribose, instead of deoxyribose, and the base Uracil instead of Thymine. RNA is the genetic material for RNA **viruses**.

RNA editing The process in which the base sequence of the **RNA transcript** is changed by addition of bases to the **RNA** molecule or by chemical transformation of one base to another. This subverts the genetic information carried in the genes.

RNA polymerase An enzyme that makes **RNA**.

Somaclonal variation Genetic variations of plant cells arising in cell culture, due to enhanced genetic instability.

Somatic cell nuclear transplant A method of **cloning** in which the **nucleus** from an adult cell is transferred to an egg cell from which the egg's own **nucleus** has been removed, which is then stimulated to develop into an embryo.

Stem cells Cells that have the potential to multiply indefinitely and become many different cell types.

Substrate A chemical substance that takes part in a chemical reaction catalyzed by an enzyme.

Terminator A piece of **DNA** that acts as a stop signal for transcription.

Terminator technology A genetic modification technique that makes plants sterile.

Transcription The process of making a complementary sequence of the **gene** sequence in the **genome**, which is either used directly, as in case of **ribosomal RNAs** (rRNAs) and **transfer RNAs** (tRNAs), or is further processed into the messenger **RNA**, which is **translated** into **protein**. The process is catalyzed by the **enzyme** DNA-dependent RNA polymerase.

Transcription factors **Proteins** in **eukaryotes** that regulate the **transcription** of other **genes** by binding to regulatory sequences of the **gene**, interact with one another and with the **RNA** polymerase.

Transduction In genetics, the transfer of **genes** by viruses from one organism to another.

Transfer RNA RNA molecules which transfer specific amino acids to the messenger RNA so that the polypeptide it encodes can be synthesized.

Transformation In genetics, the transfer of **genes** by one organism taking up **DNA** belonging to another organism of the same or different species.

Transgene A **gene** that's transferred into an organism in **genetic modification**.

Transgenic line A genetically uniform variety of plants or animals bred from a transgenic organism.

Transgenic organism An organism created by **genetic engineering**, in which one or more foreign **genes** have been incorporated into its genome, also referred to as **genetically modified organism**.

Translation The step in **protein** synthesis in which the **messenger RNA** directs the synthesis of a **polypeptide** of a particular amino-acid sequence by "decoding" the **genetic code**.

Transposon See **Mobile genetic element**.

Vector A carrier or transmitter, of **genes** or of disease. Artificial vectors are made in **genetic engineering** for multiplying and transferring **genes** into **genomes**.

Virus A parasitic genetic element enclosed in a **protein** coat that can replicate in cells, and form infectious particles, or remain dormant in the cells. Its genetic material can become integrated into the cell's **genome** to form **provirus**.

Virulence Ability (of **pathogens**) to infect organisms and cause disease.

X-Chromosome One of the sex-determining **chromosomes**. The human female carries two X chromosomes, whereas the male carries one X and one Y chromosome.

Xeno-tranplantation Organ, cell or tissue transplant between species.

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